

BASE24 Release 6.0 Version 6

Implementation Guide



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Preface

Document Control

A description of the document revisions is shown below. Any changes of text from one version to another are indicated with the symbol “|” in the outside margin.

Release 6.0 Version 6 (C. Cooney, D. Barton, J. Dargy, L. Falk, K. Koeppen, R. Schieuer, P. Spanke, J. Steffes, C. Rink, J. Urbanec, L. Watson, J. Wishart)
– Nov-2005

Information was added to Section 4, General Enhancements, General Changes, File Conversions. A note was added regarding the Store-and-Forward (SAF) file for BIC ISO (M. Ahrens)

Added information to Section 4, Optional Enhancements, Complex Passwords. The information was added to User/Operational Impacts. (C. Rink)
- November 2006

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1: Introduction

The ***BASE24 Release 6.0 Version 6 Implementation Guide*** is the primary reference document for information needed to upgrade existing BASE24 release 6.0 version 5 systems to BASE24 release 6.0 version 6.

This section provides an overview of the planning requirements for BASE24 users migrating to release 6.0 version 6. It also provides guidance for using this manual.

Also included in this section is a listing of other documents that a BASE24 user should obtain for implementation planning, the general software requirements for release 6.0 version 6, and a glossary of terms used throughout the guide.

Implementation Planning

BASE24 users must develop their own project plans for migrating their BASE24 systems to release 6.0 version 6. Project plans should address such issues as staffing and training requirements, host external message changes, device programming enhancements, hardware and software requirements, and the migration method to be used.

ACI has provided BASE24 users with the ***BASE24 Release 6.0 Version 6 Implementation Guide*** to assist in preparing project plans. This guide includes a list of services available to the BASE24 user and information that existing BASE24 users will need to upgrade to release 6.0 version 6.

BASE24 Release 6.0 products include the following:

- BASE24-atm
- BASE24-pos
- BASE24-teller
- BASE24-remote banking
- BASE24 Interchange Interfaces
- BASE24 Value-Added Products:
 - BASE24-card
 - BASE24-customer service
 - BASE24-InfoBase
 - BASE24-inventory

Who should use this guide?

There are two audiences for this guide: those who plan and manage the migration process and those who implement the migration to release 6.0 version **6**. In either case, the reader should be experienced with BASE24 software, HP NSK products, and the BASE24 user's current configuration. This guide is not directed at new BASE24 users who will be installing a BASE24 application for the first time. New users should refer to the formal BASE24 publications listed under the Publications heading in Section 3.

To migrate to release 6.0 version **6**, users must be on release 6.0 version 5. Refer to the ***BASE24 Release 6.0 Version 5 Implementation Guide*** to assist in planning the tasks necessary to get to release 6.0 version 6.

Using this guide, you can do the following:

- Determine what ACI services are available to assist in the migration to release 6.0 version 6.
- Develop a project plan to migrate to release 6.0 version 6.
- Determine staffing and training needs.
- Gain a detailed understanding of the release 6.0 version 6 enhancements.

Software Requirements

HP NSK Software Requirements

ACI recommends that the BASE24 user be on a current HP NSK-supported operating system to obtain maximum performance and support. However, BASE24 release 6.0 is currently compatible with the HP NSK Guardian D45 Operating System. Customers choosing to implement Format 2 File support are required to be on a minimum Operating System level of D46. In addition, BASE24 release 6.0 requires TMF for several files.

XPNET

ACI requires that the BASE24 user be on an XPNET 3.0 or 3.1 system.

Conversion Planning

BASE24 users must develop their own project plans for upgrading their systems to release 6.0 version 6. Project plans should address such issues as staffing and training requirements, host external message changes, device programming enhancements, hardware and software requirements, and the migration method to be used.

In general, all changes for release 6.0 version 6 can be obtained by requesting the latest fixes from ACI and then retrofitting all fixes to the user's release 6.0 version 5 modules.

Conversion Program Guidelines

Users of previous releases of BASE24 and previous versions of BASE24 Release 6.0 must run file conversion programs in order to migrate to the latest version of release 6.0. Information about specific conversion programs can be found in subsequent sections of this document under the applicable enhancements. Following are general conversion program guidelines.

The *UC04 subvolumes (e.g., AT60UC04, BA60UC04, PS60UC04) contain the conversion programs for users migrating from release 5.3 to the latest version of release 6.0. The conversion programs on these subvolumes are kept up-to-date with each new version of release 6.0.

The *UC05 subvolumes (e.g., AT60UC05, BA60UC05, PS60UC05) contain the conversion programs for users migrating from release 6.0 version 1 to release 6.0 version 2. No changes are made to these programs other than necessary fixes. Also residing on these subvolumes are the Transaction Security Services (TSS) conversion programs, which are used to move key data to the new TSS database. The TSS conversion programs are required only if the user is implementing TSS.

The *UC06 subvolumes (e.g., AT60UC06, BA60UC06, PS60UC06) contain the conversion program for users migrating from release 6.0 version 2 to release 6.0 version 3. No changes are made to these programs other than necessary fixes.

The *UC07 subvolumes (e.g., AT60UC07, BA60UC07, PS60UC07) contain the conversion program for users migrating from release 6.0 version 3 to release 6.0 version 4. No changes are made to these programs other than necessary fixes.

The *UC08 subvolumes (e.g., AT60UC08, BA60UC08, PS60UC08) contain the conversion program for users migrating from release 6.0 version 4 to release 6.0 version 5. No changes are made to these programs other than necessary fixes.

The *UC09 subvolumes (e.g., AT60UC09, BA60UC09, PS60UC09) contain the conversion program for users migrating from release 6.0 version 5 to release 6.0 version 6. No changes are made to these programs other than necessary fixes.

If a release 5.3 user wants to convert to the latest version of release 6.0, the conversion programs on *UC04 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run.

If a release 6.0 version 1 user wants to convert to release 6.0 version 2, the conversion programs on *UC05 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run.

If a release 6.0 version 2 user wants to convert to release 6.0 version 3, the conversion programs on *UC06 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run.

If a release 6.0 version 3 user wants to convert to release 6.0 version 4, the conversion programs on *UC07 should be run.

If a release 6.0 version 4 user wants to convert to release 6.0 version 5, the conversion programs on *UC08 should be run.

If a release 6.0 version 5 user wants to convert to release 6.0 version 6, the conversion programs on *UC09 should be run.

If a release 6.0 version 1 user wants to convert to release 6.0 version 6, the conversion programs on *UC05 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run. Following this, the user should run the conversion programs on *UC06, *UC07 and on *UC08. Finally, the user should run the conversion programs on *UC09.

Other Documents

In addition to this guide, the reader can obtain other BASE24 and NET24-XPNET publications that cover operational and product-specific information required to fully understand and implement the release 6.0 version 6 software. Such manuals include the formal publications mentioned under the Publications heading in Section 3, the event message documentation located on the BA60LOGM and OK60LOGM subvolumes, and the following documents:

BASE24 Classic 6.0 Installation Guide (Informal) - BA-AE000-60

A primary reference document for information about the installation of a BASE24 release 6.0 system from scratch. This document is included as a Microsoft Word document and is shipped on the BASE24 6.0 CD-ROM.

The BASE24 Classic 6.0 Installation guide is also provided on the software tape with the code. The document is located on the BA60UD04 subvolume as a file called BASE24. On this subvolume there is also a readme file called AAREAD04, which details how to move the install document to your hard drive and open it with Microsoft Word.

BASE24 Release 6.0 Interchange Interface Implementation Guide (Informal) - SW-DS0000-60

Describes the function and purpose of ACI's Switch 6.0 interfaces. It explains how customers can implement Switch 6.0 interfaces in their existing networks as well as in a BASE24 6.0 network. It also explains how customers can implement their Fortify.1/Fortify.3 Interfaces in a 6.0 network. This document is intended to assist customers in configuring and controlling their Switch 6.0 Interchange Interfaces in all BASE24 release networks and their Fortify.1/Fortify.3 Interchange Interfaces in a BASE24 release 6.0 network. This document is included as a HP NSK TGAL document and is shipped with the release 6.0 Interchange Interface software. This document is named NSTLSW60 and is located on the SW60UD04 subvolume.

Note: This guide is to be used with BASE24 releases 4.0, 5.0, 5.1.2, 5.3, and 6.0.

User Bulletin for BASE24 Release 6.0 Version 6 (Informal)

A primary reference for information about all changes made to each BASE24 release 6.0 version 6 source module. This documentation is included as a set of HP NSK TGAL documents and is shipped with BASE24 release 6.0 version 6 software. This documentation consists of a number of files located on the BA60UB04 subvolume. All files ending in FH contain fix history sections for all changes made. All files ending in FP contain a list of all fixed Procedures for each change made to all modules, except those files that have been resequenced.

Release 6.0 Data File Impacts Document (Informal)

A primary reference document for information about the data file impacts and the methods that can be used to upgrade to release 6.0 version 6. This document is included as a HP NSK TGAL document and is shipped with the release 6.0 version 6 software. This document is named AAREAD09 and is located on the BA60UC09 subvolume.

HELP24 and HELP24 InfoLink

HELP24 and HELP24 InfoLink are excellent sources of assistance and information. The following access options appear on the support page of the ACI Worldwide Inc. Internet website (<http://www.aciworldwide.com/support>).

- A current listing of HELP24 phones worldwide.
- An option to send an e-mail message to HELP24.
- An option to sign up for HELP24 InfoLink.
- An option to access HELP24 InfoLink once you have received a user ID and password.

Once you can log on to HELP24 InfoLink, you can do the following:

- Access the HELP24 database to open a new Clarify case.
- Access installation and fix release procedures online.
- Access all BASE24, BASE24-es, and NET24-XPNET documentation online.
- Search the site for information about BASE24.

The **HELP24 eSupport Users Guide** contains procedures for opening a new Clarify case, accessing the Clarify database, and accessing documentation on-line. Once you log on to HELP24, the path to the **HELP24 eSupport Users Guide** is Support Center> InfoLink > HELP24 eSupport Users Guide. Once you log on to HELP24, the path to the installation procedures is Support Center> InfoLink > Product Info >BASE24> BASE24 6.0 Install Info and Guides. The menu displayed gives you three options:

- Pre-Install Checklists- for BASE24 6.0 and ITS
- Install Guides—Procedures for installing a software release or version for the first time.
- Retail Install Guides—Procedures for installing Retail 3.1 or Check Auth 3.2.

Once you log on to HELP24, the path to the Fix Information is Support Center> InfoLink > Fix Information > BASE24

Definition of Terminology

The following list has been defined to assist the reader in understanding the terms referred to within this document.

6.0 Interchange Interfaces

The BASE24 Interchange Interfaces developed in conjunction with release 6.0. These replace the Fortify.3 Interchange Interfaces and can be used with releases 4.0, 5.0, 5.1, 5.1.2, 5.3 and 6.0 of BASE24.

Accountholder

A person for whom a financial account has been established for some purpose. Accountholders may or may not use cards to initiate transactions on their accounts.

BASE24 User

An entity that has licensed BASE24 software products.

Cardholder

A person who has been issued a plastic card for the purpose of initiating transactions.

Consumer

A person who utilizes services offered by a BASE24 user who may or may not require a plastic card.

Conversion Program

A utility that reads the records from an input file, adds new fields and initializes appropriate fields and then writes the record to an output file. The input file (existing file) and output file (converted/initialized file) are different files. This method (versus the initialization program and online conversion) is required when the record structure changes (e.g., increasing the record length, adding/deleting fields from segments, etc.). This method can also be used to initialize existing user fields in a record.

This method requires system down time to convert the file (also requires an operator to manually rename the file that has been converted), move in the new objects of modules that access the file (e.g., device handler, auth, etc.) and then restart the objects. This method impacts all BASE24 users who use the file being converted even if they are not taking advantage of the enhancement for which the modifications are being made.

Dynamic Key Management (DKM)

The automated distribution of encryption keys between BASE24 and an outside entity.

Extended Memory Table file (EMT)

A data file produced by the EMT Build utility program. Once the EMT has been built with the EMT Build utility, processes that use the EMT must be warmbooted (or started if not running).

Extended Memory Table Build utility (EMT Build)

The EMT Build utility is a stand-alone program used to build tables in extended memory from information contained in disk files. This utility can build hash tables, radix tables, or custom designed tables. Refer to the **BASE24-atm Transaction Processing Manual**, **BASE24-pos Transaction Processing Manual**, or the **BASE24 ITS Kernel Configuration Manual** for additional information about this utility.

Online (Inline) Conversion

A utility that reads and updates each record in a file to initialize user fields that have been redefined as new fields for use by an enhancement. The file is converted in place (i.e., a new file is not created). The record is read, updated and then rewritten to the same file. This method (versus the conversion program and online conversion) cannot be used when the record structure changes (e.g., increasing the record length, adding/deleting fields from segments, etc.).

The online conversion code will be executed each time a record is accessed (versus the conversion and initialization programs that are run once). When a record is read, the process will determine if the value of a new field is valid. If the value is not valid, the field will be initialized to the default value. The database is gradually converted as records are read and updated.

This method requires an operator to stop a process, roll in the new object and start processes that use the files that are impacted. This method does not require any additional steps to be performed by an operator (e.g., running conversion/initialization programs, renaming files).

Operator

A person who carries out network operations functions, such as starting and stopping the system, entering ONCF commands, controlling logging, dealing with event messages, performing refreshes and extracts. The term *operator* also can refer to persons using BASE24 screens for such purposes as file maintenance, perusal, etc.

2: Functionality

This section contains an overview of the BASE24 release 6.0 version 6 enhancements.

Functionality has been organized by product, and within each product, according to whether the enhancement is a general enhancement or an optional enhancement.

- General enhancements are those enhancements that are inherent in the release 6.0 version 6 offerings. These enhancements affect all BASE24 users. BASE24 users must review the features that are part of these enhancements to ensure an understanding of the new processing and ensure proper configuration within the BASE24 environment.
- Optional enhancements are those enhancements that can be enabled or disabled in release 6.0 version 6. BASE24 users can implement these enhancements based on their own BASE24 processing and marketing requirements.

Enhancements are described in this section by product as follows:

- BASE24 Base (applicable to all products)
- BASE24-atm
- BASE24-pos
- BASE24-teller
- BASE24-remote banking
- BASE24 Interchange Interfaces
- BASE24 Value Added Products

BASE24 users should acquaint themselves with the BASE24 Base section (Section 4) as well as the applicable product-specific sections in order to obtain an understanding of the overall release 6.0 version 6 offering.

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BASE24 Base Enhancements

The BASE24 Base product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24 Base product, refer to Section 4.

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General Enhancements

The following enhancements are common to all products or provide the base for product-specific enhancements. As these enhancements are inherent in release 6.0 version 6, BASE24 users must review them to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

General Changes

- Changes for all BASE24 release 6.0 version 6 enhancements affecting the DDL files and the BASE24 AFT Subsystem files (i.e., screens, requesters, servers) have been merged and cataloged under one SCN for each affected file. Applying changes for individual enhancements to the screen files does not allow BASE24 users to identify which changes go with a particular enhancement, as screen files do not have sequence numbers at the end of each line. Since the screen file changes usually involve new fields being added, all associated requester, server and DDL changes are required. Even though requester, server and DDL files do contain sequence numbers, all SCNs would need to be applied in order to accept the changes to the screen files. Therefore, all requester, server and DDL file changes, for all enhancements, were done under one SCN per file as well.
- The Transaction Based Pricing (TBP) reports have been modified to support Mobile Top-Up transactions for BASE24-pos. These changes impact all BASE24 TBP customers. All BASE24 TBP customers must implement the updated version of the TBP report objects and TBP documentation. For further information, see Mobile Top-Up for POS in Section 6 where the impacts of this enhancement are described.
- The Security File (SEC) has been expanded to include space to store up to 10 passwords (current and 9 previous). The Security File requester and server have been modified to accommodate the larger file and to store passwords in the new data fields. Other changes implemented in the Security File requester and server are discussed under Complex Passwords in Pathway in Section 4.
- The Card Prefix File (CPF) has been modified to include additional fields on screen 8, for the processing of manually-entered transactions. The Card Prefix File requester and server have been updated to process these new fields. For further information, see Issuer Declined Manually Entered Transactions in Section 6 where the impacts of this enhancement are described.
- The Super Extract process has been modified to avoid potentially monopolizing CPU time during the ending of daylight savings time. Previously, an extract timer expiring at the end of daylight savings time could lock up a CPU for up to one hour, while the system clock caught up with the timer's expiration time. The Super Extract process no longer enters into this situation, unless the wait time is less than one second.

- DDLGISO has been modified to include the Scoring Engine Master File (SEMF) definition for use with BASE24-prm version 7.0 and BASE24-fhm version 6.0.

Optional Enhancements

The following BASE24 Base release 6.0 version 6 enhancements are optional. BASE24 users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

Complex Passwords

Recent legislative initiatives and card association mandates require enforcement of complex passwords. For example, Visa and MasterCard's Payment Card Industry (PCI) Data Security Standards require passwords to be case-sensitive, a mix of alpha, numeric, upper/lower case and special characters, and restrictions against re-use of passwords. The password must be stored using triple DES or SHA-1 hash encryption.

Current functionality related to passwords in BASE24 has been in place for a number of years. However, to comply with all of the PCI standards, enhancements have been made to enforce the following password attributes during password selection:

- Case sensitive
- Number of alpha characters required
- Number of numeric characters required
- Number of special characters required
- Number of times in one day a password can change
- Number of old passwords to store
- No reuse of up to 10 previously used passwords

Passwords are now stored using an irreversible hashing algorithm (SHA-1).

Security Audit Reports

Recent legislative initiatives and Card Association mandates such as Sarbanes-Oxley, Gramm-Leach-Bliley Act, and the Payment Card Industry (PCI) Data Security Standards require tracking user's access and file maintenance activities. As a result, auditors require regular review of users' access and users' access rights. To help address these requirements, new COBOL reports have been created to extract information from the Online Maintenance File (OMF), Security (SEC), and User/FI Relationship (UFIR) files.

The *Logon/Logoff Report* tracks users who have logged on to the BASE24 PATHWAY system, and who have exited through the Logon screen (using SF16). Configuration options exist to display the entire OMF record image, including User, Group, LNET, FIID, REGN, BRCH, timestamp, terminal id, and reasons for failed

logon attempts. Other options exist to limit the selection of records to a single user group, or user group/number. This report enables security administrators to track and audit user's access to the system.

The *OMF File Access Report* provides information on users who have performed BASE24 file maintenance (adds, updates, and deletes) through PATHWAY. The report provides information from one day's OMF file(s), displayed in timestamp order, which can include Application ID (File ID), User, Group, LNET, FIID, REGN, BRCH, timestamp, and terminal id. Configuration options can limit the selection of records based combinations of File IDs, type of maintenance performed, and Group/User numbers. If the "detail" option is chosen at run time, the record image for each record will also be included in the report. The report enables security administrators to investigate and reconstruct user actions for forensic or troubleshooting purposes.

The *Logon/Logoff Report* and the *OMF File Access Report* use the current layout of the OMF file. No changes have been made to the OMF, and no changes were made to the current BASE24 PATHWAY system to write additional OMF records or include additional data. The reports will reside in a new BA60RPT subvolume added in Release 6.0 version 6.

Security administrators and auditors can regularly review user access rights with the new *Security File Last Logon Report* and the *Security File Access Report*. These reports extract information from the Security (SEC) and User/FI Relationship (UFIR) files. The reports display all users defined in the BASE24 Pathway environment, their most recent logon status, and the users' access rights. The *Security File Last Logon Report* sorts all of the SEC records by last logon date; the *Security File Access Report* can be run for a group, a user, or the whole file. If the option to include detail information is specified for either sort option, the following information is included below each user's summary record:

- A listing of each FIID, Region and Branch to which the user has been granted access;
- Start and end times, XPNET nodes, and stations;
- A table of the user's RADU (Read, Add, Delete and Update) access privileges for each file and screen number.

BASE24-atm Enhancements

The BASE24-atm product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-atm product, refer to Section 5.

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General Enhancements

There are no general BASE24-atm enhancements inherent in the release 6.0 version 6 offerings.

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Optional Enhancements

The following BASE24-atm release 6.0 version 6 enhancements are optional. BASE24-atm users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

Cutover Performance Improvements

BASE24 has been modified to change ATM Authorization to process dates differently at cutover. ATM Authorization has been modified to cutover the IDF dates similar to Settlement, instead of reading the IDF during cutover. This change should provide improved performance during BASE24-atm cutover.

Diebold Bulk Note Acceptance

The BASE24-atm Diebold 10xx/478x Device Handler has been enhanced to support Diebold's Bulk Note Acceptor (BNA) module.

The Bulk Note Acceptor unit utilizes sophisticated note recognition technology, which counts and verifies bundles of notes, allowing the customer to deposit large amounts of cash very quickly at their convenience. Unlike traditional deposits via the ATM, the deposit can be credited on-line giving the customer an immediate proof of deposit. With the capability to handle different denominations of notes, the Bulk Note Acceptor can also be configured for multiple currencies. Deposited bills will be transported into a storage bin contained in the security enclosure. This enhancement also provides for all the new state-of-health messages that may be generated by this new module.

Prerequisites: The Diebold 10xx/478x Device Handler is required

The above enhancement is considered a billable item. Please contact your Customer Manager for further information.

Diebold EMV Offline PIN Synchronization

BASE24-atm has been enhanced to provide the functionality of sending an EMV "PIN Update" script back to the EMV chip card when a "Change PIN" transaction is performed at an ATM.

EMV cards have the capability to hold the cardholder's PIN within the Integrated Chip Card (ICC). This allows "offline PIN verification" (i.e. where the PIN entered by the cardholder is verified by the ICC) to be offered as a Cardholder Verification Method (CVM) for electronic transactions. Many payment terminals will not be enhanced to offer offline PIN verification, and so "online PIN verification" (i.e., where the PIN

entered by the cardholder is verified by the issuer's system) will continue to be widely used.

Where both online and offline PIN verification can be used as the CVM for an EMV card, cardholder confusion would result if the PIN held on the card's ICC differed from the PIN held on the issuer's system. Also, if a cardholder exceeds the allowable bad PIN tries when entering the PIN, a mechanism is required to allow the bad PIN try counter to be reset. Therefore, it is critical that card issuers provide offline PIN management functionality.

The Diebold 10xx/478x Device Handler EMV Extension has been modified to send an EMV PIN Update script down to the ATM/card, if a PIN Change transaction is performed by an EMV card, and to support an EMV PIN Unblock script. In addition, the Diebold EMV Extension has been enhanced to handle error conditions resulting from the above two situations. These error codes are carried in the Issuer Script Results data in either an explicit Solicited Status Message or in the Last Transaction Status buffer of the next request message. The Device Handler then parses this data and if the script results indicate that a script has failed, the Device Handler makes a decision as to whether a reversal is appropriate.

For more information on EMV processing, please refer to "EMV Support" in the BASE24 Base Optional Enhancements of the BASE24 Release 6.0 Implementation Guide (Version 2).

Prerequisites: The BASE24-atm ATM Auth EMV Extension, the Diebold 10xx/478x Device Handler, and the Diebold 10xx/478x Device Handler EMV Extension.

NDC+ 5XXX Device Handler AKDS Certificate Support

The BASE24-atm NCR 5XXX Device Handler has been enhanced to support Automated Key Distribution (AKDS) functionality in those cases where the ATM software vendor utilizes the NDC+ specification to facilitate remote key downloading to ATMs equipped with certificate-based encrypting PIN pads (EPPs). The device handler was previously enhanced to support signature-based EPPs.

AKDS automates the otherwise manual process of initializing and re-keying ATM master keys by eliminating the manual processes related to key generation, dual custodianship and manual key loading at the ATM. AKDS also facilitates compliance with "unique key per ATM" guidelines specified by ANSI X9 TG-3 key management standards mandated by networks and card associations. The AKDS key distribution protocol uses public-key cryptography (RSA encryption) to generate and transport the initial master key (the "A-key"). Use of digital certificates and digital signatures is also used for authentication and to protect against tampering of data.

Prerequisites: The BASE24 Automated Key Distribution System (AKDS) and the NCR 5xxx Device Handler are required.

Shared (Wincor) 912 AKDS Signatures

BASE24-atm has been enhanced to support Automated Key Distribution (AKDS) functionality in those cases where the ATM software vendor utilizes the Shared 912 Remote Key Load (RKL) specification to facilitate remote key downloading to ATMs equipped with signature-based encrypting PIN pads (EPPs). Shared 912 certificate-based ATMs will be supported in a later version.

This specification is managed and used by Wincor Nixdorf, although it is open to all vendors. AKDS automates the otherwise manual process of initializing and re-keying ATM master keys by eliminating the manual processes related to key generation, dual custodianship and manual key loading at the ATM. AKDS also facilitates compliance with “unique key per ATM” guidelines specified by ANSI X9 TG-3 key management standards mandated by networks and card associations. The AKDS key distribution protocol uses public-key cryptography (RSA encryption) to generate and transport the initial master key (the “A-key”). Use of digital certificates and digital signatures is also used for authentication and to protect against tampering of data.

Prerequisite: The BASE24 Automated Key Distribution System (AKDS) and the Diebold 10xx/478x Device Handler are required.

Shared (Wincor) Diebold 10xx/478x EMV Extension

BASE24-atm has been enhanced to support the EMV chip-based transaction processing for those vendors that utilize the Shared 912 EMV specification.

A new Shared 912 EMV extension module for the Diebold 10xx/478x Device Handler has been developed to handle all EMV processing. The Shared 912 EMV module is responsible for scanning request messages for EMV data, responses and formatting the appropriate message sent back to the ATM.

In addition, the Shared 912 EMV extension module processes the EMV data in solicited and unsolicited status messages and any special reversal processing needed because of a failure from the device’s EMV Smart Card Application.

For more information on EMV processing, please refer to “EMV Support” in the BASE24 Base Optional Enhancements of the ***BASE24 Release 6.0 Implementation Guide (Version 2)***.

Prerequisites: The BASE24-atm ATM Auth EMV Extension and the Diebold 10xx/478x Device Handler are required.

The above enhancement is considered a billable item. Please contact your Customer Manager for further information.

Shared (Wincor) NDC+ AKDS Signatures

BASE24-atm has been enhanced to support Automated Key Distribution (AKDS) functionality in those cases where the ATM software vendor utilizes the Shared NDC+ Remote Key Load (RKL) specification to facilitate remote key downloading to ATMs equipped with signature-based encrypting PIN pads (EPPs). Shared NDC+ certificate-based ATMs will be supported in a later version.

This specification is managed and used by Wincor Nixdorf, although it is open to all vendors. AKDS automates the otherwise manual process of initializing and re-keying ATM master keys by eliminating the manual processes related to key generation, dual custodianship and manual key loading at the ATM. AKDS also facilitates compliance with “unique key per ATM” guidelines specified by ANSI X9 TG-3 key management standards mandated by networks and card associations. The AKDS key distribution protocol uses public-key cryptography (RSA encryption) to generate and transport the initial master key (the “A-key”). Digital certificates and digital signatures are also used for authentication and to protect against tampering of data.

Prerequisites: The BASE24 Automated Key Distribution System (AKDS) and the NCR 5xxx Device Handler are required.

Support for Response Code 85 in ARPC

BASE24-atm processing supports the unblocking and changing of the offline PIN on an EMV card. This functionality is collectively known as offline PIN management. Offline PIN management transactions are also supported by some interchanges, e.g. VisaNet, EPS-Net, and LINK.

When an offline PIN management transaction is approved, Visa requires that the issuer returns an ISO response code of ‘85’ to the acquirer in field P-39 of the VisaNet response message. Visa has two further requirements for the ISO response code of ‘85’:

- When the issuer generates the Authorization Response Cryptogram (ARPC) for an approved offline PIN management transaction, the ISO response code used in the ARPC generation must be set to ‘85’. The issuer then returns this value to the acquirer in the VisaNet ISO Interface message.
- When the acquirer returns the response to the ATM for an approved offline PIN management transaction, the Authorization Response Code (ARC), used by the ATM to determine whether the transaction has been approved, must be set to ‘85’. The acquirer receives this value from the issuer in field 39 of the VisaNet ISO Interface message.

This enhancement adds support for Visa’s offline PIN requirements.

Prerequisites: Corresponding changes have been made to the VisaNet ISO Interface to support response code '85'.

Tidel EMV

BASE24-atm has been enhanced to support the Tidel implementation of EMV.

A new Tidel EMV extension module for the Tidel Device Handler has been developed to handle all EMV processing. The Tidel EMV extension module is responsible for handling request messages with EMV data and formatting the appropriate message sent back to the ATM.

In addition, the Tidel EMV extension module processes the EMV data in solicited and unsolicited status messages and any special reversal processing needed because of a failure from the device's EMV Smart Card Application.

For more information on EMV processing, please refer to "EMV Support" in the BASE24 Base Optional Enhancements of the ***BASE24 Release 6.0 Implementation Guide (Version 2)***.

Prerequisites: The BASE24-atm ATM Auth EMV Extension and the Tidel Device Handler are required.

The above enhancement is considered a billable item. Please contact your Customer Manager for further information.

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BASE24-pos Enhancements

The BASE24-pos product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-pos product, refer to Section 6.

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General Enhancements

The following BASE24-pos enhancements are inherent in the release 6.0 version 6 offerings. All BASE24-pos users must review these enhancements in order to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

Expand the Number of RTBL and AST Entries

The number of allowable entries in the Routing Table File (RTBL) has been increased from 28 to 42 and the Authorization Selection Table (AST) has been increased from 36 to 48 authorizers to provide increased routing flexibility within BASE24-pos.

Prerequisites: BASE24-pos Authorization.

General Changes

- Prefix Routing Changes

Routing changes have been made to support new prefix ranges for Discover and MasterCard branded Diner's cards. Discover has added additional Discover prefix ranges in the 650000-650999 range. MasterCard now allows Diner's Club International cardholder transactions to be delivered via a MasterCard interface for U.S. and Canadian merchants. This change affects the routing of Diner's Club cards that begin with a prefix of '36'. All other Diner's Club prefixes will still be routed to Diner's via another interface. The BASE24 algorithms (ALGO) have been modified to support the new Discover prefix ranges and MasterCard card ranges. SPROUTE was also modified to recognize the new BIN ranges.

Prerequisites: BASE24-pos Authorization.

- Authorization for a Lesser Amount

BASE24-pos has been enhanced to support issuer authorizations for a lesser amount for a purchase, pre-authorization purchase, mail-phone order purchase, purchase with cash back and cash advance transactions. With this enhancement, issuers are now able to respond to authorization messages with an approval for a portion of the amount requested. This functionality enables the remaining transaction amount to be paid by other means, thereby reducing decline rates and enhancing the consumer and merchant experience at the point of sale.

SPDH has also been enhanced to support these same transactions.

Prerequisites: BASE24-pos Authorization.

Optional Enhancements

The following BASE24-pos release 6.0 version 6 enhancements are optional. BASE24-pos users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

Alternate Host Routing

The ISO Host Interface has been enhanced to manage an alternate authorizer destination when the ISO Primary Issuer destination is not available. If the HISO Primary Issuer destination is not available, the transaction will be sent to an alternate authorizer (either a BASE24 or a non-BASE24 solution) rather than using stand-in processing.

Prerequisites: The BASE24-pos ISO Host Authorization.

CRT Auth Override During Settlement / Cutover

The BASE24-pos CRT Auth Device Handler has been enhanced to allow referrals to be processed reflecting the correct date and time, regardless of other system events that have occurred such as Settlement or Cutover.

Prerequisites: The BASE24-pos CRT Administration System and CRT Auth Device Handler.

Cutover Performance Improvements

BASE24 has been modified to change POS Router Authorization to not close and reopen the LCONF at cutover. BASE24 has also been modified to change POS Router Authorization to process dates differently at cutover. POS Router Authorization has been modified to cutover the IDF dates similar to Settlement, instead of reading the IDF during cutover. This change should provide improved performance during BASE24-pos cutover.

Prerequisites: BASE24-pos.

Hypercom Enhancements

The Hypercom Device Handler (HPDH) has been upgraded to support several features of the latest Hypercom specification, the Hypercom Message Specification "The System", v3.42, Aug 4, 2003. The HPDH now supports the following:

- Additional Data Request / Response Values of C04, C05, C06, and C07

- Additional Processing Codes for EBT Cash Card Purchase, EBT Cash w/Cash Back, EBT Cash Only, EBT Food Stamps, Prior EBT, and EBT Food Stamp Return
- New CVV2 data table which enables the device to ensure the CVV2 is forwarded to the authorizer, and the result code logged
- EBT Terminal Data Table which is described in detail in section 6.28.4.34 of the Hypercom Message Specification document.
- EBT Response Data Table which is described in detail in section 6.28.4.35 of the Hypercom Message Specification document
- New EBT transaction formats of EBT Cash Only, EBT Sale and EBT Sale and Cash. These message formats are described in detail in sections 7.13, 7.14 and 7.15 respectively, of the Hypercom Message Specification document
- Level II Commercial Card Indicator Table
- CPS 2000 Data Table
- Surcharge Amount Table
- Payment Plan Table

For further information, see **HPDH Upgrade to Latest Initialization Specifications** and **HPDH Upgrade to Latest Initialization Specifications** in Section 6 where the impacts of these enhancements are described.

Prerequisites: The BASE24-pos Hypercom Device Handler.

Interac Online Payment

BASE24-pos has been enhanced to support the Interac Online Payment (IOP), previously known as iDebit, which is an Internet payment service that allows customers to securely pay for online shopping over the Internet, directly from their bank account. Traditionally, payment by credit card was the only option available at most online merchants. IOP offers an alternative payment method to Canadian Interac customers who do not wish to provide card information over the Internet due to security and privacy concerns.

Prerequisites: BASE24-pos, Interac Interface and the BASE24-pos SPDH Device Handler.

ISO Host Interface Enhanced Reversal Routing

The BASE24-pos ISO Host Interface has been enhanced to optionally route a reversal transaction to the same interchange interface process that authorized the original transaction, when the host is the transaction acquirer. This enhancement enables the reversal to be successfully processed by the correct interchange interface process, when multiple copies of the interchange interface are configured in a system. Without this enhancement, the reversal may be routed to an interchange interface process with no record of the original transaction in its Interchange Log File (ILF) and subsequently be dropped. This enhancement was added to the BASE24-pos Router module in BASE24-pos Release 6.0 Version 1.

Prerequisites: BASE24-pos and ISO Host Interface.

ISO Host Interface Stored Value Support

The BASE24-pos ISO Host Interface has been enhanced to support stored value transactions acquired by the ISO Host Interface. The BASE24-pos ISO Host Interface can now map the host transaction code to the BASE24 transaction code. The new stored value transaction codes are show in the following table.

Transaction	BASE24 Transaction Code	Host Transaction Code
Add Card Activation	"26"	"99"
Card Activation	"25"	"98"
Full Redemption	"28"	"17"
Replenishment	"27"	"28"

Prerequisites: BASE24-pos and ISO Host Interface.

Issuer Declined Manually Entered Transactions

BASE24-pos has been enhanced to support two extra configuration options to issuers when processing manually-entered transactions. Issuers will now be able to configure BASE24-pos to set the conditions under which CVD2 must be present in manually entered transactions and the conditions under which manually entered CVD2 transactions will be declined.

Prerequisites: BASE24-pos.

MasterCard Secure Code Validation

The BASE24-pos Secure Internet Validation (SIV) module has been enhanced to support MasterCard Secure Code validation using the 'calculation' method. This allows the SIV module to calculate the AAV value, using an HSM, and compare it to the AAV value sent in the authorization request, similar to Visa. SIV currently supports a 'compare' method for MasterCard.

Prerequisites: The BASE24-pos Secure Internet Validation module.

Merchant Host Interface Compatibility with TSS

The Merchant Host Interface was enhanced to be compatible with Transaction Security Services (TSS). TSS compatibility required all KEY fields in the Merchant Host Configuration File (MHCF) records to be updated to contain key locators. The key locators are used by TSS to retrieve the actual encryption/decryption keys used by the MHI terminals.

Prerequisites: BASE24-pos and Merchant Host Interface.

Mobile Top-Up for POS

BASE24-pos has been enhanced to support mobile top-up transactions acquired on BASE24. Using the new POS Transaction Context Manager (TCM), mobile top-up transactions can be acquired through the SPDH device handler for routing to two authorization entities. The TCM manages the mobile top-up transaction which typically involves routing an incoming request to two authorizers: one for the mobile phone operator and one for the issuer financial institution. In addition, the Generic XML Interface for Mobile Top-Up transactions is now available as a method for communicating with mobile phone top-up authorizers.

Prerequisites: The POS Transaction Context Manager, BASE24-pos SPDH Device Handler, Mobile Top-Up product and an interface.

The above enhancement is considered a billable item. Please contact your Customer Manager for further information.

SPDH Enhancements

The SPDH device handler has been enhanced to support offline Transaction Certificates (TC). A TC is generated during online and offline EMV transactions and may be beneficial in EMV related disputes. With this enhancement, SPDH will accept an EMV log-only transaction, identified by a new Message Subtype 'E' in the SPDH message header and will log it to the PTLF as a '9920'.

For more information on EMV processing, please refer to "EMV Support" in the

BASE24 Base Optional Enhancements of the BASE24 Release 6.0 Implementation Guide (Version 2).

The System Trace Number (STAN), Retrieval Reference Number, and Debit Sharing/Network ID Code were added to the response message for SPDH to improve PIN Debit reconciliation. The following interfaces have been updated to support the new “BY” token that carries the new data: Star ISO, VISANET ISO, Plus ISO, and VISA DPS ISO Interfaces.

For further information, see **SPDH EMV Transaction Certificates** and **SPDH PIN-Debit Reconciliation** in Section 6 where the impacts of these enhancements are described.

Prerequisites: The BASE24-pos EMV Extension, the SPDH Device Handler, and the SPDH Device Handler EMV Extension.

Suppression of Issuer Discretionary Data

In 2004, Visa, MasterCard, Discover, American Express, and JCB announced the Payment Card Industry Data Security Standard (PCI) to protect consumer data in acquirer and issuer transactions. This enhancement allows the suppression of magnetic strip discretionary data in issuer transactions on the BASE24-pos Transaction Log File (PTLF). Previously, in October 2002, BASE24-pos was enhanced to allow optional suppression of card verification data, address verification data and discretionary data in acquiring transactions

Prerequisites: BASE24-pos.

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BASE24 Interchange Interface Enhancements

All BASE24 Interchange Interface product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24 Interchange Interfaces, refer to Section 9.

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General Enhancements

There are no general enhancements to Interchange Interfaces.

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Optional Enhancements

The following BASE24 Interchange Interface release 6.0 version 6 enhancements are optional. BASE24 Interchange Interface users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

Boston Communications Group Interface (*b CGI*) Restructure

The interface between BASE24 and the Boston Communication Group, Inc. (*b CGI*), a third party telecommunications processor, which enables Mobile Top-Up transactions for a consumer's mobile phone has been restructured. The interface has been modified to integrate with the Generic Switch and Generic XML Mobile Top-Up interfaces eliminating replication of code and providing easier implementation and maintenance.

Prerequisites: Requires Mobile Top-Up ATM Transaction Context Manager and Non-Currency Dispense. XPNET Http Client Support is also required to support WebGate.

Generic XML Mobile Top-Up Interface

This XML Interface is used specifically for communicating with Mobile operators. It provides the interface between BASE24 and a Mobile Top-Up provider using an XML message format. Approved, denied and reversed transactions are logged for reporting, reconciliation, and auditing purposes.

This application requires customization (services development) for each customer / site to match specific XML messages required. This interface is sold in combination with CSMs to specific mobile providers.

Prerequisites: Requires Mobile Top-Up ATM Transaction Context Manager or Mobile Top-Up POS Transaction Context Manager. A device handler and a Mobile Top-Up device extension are also required.

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BASE24 Value Added Products

All BASE24 value added product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24 Value Added Products, refer to Section 10.

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General Enhancements

There are no enhancements affecting the following BASE24 value-added product users:

- BASE24-card
- BASE24-customer service
- BASE24-InfoBase

There are no general enhancements to Value-added products.

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Optional Enhancements

BASE24 release 6.0 version 6 provides the initial release of BASE24 Inventory Voucher Manager.

BASE24-inventory Voucher Manager

The initial release of the BASE24-inventory Voucher Manager product provides the software infrastructure to manage an inventory of stock items supplied by the inventory supplier and sell them to customers through the financial institution's existing delivery outlets. The delivery outlets could be ATM, POS or Internet.

BASE24 Inventory Supply – The Inventory Supply module manages the stock inventory and responds to authorization requests. This module houses the inventory database that contains the stock files, transaction log files and configuration files.

The Inventory Supply module runs processes for authorization of inventory transactions, settlement and reporting and interfacing with the delivery bank, if required. User interaction is facilitated through inventory screens. A file refresh program is available to load the stock files transferred from other systems and refresh the inventory database in a secured environment.

The Inventory Supply module of the BASE24-inventory Voucher Manager runs as a satellite process to XPNET 3.0 or 3.1 and BASE24.

There is no online interchange arrangement between the delivery bank and the inventory supplier. Reconciliation and settlement is manual and is outside the scope of this document.

Prerequisites: The BASE24-atm Mobile Top-Up Acquirer Package or BASE24-pos Mobile Top-Up for POS Package with at least one device handler. For more information on the BASE24-atm Mobile Top-Up Integration optional enhancement, refer to the ***BASE24 Release 6.0 Implementation Guide (Version 5)***. For more information on the BASE24-pos Mobile Top-Up for POS optional enhancement, refer to **Optional Enhancements** in Section 6.

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3: Support Services

This section summarizes the various support services available from ACI. These support services are described as follows:

- Product consulting and project management
- Publications
- Education
- HELP24
- ASTech services

Other services, besides the traditional Installation service, such as go-live support, onsite conversion assistance, and software methodology and utility training are available or are being planned; please contact your ACI Customer Manager for more details of the services ACI can offer.

Product Consulting and Project Management

No formal service programs have been created by ACI for upgrading customers to BASE24 release 6.0 version 6. Customers need to review this entire document and any referenced manuals, to determine what they will need to do to upgrade their BASE24 system.

After reviewing, customers should contact their ACI Customer Manager to discuss any services, such as project management, custom code upgrades, onsite staff, etc., required for upgrading, after reviewing all materials.

Project Planning Guide

The following Project Planning Guide is designed to assist in the successful upgrade of a BASE24 system to release 6.0 version 6. Information contained in this guide provides insight into required planning and control of the entire project.

Following this project guide helps ensure the realization of the goals and business objectives of the BASE24 user.

1. IDENTIFY PROJECT TEAM

One of the initial tasks for the BASE24 upgrade is for the user to assign a project team who will be responsible for the success of the project. A project manager and team members will need to be assigned and specific responsibilities established.

2. DETERMINE HARDWARE REQUIREMENTS

An ACI sales representative can assist in determining whether additional hardware will be required for the BASE24 upgrade. It is the BASE24 user's responsibility to order the hardware and establish the delivery date with HP NSK. HP NSK will perform diagnostic testing on the hardware when it is installed at the site. Should hardware encryption be required, it will be necessary to determine the hardware requirements. Additional hardware should be ordered for installation before testing can occur.

If applicable, the BASE24 user should order any new device firmware/software required to implement the enhancements they have chosen to support. The delivery date for the device firmware/software should be agreed upon with the vendor, as this time frame will affect the entire project.

3. IDENTIFY COMMUNICATION REQUIREMENTS

If the BASE24 user has identified additional protocol requirements, the communications lines and modems should be ordered. The customer must coordinate this effort with the local telephone company to ensure that the lines are configured in the least costly manner. The lines and modems must be installed and tested prior to the scheduled software installation date.

4. UPDATE HP NSK SOFTWARE REQUIREMENTS

ACI recommends that the BASE24 user be on a current HP NSK supported operating system to obtain maximum performance and support. BASE24 release 6.0 requires a minimum of HP NSK Guardian D45 Operating System. Customers choosing to implement Format 2 File support are required to be on a minimum

Operating System level of D46. In addition, BASE24 release 6.0 requires TMF for several files. Customers choosing to implement BASE24-es Transaction Security Services (known as TSS) with BASE24 release 6.0 must review the BASE24-es Hardware and Software Requirements for the BASE24-es release they will deploy.

5. UPGRADE TO XPNET 3.0 or 3.1

Customers should note that XPNET 3.0 is the minimum requirement for BASE24 release 6.0.

6. PROJECT SCOPE MEETING

ACI does not believe this step is necessary for customers upgrading to BASE24 release 6.0 version 6.

7. MIGRATION PLANNING

Users will need to use this document, and any references within it, for detailed analysis when creating their migration plan.

8. DEVELOP PROJECT SCOPE DOCUMENT

ACI does not believe this step is necessary for customers upgrading to BASE24 release 6.0 version 6.

9. DEFINE REQUIREMENTS DEFINITION

The customer will be required to review existing CSMs to determine which ones will require retrofitting to the new version. With ACI's Fix Release methodology, BASE24 users should evaluate their CSMs to determine if they are handled as standard product in release 6.0 version 6 or if they are specific to the customer's environment and will need to be upgraded to the new version.

10. DEVELOP PROJECT PLAN

The customer's project manager is responsible for developing an internal project plan. A review of the ACI project guidelines will help this person identify job responsibilities and establish time frames. A tracking mechanism should be included in the plan to ensure that the project is proceeding on schedule.

11. IDENTIFY CUSTOMER SPECIFIC MODIFICATION DEVELOPMENT

If the customer elects to have ACI develop modifications identified as required for the new version, a request will be submitted to ACI Software Development for a cost estimate. The functional descriptions and estimates will be sent to the

customer for review and approval. Resources will be assigned and the final delivery dates established upon receipt of the signed contract.

Should the customer decide to develop any modifications in-house, resources will need to be assigned and scheduled. Once delivery dates are established, impacts to system testing can be determined.

12. DEFINE CERTIFICATION REQUIREMENTS

Customers currently on Fortify.3 will need to verify with each of the interchanges they connect to, to determine if they need to re-certify with the Interchanges with release 6.0 version 6 code.

If re-certification is required, preliminary test times should be determined and a copy of the interchange test script secured.

The customer needs to set up test time with the interchange well in advance in order to insure time frames are met.

13. ESTABLISH TEST FACILITY

It is recommended that an area be designated specifically for testing the BASE24 release 6.0 version 6 software. The following items are recommended for the test area:

- 3 PCs
- Test devices
- Hard copy printer
- Test cards

14. INSTALL PRODUCTION HARDWARE

Installation of any additional computer hardware or peripherals is generally performed by the local hardware service organization or customer staff.

15. ATTEND BASE24 EDUCATION CLASSES

ACI offers a full suite of instructor led courses for BASE24. An enhancement course is also available upon request. The course descriptions and outlines can be viewed on the ACI Worldwide web site at www.aciworldwide.com.

16. DEVELOP INTERNAL PROCEDURES

There are no new internal procedures that ACI is aware of for BASE24 release 6.0 version 6 enhancements. User/Operational impacts are noted in the detailed sections for each enhancement described within this document.

17. DEVELOP SYSTEM TEST PLAN

The customer is responsible for developing a system test plan. Included in the test plan should be a description of each step specifying the test card to be used, the account to be used, the transaction to be executed, and the expected results. A procedure for retaining the results of the transaction processing should also be specified. Controlled testing and result verification procedures are key elements for successful system testing.

The execution of the system test plan will occur over a period of weeks to fully test all aspects of the system. BASE24 product functionality should be the first application tested. Any system modifications should then be thoroughly tested. The interface between the host and BASE24 should also be tested. This includes testing on-line authorization and the refresh and extract programs.

18. DEVELOP HOST SOFTWARE

During the course of a business day, BASE24 logs all completed and declined transactions to a transaction log file. After network settlement, the transaction log file can be extracted to tape and used as input into the customer's nightly processing cycle on the host. If the customer would like to take advantage of the new features in release 6.0 version 6, a host program will need to be developed to interpret the new token data that can be extracted with release 6.0 version 6.

After the nightly processing cycle on the host or periodically during the business day the host may need to update card and account files used by BASE24 for pre-authorization or stand-in processing. If the customer would like to take advantage of the new features in release 6.0 version 6 in these files, a host program will need to be developed to initialize the new data fields in BASE24 files.

19. INSTALL BASE24

Release 6.0 version 6 software will need to be installed at the customer site. Licensed release 6.0 version 6 software modules will be provided to the customer upon request for installation on their system.

20. CONVERT DATA FILES

ACI provides database conversion programs for selected BASE24 files. These files are discussed in the detail sections for each enhancement described within this document.

21. CERTIFY INTERCHANGE

This task involves implementing the interchange test script and receiving notice of certification from the interchange. Current BASE24 users should not have to re-certify their Interchange Interfaces, unless they are implementing new functionality not previously tested or as directed by the interchange.

22. PERFORM SYSTEM TESTING

Full business day cycle testing of the system should be performed as determined by the customer testing plan and procedures.

23. DEVELOP CONVERSION PLAN

The customer should develop an implementation plan for converting existing devices and interchange links to release 6.0 version 6. The installation of new devices and interchanges should also be included in this plan. A method of informing users of any changes to existing procedures should also be developed. The conversion schedule should be routed to all departments.

24. GO-LIVE

At the customer's request, an operational support analyst will visit the site one day prior to the "go-live" date to review the current software status and the BASE24 facilities. This individual will remain onsite to assist the customer with production cutover. The migration plan developed by the customer will be utilized in production cutover. On-site go-live support is a billable service.

25. COMPLETE POST IMPLEMENTATION REVIEW

In general, the purpose of this meeting is to ensure overall satisfaction of the customer's project.

26. TURN OVER TO CUSTOMER MANAGEMENT

If an ACI Project Manager has handled the BASE24 upgrade, a formal turnover meeting is held between the ACI Project Manager and the ACI Customer Manager to ensure a smooth transition in turnover of responsibilities for on-going customer support.

Publications

The BASE24 documentation set has been updated to include the release 6.0 version 6 enhancements. The updated BASE24 documentation is available in three formats.

- **HELP24 eSupport InfoLink.** All files for the manuals in the BASE24 6.0 documentation set are located on the HELP24 eSupport InfoLink database. You have the option of copying the files for individual manuals from the HELP24 eSupport InfoLink database to your PC.

In addition, the HELP24 eSupport InfoLink database contains documentation that is added or revised between ZIP files or CD-ROMs.

- **Software Management Electronic Distribution.** ZIP or ZIP Executable files that contain the same contents as the CD-ROM are available through the ACI electronic distribution system.
- **CD-ROM.** The CD-ROM contains all PDF files for the manuals in the documentation set.

The BASE24 6.0 documentation set contains the most current version of each document. If a BASE24 manual has not been updated for release 6.0, the manual for the prior release appears on the release 6.0 menu with a note identifying the prior release.

Documentation for XPNET 3.0 and XPNET 3.1 is available in a separate set titled NET24-XPNET 3.1 that is available in the same formats as the BASE24 documentation.

BASE24 customers can download files from the HELP24 eSupport InfoLink database or order the documentation ZIP files or CD-ROM from their ACI Customer Manager.

Education

BASE24 Technical Education offers a complete set of one- to four-day modules covering the components of the BASE24 system. BASE24 courses are structured into a modular format that eliminates repetitious material, provides information specific to certain job functions, offers hands-on lab experience, and enhances retention of information taught.

A BASE24 Enhancement course is also available upon request for existing customers upgrading to BASE24 6.0 version 6.

The modular format enables BASE24 users to choose modules applicable to their employees' job responsibilities. This provides consolidated training time and offers opportunities for additional employees to receive education.

The course descriptions, outlines and current education schedule can be viewed on the ACI Worldwide web site at www.aciworldwide.com.

HELP24

Please use ACI Worldwide's eSupport site to report problems or ask questions about BASE24 release 6.0 version 6. eSupport is an Internet Web based application that interfaces with the HELP24 Clarify database. Using eSupport, you can search our KnowledgeBase Solutions for information and resolutions to problems you are experiencing. You can also directly open Clarify cases. An eSupport option, InfoLink, is your link to a wealth of information that can help you in your day to day business. It allows you on-line access to ACI Publications, Fix History Information, HELP24 Hotlines and much more. Contact your local Help24 office for an eSupport logon.

As a result of reviewing new troubleshooting methods, HELP24 has acquired a tool called WebEx. This is a state of the art web-based technology that allows us to troubleshoot issues in a secure, online environment. Compared to the dial-up environment we have used in the past, WebEx provides convenient access, a more stable connection and most importantly, it provides a collaborative environment which also allows you to actively participate in the session by viewing and asking questions about everything that goes on.

To address audit security requirements, WebEx provides the option to record the session so all steps can be documented in a file that can be accessed after the fact if necessary. WebEx has a file transfer option that provides the ability to transfer files from your PC to ours, or vice versa. This feature can be used in the event the file(s) are too large to email.

For more information, please refer to the WebEx User's Guide located under Product Info on Infolink. Contact your local HELP24 office to schedule your WebEx demonstration.

ASTech Services

ACI offers customer support services through our ASTech Division which allows the selection of a la carte support services to meet individual customer requirements. ASTech, ACI Support and Technical Services, is a service support team which provides the customer with a variety of on-site and remote technical support. ASTech provides customers with supplemental staffing, system performance analysis, review and/or upgrading of previous release customer-specific software development, and detailed operational technical training.

The following are example descriptions of the various services ASTech could undertake to assist you during the BASE24 upgrade. The number of staff weeks identified to accomplish these tasks is estimated and will vary depending on the amount of customer specific code to be developed and delivered by ACI.

Your ACI Customer Manager can provide you with details of services available in your region that have been tailored to the special needs of different areas of the world.

Customer Software Analysis

Effort: Two to four staff weeks

ASTech personnel will perform a complete review of the customer's BASE24 software system and determine what current modifications (CSM code) exist at the customer location. ASTech personnel will perform a detailed comparison of existing customer specific software modifications to release 6.0 version 6 functions. Upon completion of the review, the customer will be provided with a document that identifies all modules that contain code unique to their BASE24 system (file names, custom functions, custom function applicability, etc.).

Conversion Planning

Effort: One and one-half to three staff weeks depending on the size and current system configuration.

This service option provides the customer with comprehensive support during the system migration to the new version. ASTech assigns the customer a Technical Project Lead, or Tech Lead from ACI. The Tech Lead will be responsible for ensuring the success of the BASE24 upgrade project.

The Tech Lead will:

- Review the migration options and develop a detailed migration plan for the conversions project
- Review the Project Scope Document on behalf of the customer (if applicable)
- Review the customer's internal procedures to accommodate release 6.0 version 6 functions.

Upgrade Custom Code

Effort: The amount of time necessary for this task varies depending on the number of modules identified as having modifications unique to the customer. Systems with a great number of customizations may take up to sixteen staff weeks. Actual estimates would be provided upon completion of the Customer Software Analysis.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

With this service, ASTech will upgrade all custom code to release 6.0 version 6. This includes unit testing of the code. Customizations that are no longer necessary will be removed. Customizations that must be carried forward can either be upgraded in their current form or be reengineered to take advantage of new functionality in release 6.0 version 6. Reengineering customizations will add time to the estimate.

System Test Planning

Effort: One to four staff weeks depending on the results of the Customer Software Analysis and the number of BASE24 products to be tested. Some of this time will be on-site to perform the System Test Plan.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

ASTech personnel will develop and execute a BASE24 System test plan designed for the customer's specific needs. This service also includes the performance and documentation of the system acceptance testing.

Switch and Host Certification Support

Effort: Two to five staff weeks depending on the results of the Customer Software Analysis and the number of BASE24 products and interfaces to be certified. All of this time will be on-site time.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

This service option allows ASTech personnel to provide complete support of the various switch and/or host communication certification testing. This includes reviewing and refining the test scripts, performance of certification testing functions, evaluation of the testing results, and documentation and presentation of the results obtained during the certification testing.

Go-Live Support

Effort: One week (on-site).

ASTech personnel will be at the customer site assisting in all conversion tasks and being present for go-live support. This includes remaining on-site for a couple of days to ensure system stability.

Post Go-Live Support

Effort: Varies.

After a successful go-live, a Tech Lead will provide on-going support which may be required by the customer. Tech Lead support can be provided at the customer location and/or remotely through ACI.

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4: BASE24 Base

Section 4 discusses the changes and enhancements made to the BASE24 Base modules in release 6.0 version 6, along with the resulting conversion and implementation requirements that existing BASE24 users must consider when upgrading from release 6.0 version 5.

Changes and enhancements to the BASE24 Base modules impact all users. Consequently, all users should review the material in this section to determine how it applies to them.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the Base modules changes that affect all BASE24 users, along with the associated conversion and implementation requirements that all BASE24 users must address.
- Optional enhancements describe the Base modules changes that affect only those BASE24 users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 users must address to implement the enhancements.

Section 4 must be used in conjunction with Sections 5 through 10 to identify the changes that BASE24 users need to consider when upgrading to release 6.0 version 6.

Standard Information Format

A standard format is used in Sections 4 through 10 to highlight and describe the various changes, conversion requirements, and implementation requirements that BASE24 users need to consider when upgrading to release 6.0 version 6. This standard format is defined below.

In situations where the following information does not apply, the information is labeled as “Not applicable”

Software Impacts

Describes the software modules impacted for a given change or enhancement and the nature of the impacts. This information includes a high level list of the modules impacted, how the modules are impacted (i.e., inline changes, hooks, dummy procs, extensions) and which customers are affected (i.e., all, only those implementing the enhancement).

CUSTMACS Changes

Describes the CUSTMACS changes required to implement a given change or enhancement. This information includes Make defines that have been added, deleted, or modified.

LCONF Assign Changes

Describes the LCONF assign changes required to implement a given change or enhancement. This information includes assigns that have been added, deleted, or modified, along with an explanation of how these assigns can or must be set.

LCONF Param Changes

Describes the LCONF param changes required to implement a given change or enhancement. This information includes params that have been added, deleted, or modified, along with an explanation of how these params can or must be set.

Database Configuration

Describes the data file changes required to implement a given change or enhancement. This information includes file fields that have been added, deleted, or modified, along with an explanation of how these fields can or must be set. It also identifies file records and record segments that must be added, deleted, or modified.

New Data Files

Identifies new data files that must be created in order to implement a given change or enhancement.

Template Changes

Identifies file templates that must be created or modified in order to implement a given change or enhancement. It also identifies COBNAMES and Device Handler Names (xxxxNAMS) file modifications that must be made.

Device Configuration

Identifies device configuration changes that are required in order to implement a given change or enhancement.

Mega Table (MEGATBL) Changes

Identifies Mega Table changes that are required in order to implement a given change or enhancement.

PI-Table Changes

Identifies Product Indicator (PI) Table changes that are required in order to implement a given change or enhancement.

SEG-Table Changes

Identifies Segment Indicator (SEG) Table changes that are required in order to implement a given change or enhancement.

Security Table (SECTBL) Changes

Identifies Security Table changes that are required in order to implement a given change or enhancement.

PATHCONF Changes

Identifies PATHCONF changes that are required in order to implement a given change or enhancement.

CRT Access Security Changes

Identifies CRT Access security changes that are required in order to implement a given change or enhancement. This includes changes that need to be made to BASE24 operator security records.

Routing Changes

Identifies routing and configuration changes that are required in order to implement a given change or enhancement. This includes required changes to files such as SPROUTE and the BASE24-pos Authorization Selection Table (AST).

Network Configuration

Identifies configuration changes that are required in order to implement a given change or enhancement. This includes NEF changes.

File Conversions

Identifies the files that need to be converted in order to implement a given change or enhancement. This implies that a given file could be impacted by multiple changes and/or enhancements. BASE24 users need to consider all of the impacts to any one specific file before the actual conversion of a file.

Host/Co-Network Impacts

Identifies impacts to a user's host or co-network as a result of a given change or enhancement. This includes changes to the external messages used to communicate with hosts and co-networks as well as changes to refresh and extract tape formats.

User/Operational Impacts

Identifies potential changes to a user's internal BASE24 operating procedures. This includes changes to operations and file maintenance screens, operator-initiated commands, and reports.

Vendor Impacts

Identifies any issues that need to be addressed with vendors in order to implement a given change or enhancement. This includes hardware, firmware, and software changes that vendors need to make in order for their products to function with the given change or enhancement.

Reference Documents

Lists any outside sources used in the design of a given change or enhancement. BASE24 users can refer to these documents for additional information.

General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24 users must address in order to upgrade to release 6.0 version 6.

- General Changes

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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General Changes

- Changes for all BASE24 release 6.0 version 6 enhancements affecting the DDL files and the BASE24 AFT Subsystem files (i.e., screens, requesters, servers) have been merged and cataloged under one SCN for each affected file. Applying changes for individual enhancements to the screen files does not allow BASE24 users to identify which changes go with a particular enhancement, as screen files do not have sequence numbers at the end of each line. Since the screen file changes usually involve new fields being added, all associated requester, server and DDL changes are required. Even though requester, server and DDL files do contain sequence numbers, all SCNs would need to be applied in order to accept the changes to the screen files. Therefore, all requester, server and DDL file changes, for all enhancements, were done under one SCN per file as well.
- The Transaction Based Pricing (TBP) reports have been modified to support Mobile Top-Up transactions in BASE24-pos. These changes impact all BASE24 TBP customers. All BASE24 TBP customers must implement the updated version of the TBP report objects and TBP documentation (subvolume BA60TBPD). Changes to the Transaction Based Pricing (TBP) reports requiring an MTAF conversion were introduced with the BASE24 release 6.0 version 5 BASE24-atm Mobile Top-Up Integration enhancement. BASE24 TBP customers who performed this MTAF conversion previously will not convert the MTAF again. For further information, see **Mobile Top-Up for POS** in Section 6 where the impacts of this enhancement are described.
- The Security File (SEC) has been expanded to include space to store up to 10 passwords (current and 9 previous). The Security File requester and server have been modified to accommodate the larger file and store passwords in the new data fields. Other changes implemented in the Security File requester and server are discussed under **Complex Passwords in Pathway** later in Section 4.
- The Card Prefix File (CPF) has been modified to include additional fields on screen 8, for the processing of manually-entered transactions. The Card Prefix File requester and server have been updated to process these new fields. For further information, see **Issuer Declined Manually-Entered Transactions** in Section 6 where the impacts of this enhancement are described.
- The Super Extract process has been modified to avoid potentially monopolizing CPU time during the fall DST change. Previously, an extract timer expiring at the DST change could lock up a CPU for up to one hour, while the system clock caught up with the timer's expire time. The Super Extract process no longer enters into this tight loop situation, unless the wait time is less than one second.

- DDLGISO has been modified to include the Scoring Engine Master File (SEMF) definition for use with BASE24-prm version 7.0 and BASE24-fhm version 6.0.

Software Impacts

All BASE24 customers:

- Inline changes to BA10UTIL.UTILB
- Inline changes to BA60AFT.COBTKN, RQSECS, SCRNMEGA, and SVISECS
- Inline changes to BA60DDL.DDLFCPF and DDLGISO
- Inline changes to BA60AFT.SCRNCPF, RQCPFS and SVCPPFS
- Inline changes to BA60EXT.BASEXS

All BASE24 Transaction Based Pricing (TBP) customers:

- New executable objects for Transaction Based Pricing (TBP) reports and updates to TBP documentation (subvolume BA60TBPD)

Note: The list of files impacted by the merged changes is included in the **Software Impacts** section of the applicable enhancement(s).

CUSTMACS Changes

Not Applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Card Prefix File (CPF)

The CPF conversion program must be run to convert the CPF. Running the conversion program is required for any EMV user. Users must create the new CPF file using the BADDLFUP file.

The CPF conversion program on the BA60UC04 subvolume can be used to convert the CPF records from the release 5.3 format to the release 6.0 version 6 format. The CPF conversion program on the BA60UC09 subvolume can be used to convert the CPF records from the release 6.0 version 5 format to the release 6.0 version 6 format. If modifications to the conversion program are necessary, the program can be modified by editing the CNVCPFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#cpfcnv.t,name $cpfm,nowait/ cnvcpfm –  
force –list –listreplace
```

Release 5.3 users should update the CNVCPFR obey file on the BA60UC04 subvolume with the appropriate information. Release 6.0 version 5 users should update the CNVCPFR obey file on the BA60UC09 subvolume with the appropriate information. The CNVCPFR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume for release 5.3 customers and to the AAREAD09 on the BA60UC09 subvolume for release 6.0 version 5 customers.

Security File (SEC)

The SEC conversion program must be run to convert the SEC. Running the conversion program is required for all BASE24 users. Users must create the new SEC file using the BADDLFUP file. Once the SEC file is converted, the modified BA60AFT.RQSECS and SVISECS must be used to maintain the file.

The SEC conversion program on the BA60UC04 subvolume can be used to convert the SEC records from the release 5.3 format to the release 6.0 version 6 format. As the SEC file format has not changed since release 6.0 version 1, the SEC conversion program on the BA60UC09 subvolume can be used to convert the SEC records from all previous versions of release 6.0 to the release 6.0 version 6 format. If modifications to the conversion program are necessary, the program can be modified by editing the CNVSECS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#seccnv.t,name $secm,nowait/ cnvsecm –  
force –list –listreplace
```

Release 5.3 users should update the CNVSECR obey file on the BA60UC04 subvolume with the appropriate information. Users for all previous versions of release 6.0 should update the CNVSECR obey file on the BA60UC09 subvolume with the appropriate information. The CNVSECR should then be obeyed to execute the conversion program.

Because of the increased length of the record, once the SEC file is converted, the modified BA60AFT.RQSECS and SVISECS must be used to access the file.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume for release 5.3 customers and to the AAREAD09 on the

BA60UC09 subvolume for release 6.0 version 5 customers.

Release 6.0 version 1, release 6.0 version 2, release 6.0 version 3 and release 6.0 version 4 customers should refer to the ***Conversion Program Guidelines*** within Section 1 of this document.

Monthly Transaction Accumulation Files (MTAF)

Changes to the Transaction Based Pricing (TBP) reports requiring an MTAF conversion were introduced with the BASE24 release 6.0 version 5 BASE24-atm Mobile Top-Up Integration enhancement. BASE24 TBP customers who performed this MTAF conversion previously will not convert the MTAF again. All BASE24 TBP customers must be running Version 9 of the TBP reports. Please refer to the ***BASE24 Release 6.0 Implementation Guide (Version 5)***.

Store-and-Forward (SAF) file for BIC ISO

SCNs SW09091 and SW09092 were applied to SW60BABI.BICIDDLs and SW60BABI.BABICIS respectively. These fixes need to be implemented at a time when the active SAF file is empty. The interface will need to be logged off and then stopped. The old SAF file will need to be removed and replaced with a new SAF file created using the BICIFUP file created by MAKE. Once the new file is in place, the BIC ISO process can be restarted and the link logged back on.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Optional Enhancements

This section identifies the Base modules enhancements that can be enabled or disabled with release 6.0 version 6, along with the changes, conversion requirements, and implementation requirements. These enhancements are the following:

- Complex Passwords in Pathway
- Security Audit Reports

BASE24 users can choose whether or not to implement the above enhancements. Users who plan to implement these particular enhancements must address the corresponding release 6.0 version 6 changes, conversion requirements, and implementation requirements. Users who do not plan to implement these particular enhancements can review the corresponding release 6.0 version 6 changes, conversion requirements, and implementation requirements for future reference.

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Complex Passwords in Pathway

This section describes the implementation requirements associated with the Complex Passwords in Pathway enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Complex Passwords in Pathway enhancement.

All BASE24 customers will receive these changes. Customers who wish to implement complex passwords must make configuration changes discussed in this section.

This enhancement is provided to meet the recent legislative initiatives and card association mandates such as Visa and MasterCard's Payment Card Industry (PCI) Data Security Standards requirements for stronger passwords. This enhancement allows for passwords to be case-sensitive. It allows LCONF parameters to be set for the minimum number of alpha, numeric, and/or special characters required to be present in every new password. It allows for the storing of up to 10 passwords (the current and 9 previous) so they cannot be reused. The password is stored using SHA-1 hash encryption rather than single DES.

Once the PASSWORD-COMPLEX LCONF parameter is set to "Y", to activate the Complex Password code, existing customers will be allowed to access the BASE24 Pathway system using their old password. The old password will be compared against the requirements set for the minimum number of alpha, numeric and special characters. If the old password meets the requirements, it will be stored as a SHA-1 hash value and the record will be changed to indicate that a complex password is in use. If the old password does not meet the complexity requirements, the user will be prompted to change their password before access is granted. The new password must meet complexity requirements and will be stored as a SHA-1 hash value. The record will also be changed to indicate that a complex password has been set.

The following LCONF parameters are currently in place to configure password use. Their use is unchanged.

- PASSWORD-CHG-INTERVAL – How often a password must be changed
- PASSWORD-LENGTH-MIN – How long a password must be (2-8 characters)
- MAX-LOGON-ATTEMPTS – Number of unsuccessful logon attempts allowed before locking out a user
- LOGOFF-INTERVAL – How long a terminal can be idle before requiring a user to log in again.

Software Impacts

All BASE24 customers:

- New executable object for BA10UTIL.UTILB
- Inline changes to BA60AFT.RQSECS, SCRNMEGA, and SVISECS
- Inline changes to BA60DDL.DDLFSEC and DDLGSEC
- Inline changes to BA60SRC.BAUTILS

A complete list of the files affected by the Complex Passwords in Pathway enhancement can be found in the SCN file BA60UD09.SCNPSWD.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

PASSWORD-COMPLEX

This parameter turns BASE24 Complex Passwords on. Other related parameters will be used to configure how Complex Passwords will work.

Y = Complex Passwords in use. Passwords are case sensitive. Other complex password parameters will be checked. Password will be stored as a SHA-1 hash value.

N = Complex passwords are not in use. Password checking and storing will continue as before. This is the default setting.

PASSWORD-MAX-CHNG-PER-DAY

This parameter sets the maximum number password changes allowed per day. Valid values are 0-10.

1-10 = Maximum number of times in a calendar day that a password can be changed.

0 = No check for this parameter. This is the default setting.

PASSWORD-NUM-ALPHA

This parameter sets the minimum number of alpha characters required in a new password. Valid values are 0-8.

- 1-8 = Minimum number of alphabetic characters required in a new password.
- 0 = No check for this parameter. This is the default setting.

PASSWORD-NUM-NUMERIC

This parameter sets the minimum number of numeric characters required in a new password. Valid values are 0-8.

- 1-8 = Minimum number of numeric characters required in a new password.
- 0 = No check for this parameter. This is the default setting.

PASSWORD-NUM-SPCL

This parameter sets the minimum number of special characters required in a new password. Valid values are 0-8. Special characters are those non-alphabetic, non-numeric characters that can be entered from a keyboard with no control characters.

- 1-8 = Minimum number of special characters required in a new password.
- 0 = No check for this parameter. This is the default setting.

PASSWORD-MAX-STORE

This parameter determines the number of passwords stored in the Security file. Valid values are 1-10.

- 2-10 = Total number of passwords stored in the Security record. This number includes the current password.
- 1 = Only the current password is stored. This is the default setting.

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Security File (SEC)

The SEC conversion program must be run to convert the SEC. Details of this conversion program can be found in the **General Changes** earlier in Section 4.

Host/Co-Network Impacts

Super Extract Process

Host code changes are required for all BASE24-atm users that use OMF Extract to accept and process information from the ATD record.

User/Operational Impacts

If complex passwords are configured, BASE24 Pathway users must be aware that passwords are case sensitive. When setting a new password, users must know the rules that have been set for the network, for how many alpha, numeric and special characters are required.

Once the parameter PASSWORD-COMPLEX is set to a "Y", it is not intended for the parameter to be reset to "N" at a later time. Changes are made to the password fields in the SEC file records to indicate that the network is using complex passwords. These changes cannot be undone. If a customer believes they may wish to turn complex passwords on for a time and then turn them off (say in a test system), a backup of the security file (SEC) should be made before the parameter is set to "Y".

The sum of the parameters PASSWORD-NUM-ALPHA, PASSWORD-NUM-NUMERIC and PASSWORD-NUM-SPCL must be less than 8. If not, all three parameters will default to zero when the server is initialized.

The value of the parameter PASSWORD-MAX-STORE must be greater than or equal to the value of the parameter PASSWORD-MAX-CHNG-PER-DAY. If not, the value of PASSWORD-MAX-STORE will be made equal to the value of PASSWORD-MAX-CHNG-PER-DAY when the server is initialized.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Security Audit Reports

This section describes the implementation requirements associated with the enhancement to provide Security Audit Reports.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Security Audit Reports enhancement.

All BASE24 customers will receive these reports in a new subvolume, BA60RPT. Customers who wish to run the reports may do so using the runfile provided for each report, as discussed in this section.

This enhancement is provided to meet the recent legislative initiatives and Card Association mandates such as Sarbanes-Oxley, Gramm-Leach-Bliley Act, and the Payment Card Industry (PCI) Data Security Standards requirements for audit reporting. Reports SEC01 and SEC02 use the Security files (SEC and UFIR) as input. SEC01 sorts the SEC file in order of last logon date, oldest date first. This report will allow customers to know which logons have not been used for some time to allow them to delete old or unused SEC records. SEC02 sorts the SEC file by group and user number. This report will allow customers to see what groups and group/user numbers have been defined in the security records. Both reports have a detail option that will provide screen access information for each file and screen for each group/user number.

Reports OMF03 and OMF04 read the OMF file(s) for the date specified in the runfile. OMF03 provides information from Logon/Logoff records. OMF04 provides information from all types of records. Both reports can be limited to a group or group/user number. The OMF04 can be limited by file identifier and type of file change (add, update, or delete) that was made. Both reports are sorted in timestamp order. Both reports have a detail option that will provide the record image that is a part of each record included in the report.

Information on how to specify the run options for each report is detailed in the runfiles provided on the new BA60RPT subvolume.

Software Impacts

All BASE24 customers:

- New BA60RPT subvolume with all report source files, Make files and runfiles.
- Inline changes to BA60SRC.BASEM, BASEFM and BASEMM

A complete list of the files affected by Base Audit Reports can be found in the SCN file BA60UD09.SCNPSWD for the SEC01 and SEC02 reports and in BA60UD09.SCNOMF for OMF03 and OMF04.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

DISK Impacts

The SEC01 and SEC02 reports require 2 sort files and a scratch file. Each file will be the size of the SEC file used as input to the report. All 3 files are purged at the end of the report run.

All reports create a spooler copy of the report. They will also create a disk copy if a filename is specified in the runfile. Care should be taken when specifying the selection criteria for OMF reports. If all records are selected and OMF files are large or consist of several OMF files in a single day, the OMF reports could be very large. Detail versions of all the reports contain much more information for each record. Care should be taken when specifying the detail option.

Performance Impacts

All of the reports for this enhancement should be run at non-peak times if possible. The SEC01 and SEC02 detail reports require the sorting of the entire SEC file. If the detail option is chosen, the screen access details must be determined for each record in the report.

The OMF03 and OMF04 reports will read through the entire OMF file(s) for the date specified in the input file. If the detail option is chosen, the entire record image is examined one byte at a time to replace unprintable characters with an “*”. This could take considerable time and resources depending on the size of the OMF file(s) and the selection criteria.

Printer Use

The reports have been designed to print on paper that is 132 characters wide. If narrower paper is used, the readability of the reports will suffer.

Vendor Impacts.

Not applicable

Reference Documents

Not applicable

5: BASE24-atm

Section 5 discusses the changes and enhancements made to BASE24-atm in release 6.0 version 6, along with the resulting conversion and implementation requirements that existing BASE24-atm users must consider when upgrading from release 6.0 version 5.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-atm users, along with the associated conversion and implementation requirements that all BASE24-atm users must address.
- Optional enhancements describe the changes that affect only those BASE24-atm users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-atm users must address to implement the enhancements.

Section 5 must be used in conjunction with Section 4 to identify the changes that BASE24-atm users need to consider when upgrading to release 6.0 version 6.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-atm users must address in order to upgrade to release 6.0 version 6.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

- General Changes

This area is summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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General Changes

- The BASE24-atm Terminal Data Files (ATD) have been modified to include additional static configuration data and dynamic data fields on screen 1 and a new screen 22. Currently these new fields are being used only by the Shared (Wincor) 912 AKDS and Shared (Wincor) NDC+ AKDS enhancements, but they may be used by other enhancements in the future. In addition, the displaying of the Native Message Format field has been moved from screen 21 to screen 1. Please note that the location of the Native Message Format data within the ATD file is unchanged. This field is now being used by the BASE24-atm EMV extensions and the Shared (Wincor) AKDS enhancements. The ATD conversion program is only required for customers wishing to implement the Shared (Wincor) 912 AKDS or Shared (Wincor) NDC+ AKDS enhancements. For more information, see **Shared (Wincor) 912 AKDS** and **Shared (Wincor) NDC+ AKDS** in the Optional enhancements section, where the impacts of these enhancements are described.
- The BASE24-atm Transaction Log File Perusal has been modified to display three new fields on the second detail screen for all settlement records. These three new fields will non-zero only for when displaying settlement records for Diebold ATMs configured with a Bulk Note Acceptor (BNA) module, when they will contain details from the Diebold BNA Counts Token. For more information, see **Diebold Bulk Note Acceptance** in the Optional enhancements section, where the impacts of this enhancement are described.

Software Impacts

All BASE24 customers:

- Inline changes to the ATD DDL files (AT60DDL.DDLFATD and DDLGATD)
- Inline changes to AT60SRC.ATDXG, ATMFM and ATMMM
- Inline changes to the ATD Requester/Server (AT60AFT.RQATDS, SCR NATD1, SVATDTBL, SVATDTG and SVATDTS)
- Minor inline changes to the device-specific ATD requesters
AT601000.RQT10XXS, AT60DCSP.RQTDSPS, AT60DIEB.RQTDIEBS,
AT60ESD.RQT1ESDS, AT60FUJ7.RQTFUJ7S, AT60IBM.RQT36XXS,
AT60N50.RQT5XXXS, AT60TIDL.RQTTIDLS and AT60TRIT.RQTTRITS

CUSTOMACS Changes

Not Applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the files that have been affected by the **General Changes**.

ATM Terminal Definition Files (ATD)

The following fields will be visible on ATD screens for all BASE24-atm customers, but will only need to be configured by customers using the Shared (Wincor) 912 AKDS signatures and Shared (Wincor) NDC+ AKDS Signatures enhancements, with the exception of the Native Message Format field, which is also used by the BASE24-atm EMV extensions.

A new field, Native Message Format, has been added to the display on screen 1 of the ATD. Please note, this field already exists in the core section of the Static data of the ATD, but was originally displayed on screen 21 of the ATD, which is an EMV-specific screen. With this enhancement, the field display has been moved to screen 1 of the ATD.

For each terminal that runs firmware that conforms to the Shared 912 or NDC+ Signature specification created by Wincor Nixdorf, ensure that the Native Message Format field is set to "1". This will ensure that Shared (Wincor) 912 AKDS or Shared (Wincor) NDC+ is performed instead of standard Diebold AKDS or standard NCR AKDS processing.

Screen Name: NATIVE MESSAGE FORMAT
Screen: ATD Screen 1
DDL Name: ATDS1-CORE.NATV-MSG-FRMT

A new field, "Hardware", has also been added to the display on screen 1 of the ATD. This field will be updated by the Diebold 10XX/478X device handler during Shared (Wincor) 912 AKDS processing and by the NCR 5XXX device handler during Shared (Wincor) NDC+ AKDS processing, and set to the value that is returned by the ATM in response to an "EPP Capabilities" request message.

Screen Name: HARDWARE
Screen: ATD Screen 1
DDL Name: ATDD1-CORE.HDWR

Valid Values are: 0 – Unknown
 1 – Wincor
 2 - NCR
 3 - Diebold
 4 - Fujitsu
 5 – De La Rue
 6 – Triton
 7 – Olivetti

Default Value: 3 if the device type is 30 (Diebold)
 2 if the device type is 22 (NCR)
 0 for all other device types.

Screen 22 has been added to the ATD and contains AKDS information. Two new fields have been defined. The “Key Signer” field is used to indicate the name to be used in the HOST field of the TSS Host Public Key Security Primary Datasource (HSPKPD). If multiple Host Public Keys are required for an individual vendor’s ATMs, the Key Signer field can be used to assign a separate HOST name for each of those Public Keys.

Screen Name: KEY SIGNER
Screen: ATD Screen 22
DDL Name: ATDS1-CORE.KEY-SIGNER

This field allows up to 16 alphanumeric characters of free-format text to be entered. Embedded spaces are allowed.

Default Value: “HOST”

The “EPP Format” field is used to indicate whether the EPP, and therefore the ATM, supports Signature- or Certificate-based AKDS processing. This field will be updated by the Diebold 10XX/478X device handler during Shared (Wincor) 912 AKDS processing and by the NCR 5XXX device handler during Shared (Wincor) NDC+ AKDS processing, and set to the value that is returned by the ATM in response to an “EPP Capabilities” request message.

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Screen Name: EPP FORMAT
Screen: ATD Screen 22
DDL Name: ATDD1-CORE.EPP-FORMAT

Valid Values are: 0 – No RKL Supported
1 – Signature
2 - Certificate
3 – Both

Default Value: 2 if the device type is 30 (Diebold)
1 if the device type is 22 (NCR).
0 for all other device types.

Transaction Log File (TLF) Perusal

The BASE24-atm Transaction Log File Perusal has been modified to display three new fields on the second detail screen for all settlement records. These fields are as follows:

Screen Name: CASHBOX
Screen: TLF Perusal detail screen 2 for Settlement records
DDL Name: DIEBOLD-BNA-CNTS-TKN.CB4(x) +
DIEBOLD-BNA-CNTS-TKN.CB3(x)
-- where x goes from 1 to the number of note types (i.e.,
DIEBOLD-BNA-CNTS-TKN.NUM-ENTRIES)

Screen Name: RETRACT
Screen: TLF Perusal detail screen 2 for Settlement records
DDL Name: DIEBOLD-BNA-CNTS-TKN.RE4(x) +
DIEBOLD-BNA-CNTS-TKN.RE3(x)
-- where x goes from 1 to the number of note types (i.e.,
DIEBOLD-BNA-CNTS-TKN.NUM-ENTRIES)

Screen Name: COUNTERFEIT
Screen: TLF Perusal detail screen 2 for Settlement records
DDL Name: DIEBOLD-BNA-CNTS-TKN.A6(x)
-- where x goes from 1 to the number of note types (i.e.,
DIEBOLD-BNA-CNTS-TKN.NUM-ENTRIES)

These three new fields will non-zero only for when displaying settlement records for Diebold ATMs equipped with a Bulk Note Acceptor (BNA) module.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Entry added to the SECTBL to support screen 22 (new screen) for the ATD requester.

PATHCONF Changes

Not applicable

CRT Access Security Changes

Access to the new ATD screen 22 must be added through the Security screen for the appropriate users.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

ATM Terminal Definition Files (ATD)

For users not wishing to take the Shared (Wincor) 912 AKDS Signatures or the Shared (Wincor) NDC+ AKDS Signatures enhancements, an inline conversion of the ATD will be performed to initialize the new data fields as follows:

Field:	Diebold device types	NCR device types
Hardware:	"3" – Diebold	"2" – NCR
EPP Format:	"2" – Certificate	"1" – Signature
Key Signer:	"HOST"	"HOST"
Native Msg Format:	"0" – Standard Native Message	"0" – Standard Native Message

For users wishing to take the Shared (Wincor) 912 AKDS Signatures or the Shared (Wincor) NDC+ AKDS Signatures enhancements, either the ATD conversion program must be run to convert the ATD, or the customer may modify the fields manually via PATHWAY for affected ATMS. If the conversion program is to be used, the customer must create the ATD file using the ATDDLUP file.

The ATD conversion program on the AT60UC04 subvolume can be used to convert the ATD records from the release 5.3 format to the release 6.0 version 6 format. The ATD conversion program on the AT60UC09 subvolume can be used to convert the ATD records from the release 6.0 version 5 format to the release 6.0 version 6 format.

For customers wishing to upgrade to the Transaction Security Services (TSS), the ATD conversion program on the AT60UC05 subvolume can be used to create Channel Security Datasource (CSECD) records while updating ATD records with key locators that point to the appropriate CSECD records. TSS data can be entered manually using the ACI Desktop Graphical User Interface instead of using the conversion programs. Key fields in corresponding ATD records would have to be updated manually as well.

By default, the conversion program will initialize the new fields as detailed above for the inline conversion.

The customer must modify the BASE24-atm ATD conversion programs in order to initialize the Native Msg Format, Key Signer, ATM Hardware and Encrypting PIN Pad (EPP) Format fields in each ATD record to a value appropriate for the customer's

environment. They may wish set certain terminal's ATD records to particular values for these new fields, depending on (for example) the Terminal ID.

The Native Msg Format field will need to be set to indicate that Shared (Wincor) 912 or Shared (Wincor) NDC+ message formats are being used.

The EPP Format field will need to be set to correctly reflect the capabilities of the terminal's installed EPP.

The Key Signer field will need to be set to the value that TSS will use as the key to the Host Public Key Datasource.

The Hardware field will need to correctly indicate the manufacturer of the ATM.

The conversion program can be modified by editing the CNVATDS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#atdcnv.t,name $atdm,nowait/ cnvatdm -  
force -list -listreplace
```

Release 5.3 users should update the CNVATDR obey file on the AT60UC04 subvolume with the appropriate information. Release 6.0 version 6 users should update the CNVATDR obey file on the AT60UC09 subvolume with the appropriate information. Customers upgrading to use the Transaction Security Services should update the CNVATDR obey file on the AT60UC05 subvolume with the appropriate information. Both a new and old CSECD file must exist before the conversion is run. The new CSECD file must be created using the FUP utility. The old CSECD file will be the output of a previous ATD conversion for Transaction Security Services or will be an empty file the first time an ATD File conversion is run.

The CNVATDR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the AT60UC04 subvolume for release 5.3 customers, to the AAREAD09 file on the AT60UC09 subvolume for release 6.0 version 5 customers and to the AAREAD05 file on the AT60UC05 subvolume for customers upgrading to the Transaction Security Services.

Release 6.0 version 1, release 6.0 version 2, release 6.0 version 3 and release 6.0 version 4 customers should refer to the **Conversion Program Guidelines** within Section 1 of this document.

Host/Co-Network Impacts

Not applicable

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User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Optional Enhancements

This section identifies the BASE24-atm enhancements that can be enabled or disabled with release 6.0 version 6, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- Cutover Performance Improvements
- Diebold Bulk Cash Acceptance
- Diebold EMV Offline PIN Synchronization
- NDC+ 5xxx Device Handler AKDS Certificate Support
- Shared (Wincor) 912 AKDS Signatures
- Shared (Wincor) Diebold 10XX/478X EMV Extension
- Shared (Wincor) NDC+ AKDS Signatures
- Support for Response Code 85 in ARPC
- Tidel EMV

BASE24-atm users can choose whether or not to implement each of the above enhancements. Users who plan to implement a particular enhancement must address the corresponding release 6.0 version 6 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding release 6.0 version 6 changes, conversion requirements, and implementation requirements for future reference.

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Cutover Performance Improvements

This section describes the implementation requirements associated with the Cutover Performance Improvements enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Cutover Performance Improvements enhancement.

The processing implemented in this enhancement is to enhance the cutover processing within the BASE24-atm Authorization module.

Support has been added for the BASE24-atm Authorization module to optionally compute the Institution Definition File (IDF) dates during cutover. If the new LCONF parameter ATM-AUTH-IDF-DATES-COMPUTE is "Y" the IDF dates will be calculated, whereas if the ATM-AUTH-IDF-DATES-COMPUTE parameter is either not present or set to "N", the IDF dates will be read from disk.

Software Impacts

All BASE24-atm customers:

- Inline changes to AT60AUTH.AUTHG and AUTHLIBS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD09.SCNCPI.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. Details of the new parameters are provided in the LCONFAT file located on BA60MISC subvolume.

ATM-AUTH-IDF-DATES-COMPUTE

This param, used by the BASE24-atm Authorization module, specifies whether the BASE24-atm Authorization module should compute the Institution Definition File

(IDF) business dates during cutover processing or if the BASE24-atm Authorization module should read the IDF from disk.

Param Default: "N".

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Performance Impacts

There will be a performance improvement if the new LCONF param "ATM-AUTH-IDF-DATES-COMPUTE" is set to "Y". This is due to the BASE24-atm Authorization module computing the IDF business dates during cutover processing and therefore reducing the number of disk i/o's, as it is no longer necessary to read the business dates from the IDF on disk.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Diebold Bulk Note Acceptance

This section describes the implementation requirements associated with the Diebold Bulk Note Acceptance (BNA) enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Diebold Bulk Note Acceptance (BNA) enhancement.

The BASE24-atm Diebold 10XX/478X device handler has been enhanced to support Diebold's bulk note acceptor. This enhancement enables customers to make cash deposits at Diebold Opteva ATMs equipped with a bulk note currency acceptor. These deposits are verified deposits because the inserted currency is scanned, validated and counted by the currency acceptor. Any invalid or unrecognizable (i.e., torn, dirty, etc.) currency is rejected and the customer must remove it before processing can continue. This is all done by the currency acceptor before it sends the request message to the device handler.

Software Impacts

All BASE24-atm DCT customers:

- Inline changes to AT60DCT.ATDCTM, ATDCTMM, SVDCTG, SVDCTM and SVDCTS
- Inline changes to Current Cash requester and screen (AT60DCT.RQDCURRS and SCNDCURR)
- Inline changes to Standard Cash requester and screen (AT60DCT.RQDSTANS and SCNDSTAN)

All BASE24-atm Diebold 10XX/478X device handler customers:

- Inline changes to AT601000.C1000G, C1000S, C100DDLML, C100DDLMS, C100NAMS
- Inline changes to Diebold 10XX DCT server and its make file (AT601000.SVD10XXS and SVD10XXM)
- Updated event (log) message manual BA60LOGM.AT1000

All BASE24-atm Diebold 10XX/478X SSB device handler extension customers:

- Inline changes to AT60D100.C100SSDG, C100SSDS

All BASE24-atm TLF Perusal customers:

- Inline changes to TLF server and screen (AT60AFT.SVTLFS and SCRNTLF)

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD09.SCNDDBNA.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Token file (TKN)

The following BASE24-atm token has been added for this enhancement:

- The Diebold BNA Counts Token (token ID “AO”) contains the note counts for the Bulk Note Acceptor (BNA) device. These counts can only be obtained from the ATM. The data in this token gets written to the BASE24-atm Transaction Log File (TLF) whenever the ATM is balanced. This token is used only by Diebold ATM’s equipped with a BNA.

If the Diebold BNA Counts Token should not to be logged to the TLF or the BASE24 Interchange Log File (ILF), the appropriate record must be configured in the Token file.

If the Diebold BNA Counts Token is to be extracted with the TLF or ILF or sent to the host, the appropriate Token records must be configured.

Transaction Log File (TLF) Perusal

The BASE24-atm Transaction Log File Perusal has been modified to display three new fields on the second detail screen for all settlement records. For details of the new fields added to the TLF Perusal for this enhancement, please see the **General Changes** earlier in this section.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Diebold 10XX/478X Configuration File

The Diebold 10XX/478X Configuration File, AT601000.CNFGIX, has been modified to include:

- Load Group 712

Load group 712 is the SSB BNA load group. It must be loaded on top of base load group 0 and SSB DH load group 12.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

A new value of '4' (Cash Deposit) has been defined for both the STM.rqst.avail_dep_typ field and the STM.rqst.dep_typ field. The value from these fields is placed into S-124 BASE24-atm Depository type for transmission to a HOST or Co-Network. Therefore, if any Host or Co-Network processes the values in this field, it must be updated to expect a possible value of '4' (Cash Deposit).

User/Operational Impacts

Device Control Terminal (DCT)

The Balance to Current Cash Screen (screen 8) and the Balance to Standard Cash Screen (screen 9) have been modified to include a new field titled COUNTS.

For a Diebold ATM equipped with a bulk note acceptor, the BNA currency counts must be obtained from the device, before the ATM can be balanced. To obtain the BNA currency counts, the operator places the cursor in the COUNTS field and presses the F1 Function key. After obtaining the BNA currency counts, the operator can then balance the terminal as usual.

Operational Impacts:

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

The Diebold Opteva ATM must be equipped with a bulk note acceptor device running Agilis 91x CSD version 1.2.4.9 (including any BNA hot fixes from Diebold) or greater.

Reference Documents

- ***Bulk Note Acceptor (BNA) for Agilis 91x Terminal Programming Supplement*** (Part Number TM-000145-001, version A, September 2004)
- ***Diebold Agilis 91x (XV) and Agilis 91x for Opteva Terminal Programming Manual*** (Part Number TP-820740-001, version C, April 2005)

Diebold EMV Offline PIN Synchronization

This section describes the changes associated with the Diebold EMV Offline PIN Synchronization enhancement for BASE24-atm. The standard BASE24-atm Diebold 10XX/478X device handler module has been modified to support this enhancement.

Section 2 contains a functional overview of the Diebold EMV Offline PIN Synchronization enhancement.

EMV cards have the capability to hold the cardholder's PIN within the Integrated Circuit Card (ICC). This allows "offline PIN verification" (i.e. where the PIN entered by the cardholder is verified by the ICC) to be offered as a Cardholder Verification Method (CVM) for electronic transactions. Many payment terminals will not be enhanced to offer offline PIN verification, and so "online PIN verification" (i.e., where the PIN entered by the cardholder is verified by the issuer's system) will continue to be widely used.

Where both online and offline PIN verification can be used as the CVM for an EMV card, cardholder confusion would result if the PIN held on the card's ICC differed from the PIN held on the issuer's system. Also, if a cardholder exceeds the allowable bad PIN tries when entering the PIN, a mechanism is required to allow the bad PIN try counter to be reset. Therefore, it is critical that card issuers provide offline PIN management functionality.

The offline PIN management introduces two new administrative functions for card issuers.

- Support for PIN change on the ICC, where the cardholder wants to change their PIN to a new value.
- Support for PIN unblocking on the ICC, where the PIN Try Counter on the card has reached zero and needs to be reset to the normal value (the PIN Try Limit).

This additional functionality is achieved by delivering a script(s) to the card during an authorization response message.

BASE24-atm Offline PIN Management support includes both PIN change and PIN unblock (offline PIN try counter reset). BASE24-atm also ensures that the issuer system is notified if the card cannot be updated with the PIN management information in the script message.

PIN Unblock

The PIN on the card may be unblocked either implicitly or explicitly.

- An EMV transaction request initiated at an ATM contains data retrieved from the EMV Integrated Circuit card. This data includes the Card Verification Results (CVR), which contains an indicator (byte 3, bit 7) of whether the cardholder has exceeded the maximum number of PIN tries allowed for offline transactions. The checking of this indicator on all EMV transactions allows BASE24 to provide an “implicit” mechanism to unblock the PIN on the card.
- A PIN Unblock transaction request may be initiated by a customer, using an EMV card at an EMV-capable terminal. This provides for an “explicit” mechanism to unblock the PIN on the ICC.

A Host system may reset the bad PIN try counter on an EMV card by including a ‘PIN Unblock’ script command in any EMV transaction response message. In BASE24, issuers are able to specify whether offline PIN unblocking is permitted.

- A parameter, configurable by card prefix, allows an issuer to specify whether “implicit” offline PIN unblocking is permitted. One setting of this parameter requires the cardholder perform an “explicit” PIN Unblock transaction or PIN Change transaction to reset the bad PIN try counter on the card.
- Another parameter, configurable by card prefix, allows an issuer to specify whether “explicit” offline PIN unblocking is permitted.

BASE24 is capable of generating a ‘PIN Unblock’ script command, either “implicitly” (if byte 3, bit 7 of the CVR is set to ‘1’) or “explicitly” (in response to a ‘PIN Unblock’ transaction request). A parameter in BASE24, configurable by card, allows an issuer to specify the conditions under which a ‘PIN Unblock’ script should be generated. A ‘PIN Unblock’ script command may be returned to the card only if the cardholder’s online PIN is verified successfully and the transaction’s Authorization Request Cryptogram (ARQC) is verified successfully.

BASE24 is also capable of generating a ‘PIN Change’ script command in response to a ‘PIN Unblock’ transaction request. For example, having forgotten the PIN, a cardholder may be issued a new PIN (but not a new card). This processing will then allow a cardholder to resynchronize the ICC held PIN with the PIN held by the issuer. (Note that a ‘PIN Change’ script command will unblock the PIN on the ICC in addition to changing the PIN.) A parameter in BASE24, configurable by card, allows an issuer to specify the conditions under which a ‘PIN Change’ script should be generated. A ‘PIN Change’ script command may be returned to the card only if the cardholder’s PIN, and the transaction’s Authorization Request Cryptogram (ARQC) are successfully verified.

PIN Change

The PIN on the card may be changed using an “explicit” mechanism only. A PIN Change transaction request (and optionally a PIN Unblock transaction request), which may be initiated by an EMV-card at an EMV-capable ATM, provides the required functionality.

A Host system may change the PIN on an EMV card by including a ‘PIN Change’ script command in a transaction response message. A parameter in BASE24, configurable by card prefix, allows an issuer to specify whether an offline PIN change is permitted.

The parameter also indicates whether the new PIN data should be logged to the BASE24 database, as this information may be required if the card subsequently has to be re-issued with the new PIN. Current PIN Change transaction logic allows the new PIN verification digits to be written to the BASE24-atm Transaction Log File (TLF). This information can then be extracted and provided to a Card Management System, which may require the new PIN verification digits if the card subsequently has to be re-issued. For Visa PVV PIN verification, however, the new PIN cannot be derived from the new PVKI/PVV; therefore, additional information (e.g. the new PIN data itself) would be required.

BASE24 is capable of generating a ‘PIN Change’ script command, if the PIN Change request is processed successfully (i.e. the cardholder’s PIN verification digits are updated successfully) in the BASE24 database. A parameter in BASE24, configurable by card, allows an issuer to specify the conditions under which a ‘PIN Change’ script should be generated. A ‘PIN Change’ script command may be returned to the card only if the cardholder’s online PIN is verified successfully and the transaction’s Authorization Request Cryptogram (ARQC) is verified successfully.

For an EMV card that does not contain an active offline PIN, it would be inappropriate to return a ‘PIN Change’ script command, as the card may not be able to process this type of script command. Conversely, for an EMV card that does contain an active offline PIN, it would be inappropriate to authorize non-EMV PIN Change transactions, as a ‘PIN Change’ script command could not be returned in the transaction response (resulting in non-synchronization of the online PIN and the offline PIN). Another parameter in BASE24, configurable by card, allows an issuer to specify whether synchronization of the online and offline PIN is required for the card. The parameter specifies the action to take when a PIN Change request (either EMV or non-EMV) is performed on the EMV card.

Script Failure Notification

Once authorized by BASE24 or the issuer Host system, it is possible for script processing to fail at the ATM.

If BASE24 is notified that the card failed to unblock the offline PIN, following “implicit” PIN Unblock processing, BASE24 will raise an event message so that appropriate action can be taken.

If BASE24 is notified that the card failed to change or unblock the offline PIN (following a PIN Change transaction or an “explicit” PIN Unblock transaction), then the PIN Change/Unblock transaction is reversed. This has the effect of either restoring the cardholder’s original PIN verification digits to the BASE24 database, or notifying the issuer that the PIN on the card has not been unblocked. Similarly, the reversal is forwarded to an issuer Host system (if external to BASE24), if the BASE24 system did not authorize the transaction.

Prerequisite: The ATM Terminal Definition Files (ATD) Infrastructure enhancement must be installed prior to implementing the Diebold EMV Offline PIN Synchronization enhancement in the Diebold 10XX/478x device handler. In the ATD Infrastructure enhancement, the ATD records were expanded and new offsets were added.

Prerequisite: The Transaction Security Services (TSS) enhancement must be installed prior to implementing the Diebold EMV Offline PIN Synchronization enhancement. For more information refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)***.

Software Impacts

All BASE24-atm Diebold 10XX/478X device handler customers:

- Hooks to EMV procedures from the main device handler module (AT601000.C1000S)
- Inline changes to AT601000.C1000G
- Inline changes to AT601000.C1000D
- Inline changes to AT601000.C100NAMS
- Updated event (log) message manual BA60LOGM.AT1000

All BASE24-atm Diebold 10XX/478X device handler EMV customers:

- Inline changes to AT601100.C100EMVG and C100EMVS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD09.SCNDPIN.

CUSTMACS Changes

Not applicable.

LCONF Assign Changes

Not applicable.

LCONF Param Changes

Not applicable.

Database Configuration

Token File (TKN)

The following BASE24 Base token is used for this enhancement:

- The EMV Issuer Script Results Token (Token ID “BJ”) is used by the Diebold 10XX ATM Device Handler. The EMV Issuer Scripts Results Token contains information about the processing of EMV Script data. This token will be created by the acquirer process (e.g. device handler or interchange). It will contain information about the results of EMV script processing.

If the token for this enhancement is not to be logged to the TLF or BASE24 Interchange Log File (ILF), the appropriate token records must be configured.

If the token for this enhancement is to be extracted with the TLF or ILF, sent to the host (via HISO), or passed through BIC ISO Interchange Interface, the appropriate token records must be configured.

Receipt File (RCPT)

A new PIN Unblock receipt template is required for this enhancement.

Please refer to AT60MISC.RCPT for an example of the new template required.

Configuration Requirements

The following configuration requirements apply when the PIN Change transaction is authorized on BASE24-atm:

- The PIN verification method must be IBM 3624-format or Visa PVV.
- The BASE24-atm authorization method must be Positive or Positive Balance, so the cardholder has a Cardholder Authorization File (CAF) record.
- PIN verification digits must be carried in the CAF.

New Data Files

Not applicable.

Template Changes

Not applicable.

Device Configuration

ATMs

Where they are directly connected to BASE24, ATMs supporting offline PIN changes must be compliant with Diebold's EMV Smart Card application:

- Diebold's EMV Smart Card Application requires a Diebold ix, Opteva Series, or foreign vendor ATM, operating in 912 mode, equipped with the following:
 - HTP or newer processor board
 - Any EMV Level 1 certified Smart Card reader
 - Microsoft Windows NT (with Service Pack 6) or Microsoft Window 2000 or Microsoft Windows XP Embedded (XPe) operating system
 - Communication Subsystem (CSS) required for TCS Plus
 - Agilis Base Communications (ABC) 4.0
 - Terminal Control Software (TCS Plus 1.4.0 or greater) and/or Agilis 91x Software, version 1.2 or greater for Opteva terminals or version 1.1 or greater for Diebold ix series terminals
 - EMV Kernel, latest version

Security Devices

In order to generate a PIN change script command, Transaction Security Services (TSS) is required to support communication with the HSMs. The following Security devices and software are required:

- Thales HSMs with version 6 software or higher (supporting the enhanced 'KU' command)
- Atalla HSMs (supporting the new '351' command)

Diebold 10XX Configuration File (CNFGIX)

The Diebold 10XX Configuration File (CNFGIX) requires new states and screens to support the new EMV PIN Unblock transaction type. For more detailed information, see the CN100DOC file on the AT601000 subvolume. A default set of screens is provided, but these may require specific modifications depending upon the institution's requirements.

Mega Table (MEGATBL) Changes

Not applicable.

PI-Table Changes

Not applicable.

SEG-Table Changes

Not applicable.

Security Table (SECTBL) Changes

Not applicable.

PATHCONF Changes

Not applicable.

CRT Access Security Changes

Not applicable.

Routing Changes

Not applicable.

Network Configuration

Not applicable.

File Conversions

Not applicable.

Host/Co-Network Impacts

Host/Co-Network Impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the BIC ISO Interface, ISO Host Interface and Super Extract processes.

BIC ISO Interface Process

In approved PIN management transaction responses, the EMV Response Token and EMV Script token (containing a 'PIN Change' or 'PIN Unblock' script) may be included in the BASE24 external message. When the BIC ISO module sends a PIN Change or PIN Unblock response message, the SEND-CRD-BLK field in the EMV Response token in field S-126 will contain either 'C' or 'U' to indicate the type of script present in the EMV Script token. When the BIC ISO module receives a PIN Change or PIN Unblock response message, the SEND-CRD-BLK field in the EMV Response token in field S-126 will contain either 'C' or 'U' to indicate the type of script present in the EMV Script token.

If the EMV Issuer Script Results Token is configured to be included in the ISO message to the co-network, this change to the message needs to be handled by the co-network programs.

ISO Host Interface Process

In approved PIN management transaction responses, the EMV Response Token and EMV Script token (containing a 'PIN Change' or 'PIN Unblock' script) may be included in the BASE24 external message. When the HISO module sends a PIN Change or PIN Unblock response message, the SEND-CRD-BLK field in the EMV Response token in field S-126 will contain either 'C' or 'U' to indicate the type of script present in the EMV Script token. When the HISO module receives a PIN Change or PIN Unblock response message, the SEND-CRD-BLK field in the EMV Response token in field S-126 will contain either 'C' or 'U' to indicate the type of script present in the EMV Script token.

If the EMV Issuer Script Results Token is configured to be included in the ISO message to the host, this change to the message needs to be handled by the host programs.

Super Extract Process

The new EMV PIN Unblock transactions are supported with an internal transaction code of "820000". Hosts should be analyzed for any impacts that may result from receiving TLF Extract records containing the new transaction code.

If the PIN Change Token (containing new PIN data) and/or new EMV Issuer Script Results Token is configured to be included in the TLF and ILF Extracts, BASE24 users must consider this additional data in host programs that use the data from the log files.

User/Operational Impacts

Disk Impacts

The TLF and ILF require additional disk space if the PIN Change Token (containing new PIN data) and/or new EMV Issuer Script Results Token is configured to be included in the transaction records logged to these files. The total impact varies for each BASE24 system depending on the transaction volume in the BASE24 system.

Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Performance Impacts

If BASE24 is configured to generate the 'PIN Change' and/or 'PIN Unblock' script command, then the additional HSM command will have an impact on system performance.

Vendor Impacts

Not applicable.

Reference Documents

- ***Agilis 91X/TCS EMV Programmers' Guide***, Version 1.0.0, Preliminary Draft Copy, November 2003
- ***Agilis 91X/TCS EMV User's Guide***, Version 1.0.0, Preliminary Draft Copy, November 2003
- ***Diebold 91X Terminal Control Software (TCS), TCS Plus, and 91X TCS CSP Terminal Programming Manual***, July 2002, edition N
- ***Europay, MasterCard, Visa (EMV) Integrated Circuit Card Specification for Payment Systems*** (Books 1–4), version 4.0 December 2000

NDC+ 5XXX Device Handler AKDS Certificate Support

This section describes the implementation requirements associated with the NDC+ 5XXX Device Handler AKDS Certificate Support enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the NDC+ 5XXX Device Handler AKDS Certificate Support enhancement.

The BASE24-atm NCR 5XXX device handler has been enhanced to support the certificate-based public key encryption scheme provided by NCR's Apra Edge software. This enhancement enables customers to perform AKDS downloads from the NCR 5XXX device handler to Diebold ATMs running Apra Edge software using the certificate-based scheme.

Prerequisite: The Transaction Security Services (TSS) version 03.1 must be installed prior to implementation of this enhancement.

Software Impacts

All BASE24 customers:

- Changes to the ACI Utilities ACIUTILS.UTILB

All BASE24-atm NCR 5XXX device handler customers:

- Hooks to AKDS procedures from the main device handler module AT60N50.N50S
- Inline changes to AT60N50.N50G
- Inline changes to AT60N50.N50D
- Inline changes to AT60N50.N50DDL
- Updated event (log) message manual BA60LOGM.AT5070

All BASE24-atm NCR 5XXX device handler AKDS customers:

- Inline changes to AT60AN50.N50AKDS
- Inline changes to AT60AN50.N50AKDG

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD09.SCNNCRT.

CUSTOMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

The Diebold ATM must be equipped with an Encrypting PIN Pad (EPP) and running NCR Aprta Edge release 2.04.00 (or greater).

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

The Transaction Security Services (TSS) must be installed in order to use this enhancement.

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Not applicable

Reference Documents

- NCR: *Support for Initial Key Loading Using Certificates in NDC* (2003)

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Shared (Wincor) 912 AKDS Signatures

This section describes the implementation requirements associated with the Shared (Wincor) 912 AKDS Signatures enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Shared (Wincor) 912 AKDS Signatures enhancement.

The BASE24-atm Diebold 10XX/478X device handler has been enhanced to support the Shared (Wincor) 912 Automated Key Distribution System (AKDS) Signatures enhancement. A new bindable extension, AT60AW10.C10WAKDS, has been developed to support the Shared (Wincor) 912 AKDS Signatures enhancement. The standard BASE24-atm Diebold 10XX/478X device handler has also been modified with “hooks” (calls) to various C10WAKDS procedures that handle the special Shared (Wincor) 912 AKDS processing.

This enhancement enables customers to perform AKDS downloads from the Diebold 10XX/478X device handler to ATMs loaded with firmware that conforms to the Shared 912 Signature specification created by Wincor Nixdorf, and Signature-based Encrypting PIN Pads (EPPs).

The Certificate Authority (CA) to be used will depend on the manufacturer of the ATMs. If the ATMs are manufactured by Wincor Nixdorf, the CA will be Wincor Nixdorf. If the ATMs are manufactured by NCR, the CA will be the NCR trusted center.

Prerequisite: The Transaction Security Services (TSS) version 05.4 must be installed prior to implementation of this enhancement.

Software Impacts

All BASE24 customers:

- Changes to BA60MAKE.CUSTOMACS and FLGSDOC
- Inline changes to the TSS Command Message DDL file (BA60DDL.DDLGTSS)
- Inline changes to the ATM Constant DDL file (BA60DDL.DDLACNST)
- Minor inline changes to the Security Utilities and its Make file (BA60SRC.SECUTILS and SECUTLLM)
- Inline changes to the AKDS Security Utilities (BA60ASRC.SECAKDS)
- Inline changes to BA60SRC.BASEMM, BASEFM and BASRCM
- Minor inline change to BA60AFT.SECTBL
- Updated event (log) message manual BA60LOGM.BASECUTL

All BASE24-atm customers:

- Inline changes to AT60SRC.ATDXG, ATMFM and ATMMM
- Minor inline changes to the device-specific ATD requesters
AT601000.RQT10XXS, AT60DCSP.RQTDSPS, AT60DIEB.RQTDIEBS,
AT60ESD.RQT1ESDS, AT60FUJ7.RQTFUJ7S, AT60IBM.RQT36XXS,
AT60N50.RQT5XXXS, AT60TIDL.RQTTIDLS and AT60TRIT.RQTTRITS
- Inline changes to the ATD DDL files (AT60DDL.DDLFATD, DDLGATD)
- Inline changes to the ATD Requester/Server (AT60AFT.RQATDS, SCR NATD1,
SVATDTBL, SVATDTG and SVATDTS)

All BASE24-atm Diebold 10XX/478X device handler customers:

- Hooks to Shared (Wincor) 912 AKDS procedures in the main device handler
module AT601000.C1000S
- Minor inline changes to AT601000.C1000G and C1000M
- Additional dummy procedures added to AT601000.C1000D
- Updated event (log) message manual BA60LOGM.AT1000

All BASE24-atm Diebold 10XX/478X device handler customers taking the new
Shared (Wincor) 912 AKDS functionality:

- A new extension subvolume, AT60AW10, with all of the files required to support
the Shared (Wincor) 912 AKDS Signatures functionality.

A complete list of the files affected by this enhancement can be found in the SCN file
BA60UD09.SCNWDAK.

CUSTMACS Changes

The following Make flag has been added to the CUSTMACS file.

- atm_dh_c1000_shrd_akd_on

It should be set to true to bind the Shared (Wincor) 912 AKDS module to the Diebold
10XX/478X device handler.

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that may be added to the LCONF to support this enhancement. Note that these params already exist for the Diebold 10XX/478X AKDS module, and are simply being used by the Shared (Wincor) 912 AKDS module, too. These params therefore already exist in the LCONFAT file located on the BA60MISC subvolume.

ATM-DH-AKDS-TSS-TIMER

This param specifies the number of seconds the device handler waits for a reply from Transaction Security Services before aborting the load. This is an optional param. Valid values are 10 through 600. If the param is not found or is invalid, the default value of 600 is used.

ATM-DH-1000-AKDS-SERIAL-NUM-VRFY

This param specifies whether or not Transaction Security Services will verify the Encrypting PIN Pad's (EPP's) serial number during the exchange of public keys between the EPP and BASE24. This is an optional param. Valid values are "Y" (yes, verify the serial number) and "N" (no, do not verify the serial number). If the param is not found or is invalid, the default value of "N" is used.

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement. The settings of each field will depend on the configuration desired by the institution.

ATM Terminal Definition Files (ATD)

For details of the new fields added to the ATD for this enhancement, please see the **General Changes** earlier in this section.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Encrypting PIN Pad

An Encrypting PIN Pad (EPP) must be installed on every ATM that is to implement AKDS.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

An entry has been added to the SECTBL to support screen 22 (new screen) for the ATD requester.

PATHCONF Changes

Not applicable

CRT Access Security Changes

Access to the new ATD screen 22 must be added through the Security screen for the appropriate users.

Routing Changes

Not applicable

Network Configuration

The Transaction Security Services (TSS) must be installed in order to use this enhancement.

File Conversions

ATM Terminal Definition Files (ATD)

For users wishing to take the Shared (Wincor) 912 AKDS Signatures enhancement, either the ATD conversion program must be run to convert the ATD, or the customer may modify the fields manually via PATHWAY for affected ATMS.

For further information on the ATD conversion program, please see the **General Changes** earlier in this section.

Host/Co-Network Impacts

Super Extract Process

Host code changes are required for all BASE24-atm users that use OMF Extract to accept and process information from the ATD record.

User/Operational Impacts

Security Device Configuration in the Transaction Security Services database

For each security device to be used for public key cryptography (PKI) functions, you must check the Enable PKI field on the Security Device Configuration window of the ACI Desktop Graphical User Interface. If this field is not checked for a security device, it is not used for PKI functions. This enables you to consolidate all PKI functions in just one or a few devices.

Setup for Transaction Security Services

Initial setup procedures must be carried out for TSS before an Authentication and Public Key Exchange sequence or a Master Key Download sequence can take place between BASE24 and an ATM.

The EPP must contain a public and private key pair, a signature of the EPP's public key, the Certificate Authority's public key, and a unique serial number. All of these are created when the EPP is manufactured.

BASE24's public/private key pair must be created. This key pair is created in a hardware security module and stored in the Transaction Security Services database in the Host Public Key Security Primary Datasource (HSPKPD). Use the "Key Generate (KEYGEN) Command" to generate and store this RSA key pair.

The public key must be sent to the appropriate Certificate Authority to be verified and signed. The Certificate Authority must return BASE24's public key signature, as well as the Certificate Authority's own public key. These values must be entered into the Transaction Security Services database.

The Certificate Authority (CA) to be used will depend on the manufacturer of the ATMs. If the ATMs are manufactured by Wincor Nixdorf, the CA will be Wincor Nixdorf. If the ATMs are manufactured by NCR, the CA will be the NCR trusted center.

For more information about these initial setup procedures, refer to the ***BASE24 Transaction Security Services Processing Guide***.

Power-Up Considerations

Prior to the Automated Key Distribution System enhancement, the Diebold 10XX/478X device handler automatically performed a full load (or a load of just the PIN encryption and MAC keys) after receiving a power-up message from an ATM. With this enhancement the device handler will continue to do so, but there is an exception: The load will not be automatically performed if the ATM has an AKDS-enabled EPP and a master key has not yet been downloaded to the EPP. Instead, the device handler will generate an event message informing the operator that the master key must be loaded. The master key must be present before a PIN encryption key or MAC key can be loaded.

Financial Institution Table (FIT) Considerations

The local PIN verification key (PEKEY), if used, is encrypted under the master key and stored in the FIT. If a new master key is downloaded to an ATM, every PEKEY in the FIT must be re-encrypted under the new master key and the FIT must be downloaded to the ATM.

Supported Keys

Only the master key (A-key) can be downloaded to an ATM using the Shared (Wincor) 912 AKDS enhancement. AKDS does not support the downloading of the power-up COMM key (B-key) because this key is not currently supported by the Diebold 10XX/478X device handler. The DES keys (e.g. PIN encryption key, MAC key) will continue to be encrypted under the master key and downloaded to the ATM as per normal procedures prior to this enhancement.

Performance Considerations

If the Shared (Wincor) 912 AKDS enhancement is implemented, the additional HSM commands may have an impact on system performance.

Event Messages

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Compatible Devices and Software

The following device platforms and software versions are compatible with BASE24's Shared (Wincor) 912 AKDS enhancement.

- ATMs running firmware compliant with Wincor Nixdorf's Shared Signature Specification – Wincor Nixdorf NCD/Diebold D91x Message Format Extension for RKL release 1.5.
- An Atalla NSP must be running a minimum of AKB version 2.40 or 1.4 software to support public key cryptography commands using the RSA scheme. The 2.40 version software is used for A10100 AKB models while the 1.4 version software is used for A10000 AKB models. AKB products require the use of a Secure Configuration Terminal (SCT), release 4.0 or later. SCT 4.0 devices support the Atalla AKB products while remaining fully backward compatible with older Atalla releases. All of the public key cryptography commands are defaulted to “off” in the security policy delivered from the factory due to security exposure. BASE24 requires a number of these commands to support the BASE24-atm Automated Key Distribution System add-on product. Some of these commands are used out-of-band (outside of the BASE24 application); nonetheless, they are all required to support the BASE24-atm Automated Key Distribution System add-on product. The following commands required for the BASE24-atm Automated Key Distribution System add-on product must be enabled in your security policy using the 108 and 109 commands:
 - Generate RSA key pair (Command 120)
 - Generate AKB for root public key (Command 122)
 - Verify public key and generate AKB of public key (Command 123)
 - Generate digital signature (Command 124)
 - Verify digital signature (Command 125)
 - Generate message digest (Command 126)
 - Generate ATM master key and encrypt with public key (Command 12F)
- Thales e-Security version 6.02 firmware and the optional DSP plug-in (for RSA functions) provides public key cryptography commands using the RSA scheme. BASE24 requires a number of these commands to support the BASE24-atm

Automated Key Distribution System add-on product. Some of these commands are used out-of-band (outside of the BASE24 application), nonetheless, they are all required to support the BASE24-atm Automated Key Distribution System add-on product. The commands are as follows:

- Generate RSA key set (Command EI)
- Load a secret key (Command EK)
- Generate a MAC on a public key (Command EO)
- Validate a certificate and generate MAC on its public key (Command ES)
- Generate a signature (Command EW)
- Validate a signature (Command EY)
- Export a DES key (Command GK)
- Hash a Block of Data (GM)

Reference Documents

- Atalla: ***Atalla Key Block Banking Command Reference Manual*** (524130-002, July 2002)
- Thales: ***Host Security Module RG7000 Programmer's Manual*** (1270A514 Issue 4, November 2001)
- ***Wincor Nixdorf NDC/Diebold D91x Message Format Extension for RKL*** (Release 1.5. 15th December 2004)

Shared (Wincor) Diebold 10XX/478X EMV Extension

This section describes the implementation requirements associated with the Shared (Wincor) Diebold 10XX/478X EMV Extension enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Shared (Wincor) Diebold 10XX/478X EMV Extension enhancement.

The Shared (Wincor) Diebold 10XX/478X EMV device handler extension is a separately bindable module, independent from, yet compatible with, the Diebold 10XX/478X EMV device handler extension. In addition, the standard BASE24-atm Diebold 10XX/478X device handler module has been modified to support this new Shared (Wincor) EMV extension.

The BASE24-atm Diebold Entry Program has also been modified to support the Shared (Wincor) Diebold 10XX/478X EMV device handler extension. An ENTIBOLD object, minimum version 6.1.13 (15NOV05), is required.

Software Impacts

All BASE24customers:

- Changes to BA60MAKE.CUSTMACS and FLGSDOC

All BASE24-atm customers:

- Inline changes to AT60SRC.ATMFM and ATMMM

All BASE24-atm Diebold 10XX/478X device handler customers:

- Hooks in the main device handler module to support the Shared (Wincor) Diebold 10XX/478X EMV device handler extension (AT601000.C1000S)
- Inline changes to AT601000.C1000G and C1000M
- Additional dummy procedures added to AT601000.C1000D
- Updated event (log) message manual BA60LOGM.AT1000

All BASE24-atm Diebold 10XX/478X device handler customers taking the new Shared (Wincor) Diebold 10XX/478X EMV device handler enhancement:

- A new extension subvolume, AT60IW10, with all the files required to support the Shared (Wincor) Diebold 10XX/478X EMV functionality.

All BASE24-atm Diebold Entry Program users:

- An ENTIBOLD object, minimum version 6.1.13

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD09.SCN1W10.

CUSTMACS Changes

The following Make flag has been added to the CUSTMACS file.

- atm_dh_c1000_win_emv_on

It should be set to true to bind the Shared (Wincor) Diebold 10XX/478X EMV module to the Diebold 10XX/478X device handler.

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes fields, within BASE24-atm files, relevant to EMV support. Field settings depend on desired configuration.

BASE24-atm Terminal Data Files (ATD)

Screen Name: EMV Terminal Capabilities
Screen: ATD Screen 21
DDL Name: ATDS1-CORE.EMV-TERM-CAP

Indicates the card data input, Card Verification Method (CVM), and security capabilities of the terminal using a bit map to represent possible options.

The ATD has a separate screen 21 to configure the EMV terminal capabilities. For additional information regarding the setting of these fields refer to the ***BASE24-atm EMV Support Manual***.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Diebold 10XX/478X Configuration File

The Diebold 10XX/478X Configuration File, AT601000.CNFGIX, has been modified to include five new EMV load groups:

- Load Group 450
Load group 450 is the basic Shared (Wincor) EMV load group. This load group is loaded on top of base load group 0.
- Load Group 454
Load group 454 is the Shared (Wincor) EMV Statement Print load group. If Statement Print is implemented, this group is loaded on top of Shared (Wincor) EMV load group 450 and load group 4 screens.
- Load Group 455
Load group 455 is the Shared (Wincor) EMV Fast Cash load group. If Fast Cash transactions are supported, this load group is loaded on top of Shared (Wincor) EMV load group 450 and load group 5 screens.
- Load Group 456
Load group 406 is the Shared (Wincor) EMV PIN Change load group. If PIN Change has been configured into the system, this load group is loaded on top of Shared (Wincor) EMV load group 450 and load group 24 screens.
- Load Group 457
Load Group 457 is the Shared (Wincor) Surcharge Load group. If Surcharge transactions are supported, this load group is loaded on top of Shared (Wincor) EMV load group 450 and load group 304 screens.
- Load Group 458
Load Group 458 is the Shared (Wincor) non-Currency Dispense Load group. If non-Currency Dispense transactions are supported, this load group is loaded on top of Shared (Wincor) EMV load group 450 and load group 500 screens.

There is a new configuration table needed for the Shared (Wincor) EMV support. This table is accessed via the Diebold Entry Program (ENTIBOLD):

- Wincor EMV AID Table

NOTE: This table must be configured by the customer. The record(s) for this table are written to the CNFGIX file. Refer to the **BASE24-atm EMV Support Manual** for additional information about these tables and their values.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Access to ATD screen 21 (EMV data screen) must be added through the Security requester for the appropriate users.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Device Control Terminal (DCT)

The following (existing) command is part of Diebold 10XX Commands Screen (screen 40) of the DCT. It is used by both Diebold and Shared (Wincor) EMV device handlers.

- LOAD EMV DATA

For Shared (Wincor) Diebold 10XX/478X EMV, when the operator places the cursor on this command and presses function key F1-ENTER, the DCT sends a command to the device handler to initiate a load of all Wincor EMV AID table records.

Operational Impacts:

New event messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the appropriate event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Shared (Wincor) NDC+ AKDS Signatures

This section describes the implementation requirements associated with the Shared (Wincor) NDC+ AKDS Signatures enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Shared (Wincor) NDC+ AKDS Signatures enhancement.

The BASE24-atm NCR 5XXX device handler has been enhanced to support the Shared (Wincor) NDC+ Automated Key Distribution System (AKDS) Signatures enhancement. A new bindable extension, AT60AWN5.N50WAKDS, has been developed to support the Shared (Wincor) NDC+ AKDS Signatures enhancement. The standard BASE24-atm NCR 5XXX device handler has also been modified with “hooks” (calls) to various N50WAKDS procedures that handle the special Shared (Wincor) NDC+ AKDS processing.

This enhancement enables customers to perform AKDS downloads from the NCR 5XXX NDC+ device handler to ATMs loaded with firmware that conforms to the Shared NDC+ Signature specification created by Wincor Nixdorf, and Signature-based Encrypting PIN Pads (EPPs).

The Certificate Authority (CA) to be used will depend on the manufacturer of the ATMs. If the ATMs are manufactured by Wincor Nixdorf, the CA will be Wincor Nixdorf. If the ATMs are manufactured by NCR, the CA will be the NCR trusted center.

Prerequisite: The Transaction Security Services (TSS) version 05.4 must be installed prior to implementation of this enhancement.

Software Impacts

All BASE24 customers:

- Changes to BA60MAKE.CUSTOMACS and FLGSDOC
- Inline changes to the TSS Command Message DDL file (BA60DDL.DDLGTSS)
- Inline changes to the ATM Constant DDL file (BA60DDL.DDLACNST)
- Minor inline changes to the Security Utilities and its Make file (BA60SRC.SECUTILS and SECUTLLM)
- Inline changes to the AKDS Security Utilities (BA60ASRC.SECAKDS)
- Inline changes to BA60SRC.BASEMM, BASEFM and BASRCM
- Minor inline change to BA60AFT.SECTBL
- Updated event (log) message manual BA60LOGM.BASECUTL

All BASE24-atm customers:

- Inline Changes to AT60SRC.ATDXG, ATMFM and ATMMM
- Minor inline changes to the device-specific ATD requesters
AT601000.RQT10XXS, AT60DCSP.RQTDCCSPS, AT60DIEB.RQTDIEBS,
AT60ESD.RQT1ESDS, AT60FUJ7.RQTFUJ7S, AT60IBM.RQT36XXS,
AT60N50.RQT5XXXS, AT60TIDL.RQTTIDLS and AT60TRIT.RQTTRITS
- Inline changes to the ATD DDL files (AT60DDL.DDLFATD, DDLGATD)
- Inline changes to the ATD Requester/Server (AT60AFT.RQATDS, SCR NATD1,
SVATDTBL, SVATDTG and SVATDTS)

All BASE24-atm NCR 5XXX device handler customers:

- Hooks to Shared (Wincor) NDC+ AKDS procedures in the main device handler
module AT60N50.N50S
- Minor inline changes to AT60N50.N50G and N50M
- Additional dummy procedures added to AT60N50.N50D
- Updated event (log) message manual BA60LOGM.AT5070

All BASE24-atm NCR 5XXX device handler customers taking the new Shared
(Wincor) NDC+ AKDS functionality:

- A new extension subvolume, AT60AWN5, with all of the files required to support
the Shared (Wincor) NDC+ AKDS functionality.

A complete list of the files affected by this enhancement can be found in the SCN file
BA60UD09.SCNWNAK.

CUSTMACS Changes

The following Make flag has been added to the CUSTMACS file.

- atm_dh_n50_shrd_akd_on

It should be set to true to bind the Shared (Wincor) NDC+ AKDS module to the NCR
5XXX device handler.

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that may be added to the LCONF to
support this enhancement. Note that these params already exist for the NCR 5XXX
NDC+ AKDS module, and are simply being used by the Shared (Wincor) NDC+
AKDS module, too. These params therefore already exist in the LCONFAT file
located on the BA60MISC subvolume.

ATM-DH-AKDS-TSS-TIMER

This param specifies the number of seconds the device handler waits for a reply from Transaction Security Services before aborting the load. This is an optional param. Valid values are 10 through 600. If the param is not found or is invalid, the default value of 600 is used.

ATM-DH-N50-AKDS-SERIAL-NUM-VRFY

This param specifies whether or not Transaction Security Services will verify the Encrypting PIN Pad's (EPP's) serial number during the exchange of public keys between the EPP and BASE24. This is an optional param. Valid values are "Y" (yes, verify the serial number) and "N" (no, do not verify the serial number). If the param is not found or is invalid, the default value of "N" is used.

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement. The settings of each field will depend on the configuration desired by the institution.

ATM Terminal Definition Files (ATD)

For details of the new fields added to the ATD for this enhancement, please see the **General Changes** earlier in this section.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Encrypting PIN Pad

An Encrypting PIN Pad (EPP) must be installed on every ATM that is to implement AKDS.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

An entry has been added to the SECTBL to support screen 22 (new screen) for the ATD requester.

PATHCONF Changes

Not applicable

CRT Access Security Changes

Access to the new ATD screen 22 must be added through the Security screen for the appropriate users.

Routing Changes

Not applicable

Network Configuration

The Transaction Security Services (TSS) must be installed in order to use this enhancement.

File Conversions

ATM Terminal Definition Files (ATD)

For users wishing to take the Shared (Wincor) NDC+ AKDS Signatures enhancement, either the ATD conversion program must be run to convert the ATD, or the customer may modify the fields manually via PATHWAY for affected ATMS.

For further information on the ATD conversion program, please see the **General Changes** earlier in this section.

Host/Co-Network Impacts

Super Extract Process

Host code changes are required for all BASE24-atm users that use OMF Extract to accept and process information from the ATD record.

User/Operational Impacts

Security Device Configuration in the Transaction Security Services database

For each security device to be used for public key cryptography (PKI) functions, you must check the Enable PKI field on the Security Device Configuration window of the ACI Desktop Graphical User Interface. If this field is not checked for a security device, it is not used for PKI functions. This enables you to consolidate all PKI functions in just one or a few devices.

Setup for Transaction Security Services

Initial setup procedures must be carried out for TSS before an Authentication and Public Key Exchange sequence or a Master Key Download sequence can take place between BASE24 and an ATM.

The EPP must contain a public and private key pair, a signature of the EPP's public key, the Certificate Authority's public key, and a unique serial number. All of these are created when the EPP is manufactured.

BASE24's public/private key pair must be created. This key pair is created in a hardware security module and stored in the Transaction Security Services database in the Host Public Key Security Primary File (HSPKPD). Use the "Key Generate (KEYGEN) Command" to generate and store this RSA key pair.

The public key must be sent to the appropriate Certificate Authority to be verified and signed. The Certificate Authority must return BASE24's public key signature, as well as the Certificate Authority's own public key. These values must be entered into the Transaction Security Services database.

The Certificate Authority (CA) to be used will depend on the manufacturer of the ATMs. If the ATMs are manufactured by Wincor Nixdorf, the CA will be Wincor Nixdorf. If the ATMs are manufactured by NCR, the CA will be the NCR trusted center.

For more information about these initial setup procedures, refer to the ***BASE24 Transaction Security Services Processing Guide***.

Power-Up Considerations

Prior to the Automated Key Distribution System enhancement, the NCR 5XXX device handler automatically performed a full load (or a load of just the PIN encryption and MAC keys) after receiving a power-up message from an ATM. With this enhancement the device handler will continue to do so, but there is an exception: The load will not be automatically performed if the ATM has an AKDS-enabled EPP and a master key has not yet been downloaded to the EPP. Instead, the device handler will generate an event message informing the operator that the master key must be loaded. The master key must be present before a PIN encryption key or MAC key can be loaded.

Financial Institution Table (FIT) Considerations

The local PIN verification key (PEKEY), if used, is encrypted under the master key and stored in the FIT. If a new master key is downloaded to an ATM, every PEKEY in the FIT must be re-encrypted under the new master key and the FIT must be downloaded to the ATM.

Supported Keys

Only the master key (A-key) can be downloaded to an ATM using the Shared (Wincor) NDC+ AKDS enhancement. AKDS does not support the downloading of the power-up COMM key (B-key) because this key is not currently supported by the NCR 5XXX device handler. The DES keys (e.g. PIN encryption key, MAC key) will continue to be encrypted under the master key and downloaded to the ATM as per normal procedures prior to this enhancement.

Performance Considerations

If the Shared (Wincor) NDC+ AKDS enhancement is implemented, the additional HSM commands may have an impact on system performance.

Event Messages

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Compatible Devices and Software

The following device platforms and software versions are compatible with BASE24's Shared (Wincor) 912 AKDS enhancement.

- ATMs running firmware compliant with Wincor Nixdorf's Shared Signature Specification – Wincor Nixdorf NCD/Diebold D91x Message Format Extension for RKL release 1.5.
- An Atalla NSP must be running a minimum of AKB version 2.40 or 1.4 software to support public key cryptography commands using the RSA scheme. The 2.40 version software is used for A10100 AKB models while the 1.4 version software is used for A10000 AKB models. AKB products require the use of a Secure Configuration Terminal (SCT), release 4.0 or later. SCT 4.0 devices support the Atalla AKB products while remaining fully backward compatible with older Atalla releases. All of the public key cryptography commands are defaulted to “off” in the security policy delivered from the factory due to security exposure. BASE24 requires a number of these commands to support the BASE24-atm Automated Key Distribution System add-on product. Some of these commands are used out-of-band (outside of the BASE24 application); nonetheless, they are all required to support the BASE24-atm Automated Key Distribution System add-on product. The following commands required for the BASE24-atm Automated Key Distribution System add-on product must be enabled in your security policy using the 108 and 109 commands:
 - Generate RSA key pair (Command 120)
 - Generate AKB for root public key (Command 122)
 - Verify public key and generate AKB of public key (Command 123)
 - Generate digital signature (Command 124)
 - Verify digital signature (Command 125)
 - Generate message digest (Command 126)
 - Generate ATM master key and encrypt with public key (Command 12F)
- Thales e-Security version 6.02 firmware and the optional DSP plug-in (for RSA functions) provides public key cryptography commands using the RSA scheme. BASE24 requires a number of these commands to support the BASE24-atm Automated Key Distribution System add-on product. Some of these commands are used out-of-band (outside of the BASE24 application), nonetheless, they are all required to support the BASE24-atm Automated Key Distribution System add-on product. The commands are as follows:
 - Generate RSA key set (Command EI)
 - Load a secret key (Command EK)
 - Generate a MAC on a public key (Command EO)
 - Validate a certificate and generate MAC on its public key (Command ES)
 - Generate a signature (Command EW)
 - Validate a signature (Command EY)

- Export a DES key (Command GK)
- Hash a Block of Data (GM)

Reference Documents

- Atalla: ***Atalla Key Block Banking Command Reference Manual*** (524130-002, July 2002)
- Thales: ***Host Security Module RG7000 Programmer's Manual*** (1270A514 Issue 4, November 2001)
- ***Wincor Nixdorf NDC/Diebold D91x Message Format Extension for RKL*** (Release 1.5. 15th December 2004)

Support for Response Code 85 in ARPC

This section describes the implementation requirements associated with the Support for Response Code 85 in Authorization Response Cryptogram (ARPC) enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Support for Response Code 85 in ARPC enhancement.

The processing implemented in this enhancement is specific to approved offline PIN management transactions performed on Visa cards.

A user field in the EMV Status Token has been redefined to indicate whether an ISO response code of '85' should be used in the ARPC generation for the transaction (for BASE24-authorized transactions) or should be returned to the ATM as the Authorization Response Code (for BASE24-acquired transactions).

BASE24 As Issuer

When BASE24-atm receives an offline PIN management request message from Visa, the VisaNet Interchange Interface will set the new field in the EMV Status Token to '1', indicating that an ISO response code of '85' should be used in the ARPC generation if the transaction is subsequently approved.

When BASE24-atm is generating the ARPC for a transaction, the Authorization process will check the value of the new field in the EMV Status Token. If the field is set to '1', and the transaction has been approved, then the BASE24-atm Authorization process will use an ISO response code of '85' to generate the ARPC.

BASE24 As Acquirer

When BASE24-atm receives an approved offline PIN management response message from Visa, the VisaNet Interchange Interface will set the redefined user field in the EMV Status Token to '9', indicating that an ISO response code of '85' should be used as the Authorization Response Code (ARC) in the response to the ATM.

When BASE24-atm is returning the ARC to an ATM, the Device Handler process will check the value of the redefined user field in the EMV Status Token. If the field is set to '9', then the BASE24-atm Device Handler process will use an ISO response code of '85' as the ARC returned to the ATM.

In BASE24-atm, the NCR 5xxx Device Handler, the Shared (Wincor) NCR 5xxx Device Handler and the Diebold 10xx/478x Device Handler have been enhanced to

support the new functionality. The Shared (Wincor) Diebold 10xx/478x Device Handler will be enhanced to support the new functionality at a later date.

Software Impacts

All BASE24 customers:

- Base Token DDL changes (DDLBATKN)

All BASE24-atm EMV customers:

- Inline changes to AT60IATH.AUTHEMVS
- Inline changes to AT60IN50.N50EMVS
- Inline changes to AT60IWN5.N50WEMVS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD09.SCNRC85.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Token file (TKN)

The following BASE24-atm token has been updated:

- A user field has been redefined within the EMV Status Token (token ID “B4”) as the ISO-RC-IND field. This field indicates that an ISO response code of “85” should be used in the ARPC generation.

If the EMV Status Token should not to be logged to the TLF or the BASE24 Interchange Log File (ILF), the appropriate record must be configured in the Token file.

If the EMV Status Token is to be extracted with the TLF or ILF or sent to the host, the appropriate Token records must be configured.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

External Interfaces

The host or co-network may require modification to recognize and process new data within the EMV Status Token. The new data will be passed in the ISO message from the BASE24-atm ISO Host Interface or BIC ISO Interface.

Super Extract Process

Host code changes may be required for BASE24 users that utilize BASE24-atm Transaction Log File (TLF) and/or BASE24 Interchange Log File (ILF) Extract files to accept and process the redefined field in the EMV Status Token.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Tidel EMV

This section describes the implementation requirements associated with the Tidel EMV enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Tidel EMV enhancement.

A new bindable extension, AT60ITDL.TIDLEMVS, has been developed to support EMV in the Tidel device handler. The standard BASE24-atm Tidel device handler has also been modified with “hooks” (calls) to various TIDLEMVS procedures that handle the special EMV processing.

Software Impacts

All BASE24 customers:

- Changes to BA60MAKE.CUSTMACS and FLGSDOC

All BASE24-atm customers:

- Inline changes to AT60SRC.ATDGX, ATMFM and ATMMM

All BASE24-atm Tidel device handler customers:

- Hooks in the main device handler module to support EMV (AT60TIDL.TIDELS)
- Minor inline changes to AT60TIDL.TIDELG, TIDELM and TIDLNAMS
- A new file for dummy procedures (AT60TIDL.TIDELD)
- Updated event (log) message manual BA60LOGM.ATTIDL
- Updated informal device handler manual AT60TIDL.TIDLMAN

All BASE24-atm Tidel device handler customers taking the new EMV functionality:

- A new extension subvolume, AT60ITDL, with all of the files required to support Tidel EMV functionality.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD09.SCNITDL.

CUSTMACS Changes

The following Make flag has been added to the CUSTMACS file.

- atm_dh_tidel_emv_on

It should be set to true to bind the EMV module to the Tidel device handler.

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes fields, within BASE24-atm files, relevant to EMV support. Field settings depend on desired configuration.

BASE24-atm Terminal Data Files (ATD/TDF)

Screen Name: EMV Terminal Capabilities
Screen: ATD Screen 21
DDL Name: TDF.ADNL-DATA.EMV-TERM-CAP

Indicates the card data input, Card Verification Method (CVM), and security capabilities of the terminal using a bit map to represent possible options.

The ATD has a separate screen 21 to configure the EMV Terminal Capabilities fields. For additional information regarding the setting of these fields refer to the ***BASE24-atm EMV Support Manual***.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Tidel devices supporting EMV must have new software and hardware configured to read Integrated Circuit Cards (ICC) and send required EMV data fields to BASE24.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Access to ATD Screen 21 (EMV Data Screen) must be added through the Security requester for the appropriate users to be able to set the EMV Terminal Capabilities for Tidel devices. This existing screen is now accessible for Tidel devices (device type B4).

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

An inline conversion for the TDF field used to store the EMV Terminal Capabilities was added with Release 6.0 version 5. No other file conversions are required for this enhancement.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Operation Impacts

New event messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Not applicable

Reference Documents

- ***Tidel ATM Communication Specification***, Part no. 201-3653-012, Revision L, Date: January 29, 2004
- ***Preliminary Tidel ATM Communication Specification***, Part no. 201-3653-013, Revision M, Date: September 10, 2004

6: BASE24-pos

Section 6 discusses the changes and enhancements made to BASE24-pos in release 6.0 version 6, along with the resulting conversion and implementation requirements that existing BASE24-pos users must consider when upgrading from release 6.0 version 5.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-pos users, along with the associated conversion and implementation requirements that all BASE24-pos users must address.
- Optional enhancements describe the changes that affect only those BASE24-pos users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-pos users must address to implement the enhancements.

Section 6 must be used in conjunction with Section 4 to identify the changes that BASE24-pos users need to consider when upgrading to release 6.0 version 6.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-pos users must address in order to upgrade to release 6.0 version 6.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

- General Changes
- Expand the Number of RTBL and AST Entries

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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General Changes

- BASE24-pos Router/Authorization supports an internal routing algorithm for routing Diner's Club transactions. This algorithm has been updated to route US and Canadian acquired Diner's Club transactions with a prefix beginning with a "36" to MasterCard.
- BASE24-pos Router/Authorization supports an internal routing algorithm for routing Discover transactions. This has been updated to route all transactions with the following prefixes to Discover:
 - "601100" – "601109"
 - "601120" – "601129"
 - "601130" – "601139"
 - "601140" – "601149"
 - "6011740" – "6011749"
 - "601177" – "601179"
 - "601186" – "601189"
 - "601190" – "601199"
 - "650000" – "650999"

This change supports the Discover April 2005 Business Enhancements requirement of the new BIN ranges.

- BASE24-pos has been modified to support acquiring purchase, pre-authorization purchase, mail-phone order purchase, purchase with cash back and cash advance transactions that can be authorized for a lesser amount. For BASE24-pos Acquirers, the existing PTD field PRE-AUTH-LESSER-AMT will be modified to read as AUTH-LESSER-AMT. Use of this field will indicate that this merchant's POS device is capable of handling a transaction that has been approved for a lesser amount. Support was also added to authorize purchase, pre-authorization purchase, mail-phone order purchase, cash advance and purchase with cash back transactions for a lesser amount when BASE24-pos is the issuer. These changes keep the BASE24-pos product in compliance with the VisaNet April 2005 Mandates.
- The Hypercom device handler has been modified to improve reversal handling. Previously, when the device handler received an 0800 Test Transaction Request, it cleared out the history of the last financial transaction. Therefore, when an 0800 Test Transaction Request/Response occurred between an 0200/0210 Financial Transaction Request/Response and its 0400 Reversal Request, the reversal may have been dropped. This enhancement eliminates saving the information from an 0800/0810 Test Request/Response message. Now when the device handler receives the 0800 Test Transaction, it will format and return an 0810 Test

Transaction Response without altering the last transaction data stored in the BASE24-pos Terminal Data Files (PTD).

Software Impacts

All XPNET customers:

- SKELB changes - SCR 3265
 - XPNET 3.0 requires version 24 or greater of SKELB
"REL3^VER0^SUTL24^051010"
 - XPNET 3.1 requires version 05 or greater of SKELB
"REL3^VER1^SUTL05^20051003"

All BASE24-pos customers:

- Inline changes to the Router/Authorization module (AUTHLIBS, ROUTERS, and RTAUG)
- Inline changes to the PTD Screen file (SCRNPTD)

All BASE24-pos customers with the ACI Standard POS Device Handler:

- Inline changes to the ACI Standard POS Device Handler module (ASPDHS)
- Inline changes to the ARSP Screen file and Server (RSPSCRNS and SVARSPS)

All BASE24-pos customers with the Hypercom Device Handler:

- Inline changes to the Hypercom Device Handler module (HPDHS)
- Updated event (log) message manual BA60LOGM.PSHYPRCM

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

BASE24-pos Terminal Data files (PTD)

The following field's screen name has been modified from PREAUTH LESSER AMT to AUTH LESSER AMT. Setting this field to a Y indicates that this terminal is capable of handling a pre-auth purchase, purchase, mail-phone order, purchase with cash back or cash advance transaction that has been authorized for a lesser amount.

Screen Name: AUTH LESSER AMT
Screen: PTD Screen 6
DDL Name: PTDS1.PTDS1-CORE.PRE-AUTH-LESS-AMT-FLG

A flag indicating whether this terminal allows a transaction request to be authorized for a lesser amount than requested. This field is used only if a usage or limit would be exceeded if the transaction were to be authorized for the requested amount. Valid values are as follows:

Y = Yes, allow transactions to be authorized for a lesser amount

N = No, do not allow transactions to be authorized for a lesser amount

BASE24-pos ACI Standard Response File (ARSP)

The following response code is used in authorizing purchase transactions for a lesser amount:

- 010 - Approved for lesser amount

If the response code description is to be returned to the terminal in FID 'g' the terminal's ARSP record must be updated to contain response display text for response code '010'.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Both the SPROUTE routing algorithms for routing MasterCard cards (INAS) and Discover cards (DSCV) and the BASE24-pos internal routing algorithms for routing Diner's Club cards and Discover cards have been updated as described earlier.

Network Configuration

Customers that use SPROUTE routing must install the new version of SKELB. Since SKELB is run as a user library, do the following only after taking appropriate action to avoid library conflicts.

1. Install SKELB on the XPNET subvolume.
2. Run the SKELMLBC step if required.
3. If on a RISC machine, accelerate SKELB as shown in the SKELAXCL file.
4. Application processes which use SKELB as a library must be stopped and restarted in order for the new library to take effect.
5. Application processes which bind in SKELB must be re-compiled or re-bound with the new SKELB.

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

A new response code, '010' – Approved for Lesser Amount, will be sent to the terminal by the BASE24-pos Standard Device Handler (SPDH) when a pre-auth purchase, purchase, mail-phone order, purchase with cash back or cash advance transaction has been approved for a lesser amount. The terminal must be able to accept this new response code and process accordingly.

Reference Documents

Not applicable

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Expand the Number of RTBL and AST Entries

This section describes the implementation requirements associated with the Expand the Number of RTBL and AST Entries enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Expand the Number of RTBL and AST Entries enhancement.

The Routing Table (RTBL) has been modified to increase the number of entries in the authorization entities array from 28 to 42. This allows additional routing destinations to be configured.

The Authorization Selection Table (AST) has been modified to increase the number of entries in the services array from 36 to 48. This allows routing parameters to be configured for additional card service/issuer routing code combinations.

Software Impacts

All BASE24-pos customers:

- Inline changes to BP60DDL.DDLFAST and DDLFRTBL
- Inline changes to BP60AFT.RQASTS, RQRTBLS, SCRNAST, SCRNRRTBL, SVASTS and SVRTBLS
- Inline changes to BA60AFT.SECTBL
- Inline changes to PS60EXT.OMFXS
- Inline changes to PS60RTAU.ROUTERS and RTAUG
- New conversion programs (BP60UC04.CNVASTS and BP60UC09.CNVASTS) to convert the AST to the required format
- New conversion programs (BP60UC04.CNVRTBLS and BP60UC09.CNVRTBLS) to convert the RTBL to the required format

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD09.SCNRTBL.

CUSTOMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Entries have been added to the SECTBL to support screens 7 and 8 (new screens) and screen 9 (formerly screen 7) for the AST requester.

PATHCONF Changes

Not applicable

CRT Access Security Changes

Access to the new AST screens must be added through the SEC requester for the appropriate users. Please note, the data formerly displayed on screen 7 is now available on screen 9.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Authorization Selection Table (AST)

The AST conversion program must be run to convert the AST. Running the conversion program is required for any BASE24-pos user who utilizes the AST in transaction processing. Users must create the new AST file using the BAPDLFUP file. Once the AST file is converted, the modified BP60AFT.RQASTS and SVASTS must be used to maintain the file.

The AST conversion program on the BP60UC04 subvolume can be used to convert AST records from the release 5.3 format to the release 6.0 version 6 format. The AST conversion program on the BP60UC09 subvolume can be used to convert the AST records from the release 6.0 version 5 format to the release 6.0 version 6 format. If modifications to the conversion program are necessary, the program can be modified by editing the CNVASTR file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#cnvast.t,name $astm,nowait/ cnvastm –  
force –list –listreplace
```

Release 5.3 users should update the CNVASTR obey file on the BP60UC04 subvolume with the appropriate information. Release 6.0 version 5 users should update the CNVASTR obey file on the BP60UC09 subvolume with the appropriate information. The CNVASTR file should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BP60UC04 subvolume for release 5.3 customers and the AAREAD09 file on the BP60UC09 subvolume for release 6.0 version 5 customers.

Release 6.0 version 1, release 6.0 version 2, release 6.0 version 3 and release 6.0 version 4 customers should follow the same procedures as release 6.0 version 5 customers, because the AST DDL was not changed until release 6.0 version 6.

Routing Table (RTBL)

The RTBL conversion program must be run to convert the RTBL. Running the conversion program is required for any BASE24-pos user who utilizes the RTBL in transaction processing. Users must create the new RTBL file using the BAPDLFUP file. Once the RTBL file is converted, the modified BP60AFT.RQRTBLS and SVRTBLS must be used to maintain the file.

The RTBL conversion program on the BP60UC04 subvolume can be used to convert RTBL records from the release 5.3 format to the release 6.0 version 6 format. The RTBL conversion program on the BP60UC09 subvolume can be used to convert the RTBL records from the release 6.0 version 5 format to the release 6.0 version 6 format. If modifications to the conversion program are necessary, the program can be modified by editing the CNVRTBLR file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#cnvrtbl.t,name $rtbm,nowait/ cnvrtblm –  
force –list –listreplace
```

Release 5.3 users should update the CNVRTBLR obey file on the BP60UC04 subvolume with the appropriate information. Release 6.0 version 5 users should update the CNVRTBLR obey file on the BP60UC09 subvolume with the appropriate information. The CNVRTBLR file should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BP60UC04 subvolume for release 5.3 customers and the AAREAD09 file on the BP60UC09 subvolume for release 6.0 version 5 customers.

Release 6.0 version 1, release 6.0 version 2, release 6.0 version 3 and release 6.0 version 4 customers should follow the same procedures as release 6.0 version 5 customers, because the RTBL DDL was not changed until release 6.0 version 6.

Host/Co-Network Impacts

Super Extract Process

Host code changes are required for all BASE24-pos users that use OMF Extract to accept and process information for the new format of the AST record.

Host code changes are also be required for all BASE24-pos users that use OMF Extract to accept and process information for the new format RTBL record.

User/Operational Impacts

Disk Impacts

The AST will require additional disk space, because the minimum record length has increased from 348 bytes to 450 bytes. The RTBL may also require additional disk space, because the minimum record length has increased from 532 bytes to 784 bytes.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Optional Enhancements

This section identifies the BASE24-pos enhancements that can be enabled or disabled with release 6.0 version 6, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- Alternate Host Routing for POS
- CRT Auth Referral Override During Settlement/Cutover
- Cutover Performance Improvements
- HPDH Upgrade to Latest Initialization Specifications
- HPDH Upgrade to Latest Message Specifications
- Interac Online Payment
- ISO Host Interface Enhanced Reversal Routing
- ISO Host Interface Stored Value Support
- Issuer Declined Manually-Entered Transactions
- MasterCard Secure Code Validation
- Merchant Host Interface Compatibility with TSS
- Mobile Top-Up for POS
- SPDH EMV Transaction Certificates
- SPDH PIN-Debit Reconciliation
- Suppression of Issuer Discretionary Data

BASE24-pos users can choose whether or not to implement each of the above enhancements. Users who plan to implement a particular enhancement must address the corresponding release 6.0 version 6 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding release 6.0 version 6 changes, conversion requirements, and implementation requirements for future reference.

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Alternate Host Routing for POS

This section describes the implementation requirements associated with the Alternate Host Routing for POS enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Alternate Host Routing for POS enhancement.

The BASE24-pos ISO Host Interface (HISO) module has been modified to allow transaction request messages to be routed to an alternate destination (configured in the Authorization Selection Table (AST)) if the Host link is unavailable.

If an alternate destination is not configured, then the ISO Host Interface will respond to a link down situation as it has done previously.

When Alternate Host Routing is configured, the BASE24-pos Router Authorization process will perform stand-in authorization (as at present) only if a HISO process is configured as the final authorization destination.

Software Impacts

All BASE24-pos HISO customers:

- Inline changes to PS60HISO.PSHISOS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD09.SCNALTH.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

It will be possible to configure more than one HISO process in the AST as an "OVER" or "UNDER" routing destination.

If more than one HISO process is configured in the AST as an “OVER” or “UNDER” routing destination, then the first of these entries must be specified as “AUTHH” (i.e. where the actual destination is configured in the IDF Routing Table).

If more than one HISO process is configured in the AST as an “OVER” or “UNDER” routing destination, then all of these processes must use an HCF record with the same DPC Number (i.e. the value configured in the IDF Routing Table).

It will also be possible to configure one or more HISO process(es) in conjunction with one or more interchange interface process(es) in the AST as “OVER” or “UNDER” routing destinations. However, the BASE24-pos Router Authorization process will perform stand-in authorization (as at present) only if a HISO process is configured as the last of the routing destinations.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

When Alternate Host Routing is configured, a HISO process will route request messages to another interface process, instead of just returning the transaction to the BASE24-pos Router Authorization process.

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Performance Impacts

When Alternate Host Routing is configured, additional inter-process message I/O will occur when a Host link is unavailable.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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CRT Auth Referral Override During Settlement/Cutover

This section describes the implementation requirements associated with the CRT Auth Referral Override during Settlement/Cutover enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the CRT Auth Referral Override during Settlement/Cutover enhancement.

Users are unable to override a referral in CRT Auth if the referral is received prior to cutover, but is processed after cutover. The CRT Authorization Device Handler/Router/Authorization process has been modified to support the overriding of referrals on the POS Referral File (PRF) for the previous day.

If the referred transaction is retrieved from the previous day's PRF, then the override advice will be logged to the current day's BASE24-pos Transaction Log File (PTLF).

Software Impacts

All BASE24-pos CRT Auth customers:

- Inline changes to PS60RTAU.RTAUG, ROUTERS and AUTHLIBS
- Inline changes to BP60CRTA.CRTAUDHS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD09.SCNCRTA.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Performance Impacts

When a referred transaction is retrieved from the PRF, an additional disk I/O may be required, in order to search the previous day's file.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Cutover Performance Improvements

This section describes the implementation requirements associated with the Cutover Performance Improvements enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Cutover Performance Improvements enhancement.

The processing implemented in this enhancement is to enhance the cutover processing within the BASE24-pos Router Authorization module. The BASE24-pos Router Authorization process has been enhanced to not close and reopen the LCONF anymore during cutover processing.

Support has been added to BASE24-pos Router Authorization to optionally compute the IDF dates during cutover. If the new LCONF parameter POS-RTAU-IDF-DATES-COMPUTE is "Y", the IDF dates will be calculated during cutover. If POS-RTAU-IDF-DATES-COMPUTE is either not present or set to "N", the IDF dates will be read from disk.

Software Impacts

All BASE24-pos customers:

- Inline changes to PS60RTAU.RTAUG, ROUTERS and AUTHLIBS
- Updated event (log) message manual BA60LOGM.PSAUTH

All BASE24-pos Secure Internet Validation (SIV) customers:

- Inline changes to PS60VRTA.RTAUSIVG

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD09.SCNCPI.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. Details of the new parameters are provided in the LCONFPS file located on BA60MISC subvolume.

POS-RTAU-IDF-DATES-COMPUTE

This param, used by the BASE24-pos Router Authorization module, specifies whether the BASE24-pos Router Authorization module should compute the Institution Definition File (IDF) business dates during cutover processing or if the BASE24-pos Router Authorization module should read IDF from disk.

Param Default: "N".

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Performance Impacts

There will be a performance improvement because the BASE24-pos Router Authorization module no longer opens and closes the LCONF during cutover processing, and therefore the BASE24-pos Router Authorization module will reduce the number of LCONF disk i/os.

There will be a performance improvement if the new LCONF param POS-RTAU-IDF-DATES-COMPUTE is set to "Y". This is due to the BASE24-pos Router Authorization module computing the IDF business dates during cutover processing and therefore reducing the number of disk i/os necessary to read the business dates from the IDF.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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HPDH Upgrade to Latest Initialization Specifications

This section describes the implementation requirements associated with the HPDH Upgrade to Latest Initialization Specifications enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the HPDH Upgrade to Latest Initialization Specifications enhancement.

The purpose of this enhancement is to provide support for downloading EMV configuration data to a Hypercom terminal. This allows the customers the ability to configure their EMV terminals via the HPDH. This will involve the creation of four new config file records, these records being four new tables.

Note that this enhancement impacts T7 and ICE Hypercom terminals only. The tables and resulting configuration file records are accessed via the HPTC. The HPDHDDLs should be consulted for the specific table definition.

The four new tables are:

- Integrated Chip Card (ICC) table
- CHIPDATA table
- KEYDATA table
- REVOKED KEYS table

ICC table: This table is used to specify a Load ICC table entry, and assign specific parameters to it. This table occurs a maximum of eight times (i.e. 8 individual records). The Table ID is '0C'.

CHIPDATA table: This table is used to specify a Load CHIPDATA table entry, and assign specific parameters to it. This table has a single occurrence. (i.e. a single record). The Table ID is '0D'.

KEYDATA table: This table is used to specify a Load KEYDATA table entry, and assign specific parameters to it. This table has a maximum of sixty occurrences (i.e. 60 individual records). The Table ID is '0E'. Note that the sixty records are actually structured as three groups of twenty occurrences each.

REVOKED KEYS table: This table is used to specify a Revoked Certificate Keys table entry, and assign specific parameters to it. This table has a single occurrence. (i.e. a single record). The Table ID is '0F'.

Software Impacts

All BASE24 customers:

- Inline changes to BA60AFT.SECTBL

All BASE24-pos HPDH customers:

- Inline changes to PS60HPDH.HPDHDDL, HPDHDDL, HPDHLDL, HPDHS, RQHPTCS, SCRNHPT and SVHPTCS
- Updated event (log) message manual BA60LOGM.PSHYPRCM

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD09.SCNHPDHI.

CUSTOMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This enhancement includes the addition of four new configuration file records. Each of these records defines one of four tables. The configuration file records are accessed and maintained via the HPTC.

Integrated Chip Card (ICC) table:

This record is accessed as record type 8 via the main menu screen of the HPTC. Up to 8 occurrences of this table are allowed.

Screen Name: TABLE ENTRY

DDL Name: ICC-TBL.ICC-TBL-DATA.ICC-TBL-NUM

Indicates which of 8 table entries this record pertains to. Valid values are 1-8.

Screen Name: ICC ISSUER ID
DDL Name: ICC-TBL.ICC-TBL-DATA.ICC-ISS-ID

Indicates the Issuer's ID for this entry. Valid values are any 2-position alpha-numeric.

Screen Name: TARGET PERCENTAGE
DDL Name: ICC-TBL.ICC-TBL-DATA.TGT-PCTG

Indicates the Target Percentage for biased random selection. Valid values are any 2-position numeric.

Screen Name: MAX TARGET PERCENTAGE
DDL Name: ICC-TBL.ICC-TBL-DATA.MAX-TGT-PCTG

Indicates the Maximum Target Percentage for biased random selection. Valid values are any 2-position numeric.

Screen Name: THRESHOLD AMOUNT
DDL Name: ICC-TBL.ICC-TBL-DATA.THRHLD-AMT

Indicates the Threshold Amount for biased random selection. Valid values are any 4-position numeric.

Screen Name: TAC DENIAL
DDL Name: ICC-TBL.ICC-TBL-DATA.TAC-DENIAL

Indicates the Terminal Action Code (TAC) Denial. Valid values are any 10-position alpha-numeric.

Screen Name: TAC ONLINE
DDL Name: ICC-TBL.ICC-TBL-DATA.TAC-ONLINE

Indicates the Terminal Action Code (TAC) Online. Valid values are any 10-position alpha-numeric.

Screen Name: TAC DEFAULT
DDL Name: ICC-TBL.ICC-TBL-DATA.TAC-DEFAULT

Indicates the Terminal Action Code (TAC) Default. Valid values are any 10-position alpha-numeric.

Screen Name: TRANSACTION CATEGORY CODE
DDL Name: ICC-TBL.ICC-TBL-DATA.TXN-CAT-CDE

Indicates the Transaction Category Code. Valid values are any single alpha-numeric.
Defaults to 'R'.

Screen Name: ALWAYS PRINT & CHECK SIGNATURES
DDL Name: ICC-TBL.ICC-TBL-DATA.ICC-OPTS-1.PRNT-CHECK-SIG

Indicates whether to always print and check signatures. Valid values are '0' (disable),
and '1' (always print and check).

CHIPDATA table:

This record is accessed as record type 9 via the main menu screen of the HPTC.
There is a single occurrence of this table.

Screen Name: TABLE ID
DDL Name: CHIPDATA-TBL.TBL-ID

Indicates the Table ID or position within the table. Valid values are any 2-position
alpha-numeric.

Screen Name: ADDITIONAL TAGS ID
DDL Name: CHIPDATA-TBL.ADDL-TAGS-ID

Indicates the Additional Tags ID within the table. Valid values are any 2-position
alpha-numeric.

Screen Name: APPLICATION ID LENGTH
DDL Name: CHIPDATA-TBL.ICCAID-LGTH

Indicates the length of the ICC AID. Valid values are any 2-position numeric.

Screen Name: APPLICATION ID
DDL Name: CHIPDATA-TBL.ICCAID

Indicates the ICC AID. Valid values are any 32-position alpha-numeric.

Screen Name: AID FLOOR LIMIT
DDL Name: CHIPDATA-TBL.FLOOR-LMT

Indicates the AID's floor limit. Valid values are any 4-position alpha-numeric.

Screen Name: AID VERSION NUMBERS
DDL Name: CHIPDATA-TBL.AID-VER-1 (2, 3)

Indicates the AID version numbers. Valid values are any 4-position alpha-numeric.

Screen Name: AID TDOL
DDL Name: CHIPDATA-TBL.AID-TDOL

Indicates the AID TDOL (data object list). Valid values are any valid data object tag(s), up to a maximum of 40 positions.

Screen Name: AID DDOL
DDL Name: CHIPDATA-TBL.AID-DDOL

Indicates the AID DDOL (data object list). Valid values are any valid data object tag(s), up to a maximum of 40 positions.

Screen Name: OFFLINE RESPONSE CODES APPROVED
DDL Name: CHIPDATA-TBL.OFFLINE-APPRV-RESP

Indicates the offline approved response code. Valid values are any 2-position alpha-numeric.

Screen Name: OFFLINE RESPONSE CODES DECLINED
DDL Name: CHIPDATA-TBL.OFFLINE-DCLN-RESP

Indicates the offline declined response code. Valid values are any 2-position alpha-numeric.

Screen Name: UNABLE RESPONSE CODES APPROVED
DDL Name: CHIPDATA-TBL.UNABLE-APPRV-RESP

Indicates the unable approved response code. Valid values are any 2-position alpha-numeric.

Screen Name: UNABLE RESPONSE CODES DECLINED
DDL Name: CHIPDATA-TBL.UNABLE-DCLN-RESP

Indicates the unable declined response code. Valid values are any 2-position alpha-numeric.

OPTIONS: Valid values are '0' (disable), '1' (enable).

Screen Name: SKIP EMV PROCESSING
DDL Name: CHIPDATA-TBL.ICC-OPTS-1.SKIP-EMV-PRO

Screen Name:	USE ADDITIONAL TAGS
DDL Name:	CHIPDATA-TBL.ICC-OPTS-1.USE-ADDL-TAGS
Screen Name:	EASY ENTRY ALLOWED
DDL Name:	CHIPDATA-TBL.ICC-OPTS-1.EASY-ENTRY-ALWD
Screen Name:	GO ONLINE IF AAR
DDL Name:	CHIPDATA-TBL.ICC-OPTS-1.GO-ONL-IF-AAR
Screen Name:	SPECIAL PROCESSING ARQC
DDL Name:	CHIPDATA-TBL.ICC-OPTS-1.SPCL-PRO-ARQC
Screen Name:	USE ISSUER FLOOR LIMIT
DDL Name:	CHIPDATA-TBL.ICC-OPTS-1.USE-ISS-FLR-LMT
Screen Name:	SELECT AID IF EXACT MATCH
DDL Name:	CHIPDATA-TBL.ICC-OPTS-1.SELCT-AID-IF-MATCH

REVOKED KEYS table:

This record is accessed as record type 10 via the main menu screen of the HPTC.
There is a single occurrence of this table.

Screen Name:	KEY RID
DDL Name:	REVOKED-CERT-KEY-TBL.KEY-RID-1 (thru KEY-RID-10)

There are 10 occurrences of the KEY RID. Indicates the revoked certificate key scheme application identifier (1 thru 10). Valid values are any 10-position alpha-numeric.

Screen Name:	KEY INDEX
DDL Name:	REVOKED-CERT-KEY-TBL.KEY-IDX-1 (thru KEY-IDX-10)

There are 10 occurrences of the KEY INDEX. Indicates the revoked certificate key index (1 thru 10). Valid values are any 2-position alpha-numeric.

Screen Name:	KEY CERTIFICATE NUMBER
DDL Name:	REVOKED-CERT-KEY-TBL.KEY-CERT-NUM-1 (thru KEY-CERT-NUM-10)

There are 10 occurrences of the KEY CERTIFICATE NUMBER. Indicates the revoked key certificate number (1 thru 10). Valid values are any 6-position alpha-numeric.

KEYDATA table:

This record is accessed as record types 11 thru 30 via the main menu screen of the HPTC. There are up to 60 occurrences of this table, accessed via the HPTC as 20 distinct record types, each containing a group of 3 records..

Screen Name: TABLE ENTRY
DDL Name: KEYDATA-TBL.KEYDATA.KEYDATA-TBL-NUM

Indicates the table number (1-3). This value is translated by the device handler into a number from 1 to 60. Valid values are any 2-position numeric.

Screen Name: PUBLIC KEY INDEX
DDL Name: KEYDATA-TBL.KEYDATA.PK-KEY-IDX

Indicates the certification authority public key scheme application identifier. Valid values are any 2-position alpha-numeric.

Screen Name: PUBLIC KEY RID
DDL Name: KEYDATA-TBL.KEYDATA.PK-RID

Indicates the certification authority public key index. Valid values are any 10-position alpha-numeric.

Screen Name: PUBLIC KEY EXPONENT
DDL Name: KEYDATA-TBL.KEYDATA.PK-EXP

Indicates the certification authority public key exponent. Valid values are any 2-position numeric.

Screen Name: PUBLIC KEY LENGTH
DDL Name: KEYDATA-TBL.KEYDATA.PK-LGTH

Indicates the certification authority public key length. Valid values are any 2-position numeric.

Screen Name: PUBLIC KEY EXPIRATION
DDL Name: KEYDATA-TBL.KEYDATA.PK-EXP-DAT

Indicates the certification authority public key expiration date. Valid date is 6-position numeric.

Screen Name: PUBLIC KEY
DDL Name: KEYDATA-TBL.KEYDATA.PK-KEY

Indicates the certification authority public key. Valid values include any valid key up to 496 positions.

Screen Name: PUBLIC KEY HASH
DDL Name: KEYDATA-TBL.KEYDATA.PK-HASH

Indicates the certification authority public key hash. Valid values include any valid hash number up to 40 positions.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

The device firmware must conform to the requirements specified in ***Hypercom T7 and ICE System Point of Sale Software Initialization Parameters*** (Version 1.24a, Document No. SPOSINI.DOC, July 2004)

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Entries added to the SECTBL to support screens 11, 13, 15 and 16 (new screens) for the HPTC requester.

PATHCONF Changes

Not applicable

CRT Access Security Changes

Access to the new HPTC screens must be added through the Security screen for the appropriate users.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Disk Impacts

The HPTC configuration file may require additional disk space when new configuration file records are added to support the 4 new tables.

Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Not applicable

Reference Documents

- ***Hypercom T7 and ICE System Point of Sale Software Initialization Parameters*** (Version 1.24a, Document No. SPOSINI.DOC, July 2004)

- ACI Worldwide Inc.

HPDH Upgrade to Latest Message Specifications

This section describes the implementation requirements associated with the HPDH Upgrade to the Latest Message Specifications enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the HPDH Upgrade to Latest Message Specifications enhancement.

This enhancement will provide for additional Hypercom device handler functionality, including support for:

- CVV2 Data processing
- EBT Transaction processing

This enhancement also includes support for the following tables:

- Level II Commercial Card Indicator (Table ID 15), and the corresponding Response table (Table ID 73)
- CPS 2000 Data (Table ID 30)
- Surcharge Amount (Table ID 44)
- Payment Plan (Table ID 45)

CVV2 Data processing: The CVV2 is a six character field used to authenticate the card when the card is not present at the time of the transaction. The device handler has been modified to ensure the CVV2 is forwarded to the authorizer. This is accomplished via Table ID 16, a redefine of personal use field 63.

EBT Transaction processing: The HPDH device handler has been upgraded to support Hypercom's Electronic Benefits Transfer transactions. This is accomplished via Table ID 47, a redefine of personal use field 63.

This has required two additional values be added to the account selection positions of the processing code: '96' (EBT Cash Benefit) and '98' (EBT Food Stamp Benefit).

Support has also been added for Hypercom's EBT Response Data table. This table is used to pass response data from the host to the terminal. This is accomplished via Table ID 48, a redefine of personal use field 63.

Level II Commercial Card Indicator: Support has been added for Hypercom's Level II Commercial Card Indicator table. This is accomplished via Table ID 15, a redefine of personal use field 63. Support has also been added for the corresponding Response table (Table ID 73).

CPS 2000 Data: Support has been added for Hypercom's CPS 2000 Data table. This is accomplished via Table ID 30, a rewrite of personal use field 63.

Surcharge Amount: Support has been added for Hypercom's Surcharge Amount table. This is accomplished via Table ID 44, a rewrite of personal use field 63.

Payment Plan: Support has been added for Hypercom's Payment Plan table. This is accomplished via Table ID 45, a rewrite of personal use field 63.

Software Impacts

All BASE24-pos HPDH customers:

- Inline changes to PS60HPDH.HPDHDDL, HPDHG and HPDHS
- Updated event (log) message manual BA60LOGM.PSHYPRCM

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD09.SCNHPDHM.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

The device firmware must conform to the requirements specified in ***Hypercom Message Specification “The System”*** (Version 3.42, Document ID. MSGSPEC.DOC, August 2003)

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Not applicable

Reference Documents

- ***Hypercom Message Specification “The System”*** (Version 3.42, Document ID. MSGSPEC.DOC, August 2003)

Interac Online Payment

This section describes the implementation requirements associated with the Interac Online Payment (IOP) enhancement. This enhancement is for customers who connect to the Interac network and need to support this type of Internet payment. The functionality was formerly known as iDebit.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 3 contains a functional overview of this enhancement.

To support this enhancement, account type '50' was added to the Base constants for an 'iDebit' account. For IOP transactions, the ASPDH will use account type '50' and add the Transaction Subtype Token (BM). The transaction will be passed through BASE24-pos Router Authorization to the POS ISO Host Interface. Here the internal account type of '50' will be mapped to an external account type of '50' to be sent to the host.

Software Impacts

All BASE24-pos customers:

- Inline changes to BA60DDL.DDLBCNST
- Inline changes to PS60HISO.PSHISOG and PS60HISO.PSHISOS.
- Inline changes to PS60RTAU.AUTHLIBS and PS60RTAU.RTAUG.
- Inline changes to PS60SPDH.ASPDHS.
- Updated event (log) message manual BA60LOGM.PSAUTH

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD09.SCNIOP.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the records that must be added to the database to support this enhancement.

Account Type Table File (ATT)

Add a new record from Screen 1 with the following information:

Screen Name: ACCOUNT TYPE
New Entry: 9D
Screen Name: ACCOUNT TYPE NAME
New Entry: IDEBIT

Acquirer Processing Code File / Issuer Processing Code File (APCF/IPCF)

On Screen 2 check the transaction allowed for Normal Purchase and Merchandise Return. These must not be set to a '0' to allow IOP to be treated as a checking account.

External Message File (EMF)

On the record for INTERF TYPE: HOST, the element p-63 must be set to 'B' for messages 0200, 0210, 0400, 0420 and 0422. This will allow/require the transaction subtype token information to be passed to and received from the host.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

The IOP transaction is supported from the ACI Standard POS Device Handler. The terminal must set the following fields in the inbound messages.

FID D Application Account Type set to "5"

FID e POS Condition Code if sent, must be set to "01"

FID q Track 2 must begin with an "M" to indicate the Track 2 was manually entered.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Changes are required to the IMNI code to support this change. The inbound Host message must include transaction subtype token in primary bit 63. The account type of IOP transactions must be a '50'.

User/Operational Impacts

Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

ISO Host Interface Enhanced Reversal Routing

This section describes the implementation requirements associated with the ISO Host Interface Enhanced Reversal Routing enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the ISO Host Interface Enhanced Reversal Routing enhancement.

The BASE24-pos ISO Host Interface has been enhanced to optionally route a reversal transaction to the same interchange interface process that authorized the original transaction, when the host is the transaction acquirer. This enhancement enables the reversal to be successfully processed by the correct interchange interface process (the process with a record of the original transaction in its BASE24 Interchange Log File (ILF)) when multiple copies of the interchange interface are configured in a system. Without this enhancement, the reversal may be routed to an interchange interface process with no record of the original transaction in its ILF and subsequently dropped.

Note: This enhancement was added to the BASE24-pos Router module in BASE24-pos Release 6.0 Version 1.

Software Impacts

All BASE24 customers:

- Inline changes to BA60DDL.DDLPSTKN
- Inline changes to BA60SRC.PSTKNID

All BASE24-pos customers:

- Inline changes to PS60HISO.PSHISOS and PSHISOG

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD09.SCNERRH.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param record that must be added to the LCONF to support this enhancement.

POS-RVSL-ROUTE

A code indicating whether the BASE24-pos Router Authorization and BASE24-pos ISO Host Interface modules uses normal or enhanced routing when sending reversal transactions to interchange interface processes acting as transaction authorizers (issuers). With enhanced reversal routing, the Router Authorization and POS ISO Host Interface modules save the symbolic process name of the authorizer when the original transaction response is received from the interchange interface process. The modules then route a subsequent reversal transaction to the symbolic process name saved from the original transaction response. This ensures that the reversal transaction is sent to the same interchange interface process that received the original transaction. Enhanced reversal routing is recommended only when online host authorization (authorization type 0 and authorization level 1) is used. Valid values are as follows:

- 0 = Use normal reversal routing logic (default).
- 1 = Use enhanced reversal routing logic.

Database Configuration

Token File (TKN)

The following BASE24-pos token has been added for this enhancement:

- The Enhanced Reversal Routing Token (token ID "CS") contains the symbolic name of the interchange interface process that authorized the original transaction and to which the reversal will be sent.

The Enhanced Reversal Routing Token must be configured to be carried in the ISO message sent by the ISO Host Interface.

If the Enhanced Reversal Routing Token should not to be logged to the BASE24-pos Transaction Log File (PTLF) or the BASE24 Interchange Log File (ILF), the appropriate record must be configured in the Token file.

If the Enhanced Reversal Routing Token is to be extracted with the PTLF or ILF, or sent to the host, the appropriate Token records must be configured.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

If the POS-RVSL-ROUTE LCONF param indicates enhanced reversal routing is enabled and if the transaction authorizer is an interchange interface process, the

POS ISO Host Interface process will include the Enhanced Reversal Routing Token (token ID "CS") in the 0210 ISO response message sent to the host. This token contains the symbolic process name of the transaction authorizer. The host application must be modified to accept this token and store its contents when it is received in a 0210 ISO response message. The host application must also be modified to add this token and its stored contents to the 0420 ISO reversal message when attempting to reverse the transaction.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

ISO Host Interface Stored Value Support

This section describes the implementation requirements associated with the ISO Host Interface Stored Value Support enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the ISO Host Interface Stored Value Support enhancement.

The BASE24-pos ISO Host Interface has been enhanced to support stored value transactions acquired by the ISO Host Interface.

BASE24 As Issuer

When BASE24-pos receives a stored value transaction request message from the host, the BASE24-pos ISO Host Interface will map the host transaction code to the BASE24 transaction code.

Software Impacts

All BASE24-pos ISO Host Interface customers:

- Inline changes to PS60HISO.PSHISOS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD09.SCNSVHI.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Host impacts are incurred by those BASE24 users implementing the changes to the BASE24-pos ISO Host Interface.

Host code changes may be required to support new transaction codes of:

HOST transaction code:	BASE24 transaction code:
------------------------	--------------------------

"17" (SV Full Redemption)	"28"
-----------------------------	------

"28" (SV Replenishment)	"27"
---------------------------	------

"98" (SV Card Activation)	"25"
-----------------------------	------

"99" (SV Additional Card Activation)	"26"
--	------

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Issuer Declined Manually-Entered Transactions

This section describes the implementation requirements associated with the Issuer Declined Manually-Entered Transactions enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Issuer Declined Manually-Entered Transactions enhancement.

The Issuer Declined Manually-Entered Transactions enhancement involves changes to BASE24-pos to allow issuers additional flexibility in processing manually entered transactions. Specifically, the following functions will be supported:

- It will be possible for issuers to configure BASE24-pos to specify the conditions under which manual card verification digits (CVD2) must be present in manually entered transactions.
- It will be possible for issuers to configure BASE24-pos to specify the conditions under which manually entered transactions must be declined.

The enhanced processing will not be performed on recurring payments, because these types of transaction should not include CVD2.

Software Impacts

All BASE24 customers:

- Base DDL changes (DDLFCPF)
- Inline changes to BA60AFT.SCRNCPF, RQCPFS, SVCDFS
- Inline changes to BA60EXT.OMFXS
- Inline changes to BA60UC04.CNVCPFS
- New conversion program (BA60UC09.CNVCPFS) to convert the CPF to the required format

All BASE24-pos customers:

- Inline changes to PS60RTAU.ROUTERS, AUTHLIBS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD09.SCNCVD2.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new and enhanced fields in the files that are required to support this enhancement.

Card Prefix File (CPF)

Screen Name: CVD PROCESSING FLAG
Screen: CPF Screen 8
DDL Name: CPF.BASE.MANUAL-CV-REQ

A flag that specifies whether manually-entered card verification digits are required for manually-entered POS transactions. This flag is not checked for recurring payment transactions (acquirers are typically not permitted to retain card verification data).

Valid values are:

- 0 = No CVD2 required (default value)
- 1 = CVD2 required for all manually-entered transactions
- 2 = CVD2 required for Cardholder Not Present (CNP) transactions only

Screen Name: DENY MANUAL CARD ENTRY
Screen: CPF Screen 8
DDL Name: CPF.POS.MANUAL-CRD-ENTRY-FLG

Indicates whether a card number can be manually entered at the terminal. This flag is not checked for recurring payment transactions (acquirers are typically not permitted to retain card verification data). Valid values are:

- 0 = Do not deny manually-entered transactions (default value)
- 1 = Deny manually-entered transactions without CVD2 where the cardholder does not provide the CVD2 value (PS51-TKN.CVD-FLD-PRESENT = '9').
- 2 = Deny manually-entered transactions without CVD2 where the cardholder does not provide the CVD2 value (PS51-TKN.CVD-FLD-PRESENT = '9'), where the CVD2 value is illegible (PS51-TKN.CVD-FLD-PRESENT = '2'), or where the merchant does not provide the CVD2 value (PS51-TKN.CVD-FLD-PRESENT = '0').
- 3 = Deny all manually-entered transactions.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Card Prefix File (CPF)

If the BASE24 user wishes to enable the Issuer Declined Manually Entered Transactions enhancement, the user must update the new fields in the CPF with the appropriate value. Otherwise, the new fields in the CPF will default to indicate that the Issuer Declined Manually Entered Transactions processing is not performed. Users must create the new CPF file using the BADDLFUP file.

The CPF conversion program on the BA60UC04 subvolume can be used to convert the CPF records from the release 5.3 format to the release 6.0 format. The CPF conversion program on the BA60UC09 subvolume can be used to convert the CPF records from the release 6.0 version 5 format to the release 6.0 version 6 format. If modifications to the conversion program are necessary, the program can be modified by editing the CNVCPFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#cpfcnv.t,name $cpfm,nowait/ cnvcpfm -  
force -list -listreplace
```

Release 5.3 users should update the CNVCPFR obey file on the BA60UC04 subvolume with the appropriate information. Release 6.0 version 5 users should update the CNVCPFR obey file on the BA60UC09 subvolume with the appropriate information. The CNVCPFR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume for release 5.3 customers and the AAREAD09 file on the BA60UC09 subvolume for release 6.0 version 5 customers.

Release 6.0 version 1, release 6.0 version 2, release 6.0 version 3 and release 6.0 version 4 customers should refer to the **Conversion Program Guidelines** within Section 2 of this document.

Host/Co-Network Impacts

Super Extract Process

Host code changes are required for all BASE24 users that utilize the OMF Extract Process to accept and process information in the CPF.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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MasterCard Secure Code Validation

This section describes the implementation requirements associated with the MasterCard Secure Code Validation enhancement. Refer to the BASE24 Release 6.0 Version 3 Implementation Guide for further information on the Secure Internet Validation enhancement in BASE24-pos.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the MasterCard Secure Code Validation enhancement.

ACI's E-Commerce products support the ability for a cardholder to register for either the Visa (3-D Secure) or MasterCard (SPA UCAF) Internet security products and to perform cardholder authentication during an online purchase. These are supported through cardholder registration and cardholder authentication during a purchase with an Access Control Server (ACS) that is compliant with Visa or MasterCard's rules.

This enhancement adds MasterCard Secure Code support to the BASE24-pos Secure Internet Validation (SIV) module. The validation of the AAV may be configured to be authenticated by the hardware security module (HSM). BASE24-pos may have this authentication performed via a purchase, preauthorization request, or preauthorization completion transaction.

Note: Preauthorization processing for a lesser amount is not supported in the Secure Internet Validation enhancement.

Software Impacts

All BASE24-pos customers:

- Inline changes to BA60DDL.DDLFCPF
- Inline changes to BA60AFT.RQCPFS and SVCDFS.
- Updated event (log) message manual BA60LOGM.PSAUTH

All BASE24-pos SIV customers:

- Inline changes to PS60VRTA.RTAUSIVG and RTAUSIVS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD09.SCNMCSIV.

CUSTOMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the fields in the files that have been modified to support this enhancement.

Card Prefix File (CPF)

Screen Name: SIV CHECK TYPE
Screen: CPF Screen 6
DDL Name: CPF.POS.SIV-CHK-TYP

Indicates whether SIV card verification is to be attempted. Two new possible values have been added for this field. New valid values are:

- 4 = SIV enabled, check either the CAVV or AVV using the CVV method
- 5 = SIV enabled, check the AAV using the CVV method

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Super Extract Process

Host code changes are required for all BASE24-pos users that use OMF Extract to accept and process information for the new values found in the POS segment of the CPF.

User/Operational Impacts

Disk Impacts

The Key Authorization File (KEYA) may require additional disk space when new KEYA records are added for a SIV KEYA group. The same KEYA record may be used when the SIV KEYA group matches the CARD VERIFICATION KEYA group.

Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Performance Impacts

When performing the AAV validation with an HSM, an Inter-Process Communication (IPC) message is added for each call to the HSM.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Merchant Host Interface Compatibility with TSS

This section describes the changes, conversion requirements, and implementation requirements associated with the Merchant Host Interface Compatibility with Transaction Security Services (TSS) enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of this the Merchant Host Interface Compatibility with TSS enhancement.

Refer to the BASE24 Release 6.0 Version 1 and 2 Implementation Guide document for further information regarding the Transaction Security Services enhancement.

Software Impacts

All BASE24-pos Merchant Host Interface (MHI) customers:

- Inline changes to PS60MHI.MHIG and MHIS
- Inline changes to PS60MHI.RQMHCFS and SVMHCFS
- New conversion program (PS60UC05.TSSMHCFS)

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD09.SCNMHI.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

MHCF

All key fields in the Merchant Host Configuration File (MHCF) records must be updated to contain key locators. The key locators are used by TSS to read Channel Security Datasource (CSECD) records to retrieve the actual encryption/decryption keys used by the merchant host interface's terminals.

Refer to the File Conversions subsection of this enhancement to see how the key fields in the MHCF can be updated programmatically, using the default terminal ID (i.e., PTDF TERMINAL ID) as the default key locator for each MHCF record, and corresponding CSECD records created.

The MHCF requester/server have been modified to read the SECURE-DEV-TYP param in the LCONF. If the SECURE-DEV-TYP param indicates TSS, the key fields maintained in the MHCF files are automatically modified to contain the default terminal ID when a new record is added. The default terminal ID is the key locator for the device records in the CSECD.

If the TSS database is partitioned across multiple logical networks, a two-character hexadecimal partition identifier, set by the LCONF param LN-IND, is used as the first two characters of the key locator. For key locators in the MHCF files, the first fourteen characters of the default terminal ID is appended to the partition identifier to create the key locator.

Channel Security Datasource (CSECD)

A CSECD record must exist for every set of encryption/decryption keys used by merchant host interface's terminals in the BASE24 network. Normally, every terminal uses unique encryption/decryption keys so there would be one CSECD record for every default terminal ID in the MHCF records.

Refer to the File Conversions subsection of this enhancement to see how CSECD records for each default terminal ID can be created programmatically.

If a default terminal ID is configured in the MHCF to send PINs in the clear to any interchanges on the BASE24 system, a CSECD record must be manually added for the encrypted intermediate key with a PIN block type of PIN/PAD.

If an interchange interface is configured to support clear PINs, a Channel Security record must be manually added for the encrypted intermediate key with a PIN block type of ANSI.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

MHI terminals supporting Triple DES encryption must have new software/hardware configured to use the Triple DES encryption algorithm with double-length or triple-length keys.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

File conversions for the following BASE24-pos Merchant Host Interface files are supplied with the TSS enhancement. There are no structural changes in any files, but TSS files can be created and data populated from BASE24-pos Merchant Host Interface files. These files are updated with key locators to point BASE24-pos processes to the correct TSS records. TSS data can be entered manually using the ACI Desktop Graphical User Interface instead of using the conversion programs.

- MHCF

BASE24-pos Merchant Host Configuration File (MHCF)

The MHCF conversion program can be run to create CSECD records while updating MHCF records with key locators that point to the appropriate CSECD records. Running the conversion program is not required. CSECD records can be entered manually using the ACI Desktop Graphical User Interface instead. Key fields in corresponding MHCF records would have to be updated manually as well.

The MHCF conversion program on the PS60UC05 subvolume can be used to create or update the CSECD. If modifications to the conversion program are necessary, the program can be modified by editing the TSSMHCFS file.

To compile the conversion program, enter the following command at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#MHCftss.t, name $MHCFm, nowait/  
TSSMHCfM -force -list -listreplace
```

Users must update the TSSMHCfR obey file on the PS60UC05 subvolume with the appropriate information. TSSMHCfR must then be obeyed to execute the conversion program.

Both a new and old CSECD file must exist before the conversion is run. The new CSECD file must be created using the FUP utility. The old CSECD file will be the output of a previous MHCF conversion for TSS or will be an empty file the first time a Merchant Host Configuration File conversion is run.

For more information on this file conversion, refer to the AAREAD05 on the BA60UC05 subvolume and TSSMHCfR files on the PS60UC05 subvolume.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Mobile Top-Up for POS

This section describes the implementation requirements associated with the Mobile Top-Up for POS enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Mobile Top-Up for POS enhancement.

Standard BASE24 modules have been modified to support this enhancement. These modules include the POS Standard Device Handler, POS Settlement Reports, and POS Transaction Log File (PTLF) Perusal.

A new BASE24-pos module has been added to support this enhancement. This module is the BASE24-pos Transaction Context Manager.

Prerequisite: The BASE24-pos General Changes - SPDH Mobile Top-Up Hooks and the BASE changes in the BASE24-atm Mobile Top-Up Integration optional enhancement must be installed prior to implementation of the Mobile Top-Up for POS enhancement. For more information refer to the **BASE24 Release 6.0 Implementation Guide (Version 5)**. A complete list of the files affected by the release 6.0 version 5 mobile top-up support can be found in the SCN file BA60UD08.SCNMTOP.

Software Impacts

All BASE24 customers:

- Changes to BA60MAKE.CUSTMACS and FLGSDOC
- Inline changes to BA60DDL.DDLBATKN, DDLBCNST, DDLGPSTM (comments only), DDLPCNST, and DDLPSTKN
- Inline changes to BA60LIB.BALIBTBL
- Inline changes to BA60AFT.COBTKN and SECTBL
- Inline changes to BA60SRC.COBNames, PSTKNCVS, and PSTKNID

All BASE24 Transaction Based Pricing (TBP) customers:

- New executable objects for Transaction Based Pricing (TBP) reports and update to TBP documentation (subvolume BA60TBPD)

All BASE24-atm and BASE24-pos Mobile Top-Up customers:

- Inline changes to BA60TMTU.MOFLIBS, the Mobile Top-up infrastructure

All BASE24-pos customers:

- Inline code changes to PS60DDL.DDLFPTLF (comments only)
- Inline code changes to PS60AFT.SCRNPTL and SVPTLS
- Inline changes to PS60SRC.POSFM, POSM, POSMM and PSCOUTLS
- Inline changes to the POS Settlement Reports (e.g., PS60RPT.B24PS1S, B24PS2S, B24PS35S, SVPS21S, SVPS22S)
- Inline changes to PS60RTAU.ROUTERS and RTAUG
- Updated event (log) message manual BA60LOGM.PSROUTER

All BASE24-pos BIC ISO Host Interface customers:

- Inline changes to SW60PSBI.PSBICIS

All BASE24-pos ISO Host Interface customers:

- Inline changes to PS60HISO.PSHISOS

All BASE24-pos Standard POS Device Handler customers taking the Mobile Top-Up for POS enhancement:

- Hooks in the main device handler module to support Mobile Top-Up (PS60SPDH.ASPDHS)
- Inline changes to PS60SPDH.ASPDHG
- Additional procedures in PS60SPDH.SPDHD
- Inline changes to PS60SPDH.ASPDHM
- Inline changes to ACNF to support Mobile Top-Up (PS60SPDH.CNFSCRNS, and SVACNFS)
- Inline changes to ACI Standard Device Response to support Mobile Top-Up (PS60SPDH.RSPSCRNS and SVARSPS)
- Updated event (log) message manuals BA60LOGM.PSSPDH
- A new extension subvolume, PS60TSPD, with files required to support the SPDH device handler Mobile Top-Up functionality

All BASE24-pos customers taking the Mobile Top-Up for POS enhancement:

- New subvolume, PS60TCM, with files for the BASE24-pos Transaction Context Manager
- New subvolume, PS60TTCM, with files for the BASE24-pos Transaction Context Manager Mobile Top-Up extension
- A new event (log) message manual for the BASE24-pos Transaction Context Manager, BA60LOGM.PSTCM

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD09.SCNMTOPP.

CUSTMACS Changes

Various Make flags have been added to the CUSTMACS file for Mobile Top-Up for POS.

Set the following flag to true for the BASE24-pos Transaction Context Manager.

- pos_tcm_on

Set the following flag to true for the BASE24-pos Transaction Context Manager Mobile Top-Up extension.

- pos_tcm_mtu_on

The following Make flags were added as part of release 6 version 5 and apply to this enhancement:

Set the following flag to true to bind the Mobile Top-Up module to the ACI Standard POS Device Handler (ASPDH).

- pos_dh_spdh_mtu_on

Set the following flag to true to include the mobile top-up infrastructure subvolume (BA60TMTU).

- mtu_on

LCONF Assign Changes

The following LCONF assigns already exist. They are used by the BASE24-pos Transaction Context Manager and are required for the Mobile Top-Up for POS enhancement.

- LMCF
- MOF
- POS-PTLF
- STRF
- STRFEMT
- TKN
- TSRF

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. Details of the new parameters are provided in the LCONFBA and LCONFPS files located on BA60MISC subvolume.

POS-SPLIT-TXN-RTG-DEST

This param, used by the Standard POS Device Handler, specifies the destination of the BASE24-pos Transaction Context Manager process. The BASE24-pos Transaction Context Manager is required because of the split transaction routing of mobile top-up transactions (i.e., a Funds part and a mobile Interface part). This parameter is required for the Mobile Top-Up for POS enhancement.

POS-SETL-NOTIFY^{identifier}

Another configuration change required for the Mobile Top-Up for POS enhancement is the adding of the BASE24-pos Transaction Context Manager process name to the Settlement Notify List params in the LCONF.

The following LCONF params already exist. They are used by the BASE24-pos Transaction Context Manager and are required for the Mobile Top-Up for POS enhancement.

- LOGICAL-NET
- POS-LN-CUTOVER
- POS-NOTIFY-INTERVAL
- POS-NOTIFY-PERCENTAGE
- STRFEMT-MAX-TBL-SIZE
- STRFEMT-WARM^{identifier}

Database Configuration

ACI Standard Device Configuration File (ACNF)

Field Map records must be configured for the Mobile Top-Up Cash and Mobile Top-Up Funds transactions supported by the SPDH for the terminals configured to send the Mobile Top-Up transactions. A new screen, screen 9, was added for this purpose.

FID B (Amount 1), FID R (Card Type), and FID 7 (Mobile Top-Up data) should be included in the request Field Map records of the ACNF for terminals supporting the mobile top-up using cash transaction.

FID B (Amount 1), FID R (Card Type), FID q (Customer Track 2) or FID 2 (Customer Track 1) and FID 7 (Mobile Top-Up data) should be included in the request Field Map records of the ACNF for terminals supporting the mobile top-up using funds transaction.

FID R (Card Type) and FID 7 (Mobile Top-Up data) should be included in the response Field Map records of the ACNF for terminals supporting mobile top-up using cash and mobile top-up using funds transactions.

Acquirer Processing Code File (APCF)

The APCF profiles should be updated for each individual acquiring terminal or group of terminals supporting mobile top-up transactions.

Description	Tran Code
Mobile Top-Up using Cash	1S
Mobile Top-Up using Funds	1T
Refund Mobile Top-Up using Cash (future use)	2S
Refund Mobile Top-Up using Funds (future use)	2T

ACI Standard Device Response File (ARSP)

Customers can enter customized transaction description text in the ACI Standard Device Response File (ARSP) for each mobile top-up transaction type for use in FID s (Transaction Description) in the response.

Mobile Operator File (MOF)

The Mobile Operator File (MOF) is a required file for the Mobile Top-Up for POS Enhancement and contains one record for each telecommunications provider supplying mobile top-up services for its customers. It defines details of the mobile operators and the services provided. Information on use of the MOF file may be found in the **Base Files Maintenance Manual**.

BASE24-pos Terminal Data files (PTD)

Screen Name: CARD TYPE
Screen: PTD Screens 8, 9, 10
DDL Name: PTDD1.[ptdd1-core].SRV[x].TYP

Add the one- or two-character code used to identify the type of phone card used to initiate the mobile top-up transaction. Codes used in this field are either reserved for BASE24 products or are user-defined. Transaction counters and amount accumulators are also provided for each service entered here. Information on the codes being used may be found in the **BASE24-pos Files Maintenance Manual**.

BASE24-pos Transaction Log File (PTLF)

The following mobile top-up specific data has been added to page 2 of screen 4—format 1. It is displayed only if the information is present in the PTLF for that detail record. The message “MORE DETAIL INFORMATION AVAILABLE” is displayed on the bottom of page 1 when page 2 contains information. If page 2 does not contain any information, the message “NO MORE DETAIL INFORMATION AVAILABLE” is displayed.

Screen Name: MOBILE PHONE NUM
Screen: PTLF screen 4
DDL Name: PRE-PAY-TOP-UP-TKN.PHN-NUM

The mobile phone number to receive the top-up. This number is provided by the cardholder at the acquiring device.

Screen Name: TAX AMOUNT
Screen: PTLF screen 4
DDL Name: PRE-PAY-RESP-TKN.TAX-AMT

The tax amount applied to the top-up transaction.

Screen Name: TOP-UP REF NUM
Screen: PTLF screen 4
DDL Name: PRE-PAY-TOP-UP-TKN.REF

The reference number issued by the mobile operator in top-up response messages. This number is unique for each transaction.

Screen Name: RESP CODE
Screen: PTLF screen 4
DDL Name: PRE-PAY-TOP-UP-TKN.RESP-CDE

The mobile operator's response code. Alternatively, this field may contain a generic response code agreed upon between the mobile operator and the service provider. This code is converted by the BASE24 Mobile Operator Interface process into an internal BASE24 response code that is carried in the PSTM response code field.

Screen Name: MNO CODE
Screen: PTLF screen 4
DDL Name: PRE-PAY-TOP-UP-TKN.APPRV-CDE

The approval code if supplied by the mobile operator in the response message.

Split Transaction Routing File (STRF)

The Split Transaction Routing File (STRF) is a required file for the Mobile Top-Up for POS Enhancement. It contains one record for each Transaction Subtype in the BASE24 network that requires unique routing. The STRF enables the routing of transaction requests to two or more destinations based upon information obtained from a device for a single transaction. Information on use of the STRF file may be found in the **Base Files Maintenance Manual**.

Token file (TKN)

The following BASE24 base tokens have been added for this enhancement:

- The POS Split Transaction Routing Token (token ID "CR") carries BASE24-pos specific data that allows BASE24 to route multiple transaction requests related to a single cardholder request. It also allows BASE24 to identify and merge the multiple responses received into a single response destined for the cardholder. It is added by the BASE24-pos Transaction Context Manager process.
- The Pre-Pay Merchant Token (token ID "CJ") carries BASE24-pos specific data related to the mobile operator. It is added by the BASE24-pos Transaction Context Manager process.

The following existing BASE24 tokens are utilized for this enhancement:

- Inventory Voucher Token (token ID “N8”)
- Pre-Pay Generic Receipt Token (token id “BB”)
- Pre-Pay Original Data Token (token ID “BX”)
- Pre-Pay Receipt Token (token ID “BF”)
- Pre-Pay Response Token (token ID “BT”)
- Pre-Pay Top-Up Token (token ID “BI”)
- Split Transaction Routing Token (token ID “BR”)
- Transaction Subtype Token (token ID “BM”)

If the tokens for this enhancement should not be logged to the BASE24-pos Transaction Log File (PTLF) or the BASE24 Interchange Log File (ILF), the appropriate token records must be configured in the Token file.

If the tokens for this enhancement are to be extracted with the PTLF or ILF, sent to the host, or passed through the BIC ISO Interchange Interface, the appropriate records must be configured in the Token file.

Transaction Code/Subtype Relationship File (TSRF)

The Transaction Code/Subtype Relationship File (TSRF) is a required file for the Mobile Top-Up for POS Enhancement. It contains one record for each transaction subtype supported in the network. This file is used to assign an alphanumeric description to a transaction subtype as well as to associate the transaction subtype to one or more transaction codes. Information on use of the TSRF file may be found in the **Base Files Maintenance Manual**.

New Data Files

Not Applicable

Template Changes

COBNAMES

Mobile top-up phone card types and their descriptions must be entered into the CARD-TYPE-TABLE within the COBNAMES file. The number of occurrences must also be incremented appropriately in the COBNAMES file if the card descriptions are to be displayed on the BASE24-pos Terminal Data file (PTD).

Mobile Top-Up SPDH Names File

Supported Issuer Identification Numbers (IIN) and transaction subtypes must be entered into the PS60TSPD.TSPDNAMS file to support mobile top-up transactions. This file is used by the device handler during compilation. The IIN values will match the Operator IIN values in corresponding Mobile Operator File (MOF) records

Device Configuration

To utilize this enhancement, terminals must run a version of firmware that supports:

- Setting the standard message header transaction type field to 19 for a mobile top-up using cash transaction request message.
- Setting the standard message header transaction type field to 20 for a mobile top-up using funds transaction request message.
- The new or modified FIDs and SubFIDs listed in the Vendor Impacts section of this enhancement.

Acquirers and Acquirer Processors may need to work with merchants to make changes to terminal software for support of mobile top-up transactions.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

A SECTBL entry for the new ACNF screen 9 has been added.

PATHCONF Changes

Not applicable

CRT Access Security Changes

Access to the new ACNF screen 9 must be added through the SEC requester for the appropriate users.

Routing Changes

In a standard BASE24-pos system, the device handlers always route transactions directly to the BASE24-pos Router Authorization module. In a mobile top-up environment, the device handler must accommodate the possibility of multiple authorization destinations. The ACI Standard POS Device Handler (SPDH) will route mobile mop-up transactions to the destination specified in the LCONF parameter POS SPLIT-TXN-RTG-DEST. This destination should be the process name of the BASE24-pos Transaction Context Manager (TCM).

The TCM will utilize the new Split Transaction Routing File (STRF) for purposes of routing.

When the transaction is not a mobile top-up transaction, normal transaction processing and routing occurs.

Network Configuration

A new process for the BASE24-pos Transaction Context Manager must be added to any XPNET Node that processes BASE24-pos mobile top-up traffic. The following is an example of a process under XPNET 3.0. This example is intended to aid BASE24 users in their initial setup and should be altered to fit specific environments.

```
RESET PROCESS
SET PROCESS BCPU 0
SET PROCESS LIBRARY $VOLUME.XPNET.SKELB
SET PROCESS PROGRAM $VOLUME.PS60OBJ.PTCM
SET PROCESS PPD $T1P1
SET PROCESS PRIORITY 145
SET PROCESS CPU 1
SET PROCESS STARTUP AUTOMATIC
SET PROCESS QAT 64
SET PROCESS SERVICE P1A^QUEUE
ADD PROCESS P1A^PTCM1, UNDER SYSNAME \<sys>, UNDER NODE P1A^NODE
```

For more information about adding Processes to the XPNET system, refer to the ***NET24-XPNET Network Control Operations Guide*** or the ***NET24-XPNET Network Control Command Reference Manual***.

File Conversions

Monthly Transaction Accumulation Files (MTAF)

Changes to the Transaction Based Pricing (TBP) reports requiring an MTAF conversion were introduced with the BASE24 release 6.0 version 5 BASE24-atm Mobile Top-Up Integration enhancement. BASE24 TBP customers who performed this MTAF conversion previously will not convert the MTAF again. All BASE24 TBP customers must be running Version 9 of the TBP reports.

Refer to **General Changes** in Section 4 for additional information.

Host/Co-Network Impacts

Host impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the External Interfaces and Super Extract processes.

External Interfaces

Host code changes may be required to support a new POS reversal reason code of 40 – Split Routing Enabled; Secondary Service Not Approved which is mapped to an ISO value of U6. This value is supported by the BASE24-pos ISO Host Interface process and the BASE24-pos BIC ISO Interface process.

Super Extract Process

Standard PTLF Extract processing is not specifically impacted by the BASE24-pos Mobile Top-Up for POS enhancement but host systems must be aware of the following:

- the PTLF.AUTH.RESPONDER field can contain a new value of S – Secondary Service
- the PTLF.AUTH.RVRL-CDE field can contain a new value of 40 – Split Routing Enabled; Secondary Service Not Approved
- the PTLF.AUTH.TRAN-CDE.TC field may contain new values of 29 – mobile top-up using cash and 30 – mobile top-up using funds

If the new tokens for this enhancement are to be included in the PTLF and ILF Extracts, BASE24-pos users must consider this additional data in host programs that use the data from the log files.

Host code changes are required for all BASE24 users that wish to utilize the OMF Extract Process to accept and process file maintenance made to the MOF and STRF files using the BASE24 AFT subsystem.

User/Operational Impacts

Disk Impacts

The MOF, STRF and STRFEMT files impact disk requirements. The total impact varies for each BASE24 user depending on the number of MOF and STRF records defined in the BASE24 system.

The TSRF file impacts disk requirements. The total impact varies for each BASE24 user depending on the number of TSRF records defined in the BASE24 system.

Existing BASE24-pos users that implement this enhancement will experience a change in messaging or logging to the PTLF as follows:

- Each successful mobile top-up transaction will be recorded in two PTLF records: one for the funds authorization and one for the top-up authorization. The top-up authorization will be logged by the TCM process with a transaction code of '29' (mobile top-up using cash) or a transaction code of '30' (mobile top-up using funds). Additionally, the top-up authorization will be logged with a value of 'S' (secondary service) in the responder field.
- Declined transactions may have as many as two PTLF records.
- Reversed transactions may have an additional one or two PTLF records depending on the point of failure.
- If BASE24 is configured to do so, there may be an additional record logged to the PTLF for the logging of the Funds transaction.

The PTLF and ILF require additional disk space as the new tokens for this enhancement can be included in the transaction records logged to these files.

Personnel must understand that all Mobile Top-Up transaction will impact POS Terminal Data (PTD) clerk totals. This includes mobile top-up using cash transactions.

The following table below shows how the PTD terminal totals are impacted by the new Mobile Top-Up transactions. Personnel must understand that a Mobile Top-Up using Cash transaction does not impact the terminal totals. This is due to a Top-Up transaction being paid for with cash.

Transaction	Debit Amount	Debit Count	Credit Amount	Credit Count
Mobile Top-Up using cash				
Mobile Top-Up using funds	+Amt1	+1		
Refund Mobile Top-Up using cash (future)				
Refund Mobile Top-Up using funds (future)			+Amt1	+1
Mobile Top-Up using cash reversal				
Mobile Top-Up using funds reversal	-Amt1	-1		
Refund Mobile Top-Up using cash reversal (future)				
Refund Mobile Top-Up using funds reversal (future)			-Amt1	-1

Operational Impacts

BASE24 users must understand the concept of EMT Builds and the procedure to create an Extended Memory Table file (EMT). BASE24 users must understand the EMT Build Utility report that is generated by the EMT Build Utility. BASE24 users must also understand the concept of EMT warmboots and the procedure to warmboot an EMT. The STRFEMT is used in this enhancement.

PTLF Perusal supports data defined in the Pre-Pay Top-Up and the Pre-Pay Response tokens on the additional detail screen. Operators should be trained to recognize the information on the additional detail screen.

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the PTLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the **BASE24 Tokens Manual** provides information on assessing the impacts of tokens on log file disk space requirements.

Performance Impacts

Performance will be impacted in systems running mobile top-up transactions due to the increase in the transaction path length. The transaction path is longer because mobile top-up transactions are routed to the Transaction Context Manager for split routing. The TCM then routes the mobile top-up transaction to potentially two destinations (i.e., a funds authorizer and a mobile I/F or inventory stock manager).

Transaction Based Pricing Reports

The Transaction Based Pricing (TBP) reports have been modified to support the Secondary Service in BASE24-pos. These changes necessitate that all BASE24 TBP customers implement the updated version of the TBP report objects into their system.

Vendor Impacts

Existing devices that wish to support Mobile Top-Up transactions will need to be modified to send the SPDH messages defined below. Refer to the **ACI Standard POS Device Message Specifications Manual** and the **Standard POS Device Configuration Manual** for additional information.

ACI Standard POS Device Handler (SPDH) Message Format

The following field identifier (FID) has been modified to support an additional value. This FID is mandatory for Mobile Top-Up transactions.

FID	Picture	Length	Field Name	RQST	RESP
R	PIC X	1 byte	CARD TYPE Additional value of 'N' to indicate no card type for mobile top-up using cash transactions.	Mandatory	Mandatory

The following table shows how existing optional standard FIDs can be re-used for Mobile Top-Up transactions.

FID	Picture	Length	Field Name	RQST	RESP
J	PIC 9(18)	18 Numeric Characters	AVAILABLE BALANCE For Mobile Top-Up transactions, this field may contain the available balance returned from the Mobile Operator.	n/a	Optional
f	PIC X(400)	1 – 400 characters (variable length)	RECEIPT DATA Generic Message from Mobil Operator File and/or either the Customer Message from the Inventory Stock Configuration File (NSCF) or Marketing Message returned from the Mobile Operator Interface.	n/a	Optional

The following FIDs have been added to the ACI Standard POS Message.

FID	Picture	Length	Field Name	RQST	RESP
7		Up to 120 characters	Mobile Top-Up Data	Optional	Optional

The following optional sub field identifiers (subFIDs) have been added to the Product data field identifier (FID 7) of the ACI Standard POS Message.

SFID	Picture	Length	Field Name	RQST	RESP
a	PIC X(40)	1 to 40 characters (variable length)	MOBILE TOP-UP TRACK 2	Mandatory for Top-Up using Funds	Optional for Top-Up using Funds
b	PIC X(15)	15 characters	ORIGINAL TOP-UP REFERENCE The Mobile Operator uses this reference number for refund message to match with the original request. Currently considered for future use.	Optional	

c	PIC X(65)		MOBILE TOP-UP RESPONSE		
		1 – 16	TOP-UP REFERENCE This reference number is returned for all Mobile Top-Up transactions that are returned from the Mobile Operator. ACTIVATION NUMBER For all Mobile Top-Up transactions that are approved by the BASE24-inventory Stock Manager.		Optional
		17 – 32	OPERATOR NAME The name of the Mobile Operator as defined in the Mobile Operator File (MOF).		Optional
		33 – 47	TOP-UP APPROVAL CODE. Present on all approved Mobile Top-Up using Funds transactions as specified by the Mobile Operator.		Optional
		48 – 65	AVAILABLE BALANCE The available balance remaining on a pre-pay phone as specified by the Mobile Operator.		Optional

- The standard message header transaction type field must be set to a 19 for a Mobile Top-Up using Cash transaction request message.
- The standard message header transaction type field must be set to a 20 for a Mobile Top-Up using Funds transaction request message.
- Minimum FIDs required for MobileTop-Up using Cash

FID	SFID	Picture	Length	Field Name	RQST	RESP
B		PIC 9(18)	1 to 18 Numeric Characters	AMOUNT 1	M	M
J		PIC 9(18)	18 Numeric Characters	AVAILABLE BALANCE		O
R		PIC X	1 Character	CARD TYPE Set to "N" – To indicate that 'no card type' has been supplied (Cash).	M	M
f		PIC X(400)	1 to 400 Characters Variable Length. 1 – 200 201 - 400	RECEIPT DATA Contains a Message from the BB token (PRE- PAY-GENRC- RCPT-TKN) and/or may contain a marketing message from the BF token (PRE- PAY-RCPT-TKN).		O
7	a	PIC X(40)	1 to 40 Characters	MOBILE TOP- UP TRACK2	M	M
	b	PIC X(15)	15 Characters	ORIGINAL TOP-UP REFERENCE (future use)	O	
	c			PRE-PAY TOP-UP RESP		
		PIC X(16)	16 Characters	TOP-UP REFERENCE or ACTIVATION NUMBER		O
		PIC X(16)	16 Characters	OPERATOR NAME		O
		PIC X(15)	15 Characters	TOP-UP APPROVAL CODE		O
		PIC X(18)	18 Characters	AVAILABLE TIME		O

- Minimum FIDs required for Mobile Top-Up using Funds

FID	SFID	Picture	Length	Field Name	RQST	RESP
B		PIC 9(18)	1 to 18 Numeric Characters	AMOUNT 1 .	M	M
J		PIC 9(18)	18 Numeric Characters	AVAILABLE BALANCE		O
R		PIC X	1 Character	CARD TYPE Indicating type of card used for the purchase. Set to either "C" – Credit or "D" - Debit.	M	M
f		PIC X(400)	1 to 400 Characters Variable Length. 1 – 200 201 - 400	RECEIPT DATA Contains a Message from the BB token (PRE- PAY-GENRC- RCPT-TKN) and/or may contain a marketing message from the BF token (PRE- PAY-RCPT-TKN).		O
q		PIC X(40)	1 to 40 Characters	CUSTOMER TRACK2 Usual format for Track 2 to hold the purchase details.	M	M
7	a	PIC X(40)	1 to 40 Characters	MOBILE TOP- UP TRACK2	M	M
	b	PIC X(15)	15 Characters	ORIGINAL TOP-UP REFERENCE (future use)	O	

7	c			PRE-PAY TOP-UP RESP		
		PIC X(16)	16 Characters	TOP-UP REFERENCE or ACTIVATION NUMBER		O
		PIC X(16)	16 Characters	OPERATOR NAME		O
		PIC X(15)	15 Characters	TOP-UP APPROVAL CODE		O
		PIC X(18)	18 Characters	AVAILABLE TIME		O

Reference Documents

Not applicable

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SPDH EMV Transaction Certificates

This section describes the implementation requirements associated with the SPDH EMV Transaction Certificates enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the SPDH EMV Transaction Certificates enhancement.

The BASE24-pos Standard POS Device Handler (SPDH) has been enhanced to accept EMV transaction certificates. A transaction certificate (TC) is a card-generated cryptogram providing information that can be used if a transaction is disputed. TCs are stored in the terminal's memory and can be uploaded to the SPDH at intervals determined by the BASE24-pos customer and terminal vendor. A TC is uploaded in the form of an EMV log-only transaction, identified by a new Message Subtype, "E", in the SPDH message header. The TC is contained in the Authorization Request Cryptogram (ARQC) field within field identifier (FID) 6 sub field identifier (subFID) O (EMV Request Data). The SPDH logs the uploaded transaction to the BASE24-pos Transaction Log File (PTLF) using a new message type, 9920, which identifies the PTLF record as an EMV log-only transaction. The TC is found in the ARQC field of the EMV Request Token (token ID "B2:") within the PTLF record. Account balances are not impacted by the EMV log-only transaction. EMV checks are not performed on the transaction nor is it routed to an authorizer. The terminal totals in the BASE24-pos Terminal Data Files (PTD) are updated, however, enabling the EMV terminal to successfully balance during settlement.

The BASE24-pos PTLF Extract module has been enhanced to extract the new 9920 EMV log-only transaction records from the PTLF.

No modifications have been made to the Refresh module. The 9920 PTLF records are not used by Refresh when it performs transaction impacting.

No modifications have been made to the Transaction Based Pricing module. All 9920 PTLF records fall under "Other Record Categories".

No modifications have been made to the BASE24-pos Report programs. The 9920 PTLF records are ignored by the reports when they are run.

Software Impacts

All BASE24-pos customers:

- Inline changes to PS60EXT.PTLFXS

All BASE24-pos Standard POS Device Handler (SPDH) customers:

- Inline changes to PS60SPDH.ASPDHS and ASPDHG

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD09.SCNEMVTC.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Token File (TKN)

Configure the EMV Request Token (token ID "B2") to be logged to the PTLF.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Super Extract Process

Host code changes are required for all BASE24-pos users that utilize the PTLF Extract process and want to accept and process the new 9920 EMV log-only PTLF record.

User/Operational Impacts

Not applicable

Vendor Impacts

The terminal should upload EMV log-only transactions one at a time to the SPDH. Only approved transactions that were authorized offline by the terminal/EMV card should be uploaded. Declined transactions or transactions that were authorized online should not be uploaded.

EMV log-only transactions have the same format as their online counterparts, with all the usual header fields, FIDs and subFIDs present. The Message Subtype field in the SPDH message header contains a new value, "E" (EMV Log-Only). FID 6 subFID O (EMV Request Data) is included in the transaction request, just as it is with online transactions. The ARQC field within FID 6 subFID O contains the offline EMV transaction's TC.

The BASE24-pos customer and terminal software vendor must determine the method of uploading EMV log-only transactions. For example, some may choose to send all EMV log-only transactions during a batch close transaction. Others may choose to "piggyback" transactions, meaning the terminal sends an EMV log-only transaction immediately after an online transaction, during the same dialup session. The method chosen should not adversely affect online transactions or system performance.

Reversals of EMV log-only transactions are not allowed. If the SPDH does not respond to an EMV log-only transaction request before the terminal times out, the terminal should not send a reversal request. Rather, it should resubmit the EMV log-only transaction and continue to do so until a response is received. Resubmission should occur immediately with no intervening transactions from the terminal. If the SPDH successfully processes the EMV log-only transaction, the resubmitted transaction will be detected as a duplicate and dropped, and the SPDH will send a response to the terminal indicating it received a duplicate request. In order for duplicate detection to work, the terminal must utilize the Transmission Number field in the message header per standard SPDH message specifications.

Reference Documents

Not applicable

SPDH PIN-Debit Reconciliation

This section describes the implementation requirements associated with the SPDH PIN-Debit Reconciliation enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the SPDH PIN-Debit Reconciliation enhancement.

The BASE24-pos Standard POS Device Handler (SPDH) has been enhanced to support the reconciliation of PIN-debit transactions. This is accomplished by retrieving the System Trace Audit Number (STAN), Retrieval Reference Number and Debit Network / Sharing Group ID from the new Switch Common Data Token (token ID "BY") and sending this information in the response message to the terminal.

Software Impacts

All BASE24-pos Standard POS Device Handler (SPDH) customers:

- Inline changes to PS60SPDH.ASPDHS and ASPDHG

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD09.SCNPINDR.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

ACI Standard Device Configuration File (ACNF)

In the Response Field Map of the terminal's ACNF record, configure field identifier (FID) 6 (Product sub field identifiers (subFIDs)) to be returned to the terminal.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

ACI Standard POS Message

The following subFIDs have been added to FID 6 (Product subFIDs) of the ACI Standard POS Message.

SFID	Picture	Length	Field Name	RQST	RESP
h	PIC X(6)	6 bytes	System Trace Audit Number (STAN)		✓
i	PIC X(12)	12 bytes	Retrieval Reference Number		✓
j	PIC X(4)	4 bytes	Debit Network / Sharing Group ID		✓

Reference Documents

Not applicable

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Suppression of Issuer Discretionary Data

This section describes the implementation requirements associated with the Suppression of Issuer Discretionary Data enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Suppression of Issuer Discretionary Data enhancement.

BASE24-pos has been enhanced to protect sensitive consumer information in issuer transactions, in compliance with the Payment Card Industry (PCI) Data Security Standard. The protected information is discretionary data from track 1 and track 2, card verification data (CVC2/CVV2) and address verification data. New LCONF parameters are used to control which of these data elements are suppressed when logging issuer transaction data to the BASE24-pos Transaction Log File (PTLF).

Software Impacts

All BASE24-pos customers:

- Inline changes to PS60RTAU.AUTHLIBS, ROUTERS and RTAUG
- Updated event (log) message manual BA60LOGM.PSROUTER

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD09.SCNISDD.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. Details of the new parameter are provided in the LCONFPS file located on BA60MISC subvolume.

POS-ISS-AVS-DATA-SUPPRS

A flag indicating whether logging address verification data for issuer transactions to the PTLF should be suppressed. Valid values are as follows:

Y = Yes, suppress the logging of address verification data.

N = No, do not suppress the logging of address verification data (default).

POS-ISS-CVD2-DATA-SUPPRS

A flag indicating whether logging card verification data for issuer transactions to the PTLF should be suppressed. Valid values are as follows:

Y = Yes, suppress the logging of card verification data.

N = No, do not suppress the logging of card verification data (default).

POS-ISS-DISCR-DATA-SUPPRS

A flag indicating whether logging discretionary data for issuer transactions to the PTLF should be suppressed. Valid values are as follows:

Y = Yes, suppress the logging of discretionary data.

N = No, do not suppress the logging of discretionary data (default).

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Super Extract Process

Standard PTLF Extract processing is not specifically impacted by the Suppression of Issuer Discretionary Data enhancement but host systems must be aware that is the new LCONF params are set to a “Y”, the related data will not be present in the PTLF Extract file.

User/Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

7: BASE24-teller

Section 7 discusses the changes and enhancements made to BASE24-teller in release 6.0 version 6, along with the resulting conversion and implementation requirements that existing BASE24-teller users must consider when upgrading from release 6.0 version 5.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-teller users, along with the associated conversion and implementation requirements that all BASE24-teller users must address.
- Optional enhancements describe the changes that affect only those BASE24-teller users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-teller users must address to implement the enhancements.

Section 7 must be used in conjunction with Section 4 to identify the changes that BASE24-teller users need to consider when upgrading to release 6.0 version 6.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-teller users must address in order to upgrade to release 6.0 version 6.

There are no general enhancements inherent to BASE24-teller.

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Optional Enhancements

This section identifies the BASE24-teller enhancements that can be enabled or disabled with release 6.0 version 6, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no new optional enhancements available for BASE24-teller.

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8: BASE24-remote banking

Section 8 discusses the changes and enhancements made to BASE24-remote banking in release 6.0 version 6, along with the resulting conversion and implementation requirements that existing BASE24-remote banking users must consider when upgrading from release 6.0 version 5. This section covers changes made to the following products within the BASE24 family:

- BASE24-telebanking
- BASE24-billpay
- BASE24 Kernel/IEP

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-remote banking users, along with the associated conversion and implementation requirements that all BASE24-remote banking users must address.
- Optional enhancements describe the changes that affect only those BASE24-remote banking users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-remote banking users must address to implement the enhancements.

Section 8 must be used in conjunction with Section 4 to identify the changes that BASE24-remote banking users need to consider when upgrading to release 6.0 version 6.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-remote banking users must address in order to upgrade to release 6.0 version 6.

There are no general enhancements inherent to BASE24-remote banking.

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Optional Enhancements

This section identifies the BASE24-remote banking enhancements that can be enabled or disabled with release 6.0 version 6, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no new optional enhancements available for BASE24-remote banking.

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9: BASE24 Interchange Interfaces

Section 9 discusses the changes and enhancements made to BASE24 Interchange Interfaces in release 6.0 version 6, along with the resulting conversion and implementation requirements that existing BASE24 Interchange Interface users must consider when upgrading from release 6.0 version 5.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24 Interchange Interface users, along with the associated conversion and implementation requirements that all BASE24 Interchange Interface users must address.
- Optional enhancements describe the changes that affect only those BASE24 Interchange Interface users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 Interchange Interface users must address to implement the enhancements.

Section 9 must be used in conjunction with Section 4 to identify the changes that BASE24 Interchange Interface users need to consider when upgrading to release 6.0 version 6.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24 Interchange Interface users must address in order to upgrade to release 6.0 version 6.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

- General Changes

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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General Changes

- Changes for a new reversal code in the BASE24-pos BIC ISO Interface and the BASE24-pos ISO Host Interface to support the Mobile Top-Up for POS enhancement.

Software Impacts

All BASE24-pos BIC ISO Interface customers:

- Inline changes to SW60PSBI.PSBICIS

All BASE24-pos ISO Host Interface customers:

- Inline changes to PS60HISO.PSHISOS

CUSTMACS Changes

Not applicable.

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Host/Co-network impacts are incurred by those BASE24 users implementing the changes to the BASE24-pos BIC ISO Interface and the BASE24-pos ISO Host Interface.

External Interfaces

Host code changes may be required to support a new reversal reason code for Split Routing Enabled; Secondary Service Not Approved. The BASE24 value of "40" is mapped to an ISO value of "U6". This value is supported by the BASE24-pos ISO Host Interface process and the BASE24-pos BIC ISO interface process.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable.

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Optional Enhancements

This section identifies the BASE24 Interchange Interface enhancements that can be enabled or disabled with release 6.0 version 6, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- Boston Communications Group Interface (BCGI) Restructure
- Support for Response Code 85 in ARPC
- XML Telco Interface (XTI)

BASE24 Interchange Interface users can choose whether or not to implement each of the above enhancements. Users who plan to implement a particular enhancement must address the corresponding release 6.0 version 6 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding release 6.0 version 6 changes, conversion requirements, and implementation requirements for future reference.

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Boston Communications Group Interface Restructure

This section describes the implementation requirements associated with the enhancement to provide support for Boston Communications Group Interface (BCGI).

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Boston Communications Group Interface (BCGI) Restructure enhancement.

All customers who use the BASE24-atm Mobile Top-Up Boston Communication Group Interface (BCGI) product must implement this change.

This enhancement implements support for BASE24-atm Boston Communications Group Interface.

Software Impacts

All BASE24 customers:

- Changes to BA60MAKE.CUSTMACS and FLGSDOC
- Inline changes BA60SRC.BASEFM and BASEMM
- Software fix to BA60SRC.HISWUTLS

All BASE24 Interchange Interface customers:

- Inline changes to SW60SRC.SWIFM, SWIM and SWIMM

All BASE24 Interchange Generic Switch Interface customers:

- Inline changes to SW60SWCH.SWCHS and SWCHD

BASE24-atm customers using Mobile Top-Up functionality with the Boston Communications Group Interface:

- New subvolume, BA60XML, containing BASE XML files.
- New subvolume, SW60XML, containing files for the Generic XML Switch Interface and its documentation.
- New subvolume, SW60BCGI, containing files for the Boston Communications Group Interface (BCGI) and its documentation.

A complete list of the files affected by the BASE24-atm Mobile Top-Up Boston Communications Group Interface enhancement can be found in the SCN file BA60UD09.SCNBCGI.

CUSTMACS Changes

The following Make flags have been added to the CUSTMACS file.

Set the following flags to true when implementing the BASE24-atm Mobile Top-up and the Boston Communications Group Interface (BCGI). These flags must be set to true if the BCGI Interface is supported.

- base_xml_on
- generic_switch_xml_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

- For more information, please refer to the ***BASE24 Release 6.0 Implementation Guide (Version 5)***.

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Support for Response Code 85 in ARPC

This section describes the implementation requirements associated with the Support for Response Code 85 in Authorization Response Cryptogram (ARPC) enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Support for Response Code 85 in ARPC enhancement.

Logic has been added to the VisaNet Interchange Interface to enhance the processing of offline PIN management transactions performed on Visa cards.

A user field in the EMV Status Token has been redefined to indicate whether an ISO response code of '85' should be used in the ARPC generation for the transaction (for BASE24-authorised transactions) or should be returned to the ATM as the Authorization Response Code (for BASE24-acquired transactions).

For further information regarding this enhancement see **Support for Response Code 85 in ARPC** in Section 5.

BASE24 As Issuer

When BASE24-atm receives an offline PIN management request message from Visa, the VisaNet Interchange Interface will set the new field in the EMV Status Token to '1', indicating that an ISO response code of '85' should be used in the ARPC generation if the transaction is subsequently approved.

BASE24 As Acquirer

When BASE24-atm receives an approved offline PIN management response message from Visa, the VisaNet Interchange Interface will set the redefined user field in the EMV Status Token to '9', indicating that an ISO response code of '85' should be used as the Authorization Response Code (ARC) in the response to the ATM.

Software Impacts

All BASE24 VisaNet Interchange Interface customers:

- Inline changes to SW60IVIS.VISAEMVS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD09.SCNRC85.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Super Extract Process

Host code changes may be required for BASE24 users that utilize BASE24 Interchange Log File (ILF) Extract files to accept and process the redefined field in the EMV Status Token.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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XML Telco Interface

This section describes the implementation requirements associated with the enhancement to provide support for XML Telco Interface (XTI).

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Generic XML Interface (also known as the XML Telco Interface) enhancement.

All customers who use the BASE24-atm and/or BASE24-pos Mobile Top-Up XML Telco Interface (XTI) product must implement this change.

This enhancement implements support for BASE24-atm and/or BASE24-pos Mobile Top-up XML Telco Interface.

Software Impacts

All BASE24 customers:

- Changes to BA60MAKE.CUSTMACS and FLGSDOC
- Inline changes BA60SRC.BASEFM and BASEMM
- Software fix to BA60SRC.HISWUTLS

All BASE24 Interchange Interface customers:

- Inline changes to SW60SRC.SWIFM, SWIM and SWIMM

All BASE24 Interchange Generic Switch Interface customers:

- Inline changes to SW60SWCH.SWCHS and SWCHD

BASE24-atm customers using Mobile Top-Up functionality with the XTI:

- New subvolume, BA60XML, containing BASE XML files.
- New subvolume, SW60XML, containing files for the Generic XML Switch Interface and its documentation.
- New subvolume, SW60XTI, containing files for the XML Telco Interface (XTI) and its documentation.

A complete list of the files affected by the BASE24-atm and/or BASE24-pos Mobile Top-Up XML Telco Interface enhancement can be found in the SCN file BA60UD09.SCNXTI.

CUSTMACS Changes

The following Make flags have been added to the CUSTMACS file.

Set the following flags to true when implementing the BASE24-atm Mobile Top-up and the XTI. These flags must be set to true if the XTI is supported.

- base_xml_on
- generic_switch_xml_on
- xti_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param record that must be added to the LCONF to support this enhancement.

SW-ETUO-RETURN-0430

This param, used by the XML Telco Interface (XTI) module, specifies how the XML Telco Interface module returns a reversal completion to BASE24.

Valid values are:

- “Y” – the reversal of the AUTH funds does not happen until BASE24 gets a completion back from the TELCO.
- “N” – the AUTH finds are reversed immediately without waiting for the reversal completion from the TELCO.

Param Default: “N”.

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

10: BASE24 Value-Added Products

Section 10 discusses the changes and enhancements made to the BASE24 value-added products in release 6.0 version 6, along with the resulting conversion and implementation requirements that existing users of these products must consider when upgrading from release 6.0 version 5.

The value-added products discussed in this section include the following:

- BASE24-card
- BASE24-customer service
- BASE24-InfoBase
- BASE24-inventory Voucher Manager

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24 value-added product users, along with the associated conversion and implementation requirements that all BASE24 value-added product users must address.
- Optional enhancements describe the changes that affect only those BASE24 value-added product users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 value-added product users must address to implement the enhancements.

Section 10 must be used in conjunction with Section 4 to identify the changes that BASE24 value-added product users need to consider when upgrading to release 6.0 version 6.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all BASE24 Value-added product users must address in order to upgrade to release 6.0 version 6.

There are no enhancements affecting the following BASE24 value-added product users:

- BASE24-card
- BASE24-customer service
- BASE24-InfoBase

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

There are no general enhancements inherent to BASE24-inventory Voucher Manager.

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Optional Enhancements

This section identifies the BASE24 enhancements that can be enabled or disabled with release 6.0 version 6 that have an impact on the following BASE24 value-added products, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no enhancements affecting the following BASE24 value-added product users:

- BASE24-card
- BASE24-customer service
- BASE24-InfoBase

BASE24 value-added product users can choose whether or not to implement the enhancements described in this section. Users who plan to implement a particular enhancement must address the corresponding release 6.0 version 6 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding release 6.0 version 6 changes, conversion requirements, and implementation requirements for future reference.

BASE24 release 6.0 version 6 provides the initial release of BASE24-inventory Voucher Manager.

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BASE24-inventory Voucher Manager

This section describes the implementation requirements associated with the BASE24-inventory Voucher Manager initial release.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the BASE24-inventory Voucher Manager initial release.

The BASE24-inventory Voucher Manager product was introduced in the EMEA channel as a Channel Specific Product. The product has been taken forward into the standard BASE24 product stream.

The BASE24-inventory Voucher Manager product provides the software infrastructure that enables customers to manage stock inventory for Mobile Phone Operators and to interface with the BASE24 financial network to receive and respond to inventory authorization requests. It houses the inventory database that contains the stock files, transaction log files and configuration files. It consists of a Stock Manager, Housekeeping, Settlement, Refresh and Enform Report and Extract modules.

The Stock Manager process authorizes inventory transactions according to the stock type's pre-defined parameters and logs the transactions to the transaction log file. One process is defined for each stock type.

The Settlement process cuts over the stock types and inventory to the next business day.

The Generic Refresh process adds stock records to the stock file with input from the Mobile Operator systems. One process is defined for each stock type.

The Housekeeping process monitors the stock levels in the stock files and performs cleanup.

Enform reports are defined to provide information on the stock files and an Enform extract module will extract the transaction log file for transferring to a PC.

Prerequisite 1: The BASE changes in the BASE24-atm Mobile Top-Up Integration optional enhancement must be installed prior to implementation of BASE24-inventory Voucher Manager. For more information refer to the **BASE24 Release 6.0 Implementation Guide (Version 5)**. A complete list of the files affected by the release 6.0 version 5 mobile top-up support can be found in the SCN file BA60UD08.SCNMTOP.

Prerequisite 2: Either the BASE24-atm Mobile Top-Up Integration optional enhancement or the BASE24-pos Mobile Top-Up for POS optional enhancement must be installed in conjunction with this enhancement. For more information on the BASE24-atm Mobile Top-Up Integration optional enhancement, refer to the **BASE24 Release 6.0 Implementation Guide (Versions 5)**. For more information on the BASE24-pos Mobile Top-Up for POS optional enhancement, refer to **Optional Enhancements** in Section 6 above.

Software Impacts

All BASE24 customers:

- Inline changes to BA60AFT.MEGATBL, PITABLE, SECTBL and SVMHLPS
- Inline changes to BA60CSE.DESCTBL
- Changes to BA60DDL.DDLFOMF
- Inline changes to BA60EXT.EXTRM.
- Changes to BA60MAKE.CUSTMACS and FLGSDOC
- Addition of file BA60MISC.LCONFNV
- Changes to BA60MISC.CONTENTTS
- Changes to BA60SRC.CLSCDDL

Only BASE24-inventory Voucher Manager customers:

- New subvolumes NV60AFT, NV60DDL, NV60EXT, NV60HKP, NV60REFR, NV60RPTS, NV60SETL, NV60SMGR, NV60SRC, and NV60UC04.

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD09.SCNINV.

CUSTMACS Changes

Various Make flags have been added to the CUSTMACS file for BASE24-inventory Voucher Manager. All of the following Make flags were added with release 6.0 version 5.

Set the following flag to true for the BASE24-inventory Voucher Manager product.

- invtry_on

Set the following flag to true if the BASE24-inventory Voucher Manager Generic Housekeeping module is present.

- invtry_genrc_hkp_on

Set the following flag to true if the BASE24-inventory Voucher Manager Generic Refresh module is present.

- invtry_genrc_refr_on

Set the following flag to true if the BASE24-inventory Voucher Manager Settlement Initiator module is present.

- invtry_genrc_setl_on

Set the following flag to true if the BASE24-inventory Voucher Manager Stock Manager module is present.

- invtry_genrc_smgr_on

Set the following flag to true to include the mobile top-up infrastructure subvolume (BA60TMTU).

- mtu_on

The flag invtry_vodafone_refresh_on was added with release 6.0 version 5 and has been removed from CUSTMACS with this release.

LCONF Assign Changes

This section describes the assign records that must be added to or already exist in the LCONF to support this enhancement. Modifications for the new assigns are provided in the LCONFNV file located on BA60MISC subvolume.

INV-EXP-STOCK-CLEANUP-REP-LOC

The fully qualified file name of the report location for the BASE24-inventory Voucher Manager housekeeping expired stock cleanup task. This assign is used and required by BASE24-inventory Voucher Manager Housekeeping.

INV-GROUP-STOCK-DELETE-REP-LOC

The fully qualified file name of the report location for the BASE24-inventory Voucher Manager housekeeping group stock delete task. This assign is used and required by BASE24-inventory Voucher Manager Housekeeping.

INV-REFR-RPT

The fully qualified file name of the BASE24-inventory Voucher Manager Generic Refresh report location. This assign is used and required by BASE24-inventory Voucher Manager Generic Refresh.

INV-SOLD-STOCK-CLEANUP-REP-LOC

The fully qualified file name of the report location for BASE24-inventory Voucher Manager housekeeping sold stock cleanup task. This assign is used and required by BASE24-inventory Voucher Manager Housekeeping.

NRIF

The fully qualified file name of the BASE24-inventory Voucher Manager generic refresh input file (NRIF). This assign is used by BASE24-inventory Voucher Manager Generic Refresh and is required for this enhancement. This assign is process specific for each stock type supported.

NSCF

The fully qualified file name of the BASE24-inventory Voucher Manager Stock Configuration File (NSCF). This assign is used by BASE24-inventory Voucher Manager Stock Manager, Generic Refresh, Housekeeping and Settlement Initiator and is required for this enhancement.

NTLF

The fully qualified file name of the BASE24-inventory Voucher Manager Transaction Log File (NTLF). This assign is used by BASE24-inventory Voucher Manager Stock Manager and Settlement Initiator and is required for this enhancement.

The following LCONF assigns already exist. They are used by the BASE24-inventory Voucher Manager Stock Manager, Housekeeping, Refresh and Settlement Initiator and are required for this enhancement.

- LMCF
- SPANCTL
- MOF

LCONF Param Changes

This section describes the param records that must be added to or already exist in the LCONF to support this enhancement. Details of the new parameters are provided in the LCONFNV file located on BA60MISC subvolume.

INV-LN-CUTOVER

The BASE24-inventory Voucher Manager logical network cutover time. This param is used by the BASE24-inventory Voucher Manager Settlement Initiator process. It is a required param. Valid values are 0000 - 2359.

INV-LNET-CUTOVER-SERVICE

The service name of the processes that require a logical network cutover message from the BASE24-inventory Voucher Manager Settlement Initiator process. This param is used by the BASE24-inventory Voucher Manager Settlement Initiator process. It is a required param.

INV-SETL-ATM-CUT-SERVICE

The service name of the BASE24-atm processes that require a logical network cutover message from the BASE24-inventory Voucher Manager Settlement Initiator process. This param is used by the BASE24-inventory Voucher Manager Settlement Initiator process. It is an optional param.

INV-SETL-LN-CUTOVER-LEAD-TIME

The minimum number of minutes before BASE24-inventory Voucher Manager logical network cutover that all BASE24-inventory Voucher Manager stock manager processes must be cut over. This param is used by the BASE24-inventory Voucher Manager Settlement Initiator process. Valid values are 0 – 1440. Default is 30.

INV-SETL-NTLF-EXT-TO-ALLOCATE

The number of extents BASE24-inventory Voucher Manager Settlement Initiator allocates when creating new BASE24-inventory Voucher Manager Transaction Log Files (NTLF). This param is used by the BASE24-inventory Voucher Manager Settlement Initiator process. Valid values are 1 – 16. The default is 16.

INV-SETL-NTLF-FILE-RETENTION

The number of days the BASE24-inventory Voucher Manager Transaction Log Files (NTLF) are retained on disk. This param is used by the BASE24-inventory Voucher Manager Settlement Initiator process. Valid values are 2 – 32767. The default is 7.

INV-SETL-POS-CUT-SERVICE

The service name of the BASE24-pos processes that require a logical network cutover message from the BASE24-inventory Voucher Manager Settlement Initiator process. This param is used by the BASE24-inventory Voucher Manager Settlement Initiator process. It is an optional param.

INV-SMGR-STOCK-TYPE

The stock type that a specific BASE24-inventory Voucher Manager Stock Manager process controls. This param is used by BASE24-inventory Voucher Manager Stock Manager processes. It is a required param. The stock type must be a 4 character alphanumeric value, which needs to match the operator id field in the corresponding Mobile Operator File (MOF) record, and the stock type in the corresponding NSCF record. This param is (and must be) process specific for each Stock Manager process. A Stock Manager process can only support one stock type.

The following LCONF params already exist. They are used by the BASE24-inventory Voucher Manager Stock Manager, Housekeeping and Settlement Initiator and are required for this enhancement.

- LOGICAL-NET
- DATE-DISPLAY-TYPE

Database Configuration

Mobile Operator File (MOF)

The Mobile Operator File (MOF) is a required file for BASE24-inventory Voucher Manager and contains one record for each stock type supported. It defines details of the mobile operators and the services provided. The Operator ID of the MOF record must match the value of the LCONF param INV-SMGR-STOCK-TYPE for each of the stock types supported.

The MOF DDLs and MOF infrastructure were added with the BASE24 release 6.0 version 5 BASE24-atm Mobile Top-Up Integration enhancement. Information on use of the MOF file may be found in the **Base Files Maintenance Manual**.

New Data Files

BASE24-inventory Voucher Manager Stock Configuration File (NSCF)

The BASE24-inventory Stock Configuration File (NSCF) contains one record for each stock type maintained by the BASE24-inventory Voucher Manager system. The NSCF defines details of the stock type. The stock type defined must match the value of the LCONF param INV-SMGR-STOCK-TYPE for each of the Stock Manager processes defined. Information on use of the NSCF may be found in the **BASE24-inventory Reference Guide** publications document.

BASE24-inventory Voucher Manager Transaction Log File (NTLF)

The BASE24-inventory Voucher Manager Transaction Log File (NTLF) contains a record of every Inventory transaction (approved or denied) processed by the BASE24-inventory Voucher Manager system during a single processing day. The NTLF records are maintained by the BASE24-inventory Voucher Manager Settlement Initiator process for a set number of days.

The number of days each file is retained is defined in the INV-SETL-NTLF-FILE-RETENTION parameter in the Logical Network Configuration File (LCONF). The parameter default is seven. With a parameter value of seven, there are six NTLFs from the previous six days and the NTLF from the current day. The oldest NTLF is automatically purged by the BASE24-inventory Voucher Manager Settlement Initiator process once a new NTLF has been created.

The NTLF is used by the BASE24-inventory Voucher Manager report programs to produce their reports and can also be extracted daily to provide detailed transaction data for processing by a host.

Records are written to the NTLF sequentially, and each NTLF record has three alternate keys which allow for perusal of the file via BASE24-inventory Voucher Manager file maintenance functions.

A new NTLF is created by the BASE24-inventory Voucher Manager Settlement Initiator process at logical network cutover. Once a new NTLF is created, the previous NTLF is closed and is available for reporting and extract processing.

Information on the use of the NTLF may be found in the **BASE24-inventory Reference Guide** publications document.

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

The NSCF requester name was added to the MEGATBL file.

PI-Table Changes

The Inventory product indicator was added to the PI-TABLE section of the PITABLE file. This product indicator must be set to Y for this enhancement.

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Entries were added to the SECTBL to support the new NSCF and NTLF screens. These files will appear on the INVENTORY Product Menu.

PATHCONF Changes

The new NSCF server must be defined in the PATHCONF file as SERVER-NSCF. The new NTLF server must be defined in the PATHCONF file as SERVER-NTLF. These servers can be added to the system online through the PATHCOM utility. After adding the servers to the PATHCONF file, the BASE24 user must perform a PATHCOLD operation to actually gain access to the newly defined server.

CRT Access Security Changes

Access to the new NSCF and NTLF screens must be added through the SEC requester for the appropriate users.

Routing Changes

Not applicable

Network Configuration

New processes for the BASE24-inventory Voucher Manager Settlement Initiator, Housekeeping, Refresh, and Stock Manager processes must be added to any XPNET Node that processes BASE24-inventory Voucher Manager traffic. One BASE24-inventory Voucher Manager Refresh and Stock Manager process must be added for each stock type supported. The Stock Manager processes must have the service name defined as the value in the LCONF parameter INV-LNET-CUTOVER-SERVICE to correctly cutover to the next business day.

The following are examples of the required processes defined under XPNET 3.0 or XPNET 3.1. These examples are intended to aid BASE24 users in their initial setup and should be altered to fit specific environments.

```
RESET PROCESS
SET PROCESS BCPU 0
SET PROCESS LIBRARY $VOLUME.XPNET.SKELB
SET PROCESS PROGRAM $VOLUME.NV60OBJ.NSETL
SET PROCESS PPD $T1P1
SET PROCESS PRIORITY 145
SET PROCESS CPU 1
SET PROCESS STARTUP AUTOMATIC
SET PROCESS QAT 64
ADD PROCESS P1A^NSETL, UNDER SYSNAME \<sys>, UNDER NODE P1A^NODE
```

```
RESET PROCESS
SET PROCESS BCPU 0
SET PROCESS LIBRARY $VOLUME.XPNET.SKELB
SET PROCESS PROGRAM $VOLUME.NV60OBJ.HKP
SET PROCESS PPD $T1P2
SET PROCESS PRIORITY 145
SET PROCESS CPU 1
SET PROCESS STARTUP AUTOMATIC
SET PROCESS QAT 64
ADD PROCESS P1A^NHKP, UNDER SYSNAME \<sys>, UNDER NODE P1A^NODE
```

```
RESET PROCESS
SET PROCESS BCPU 0
SET PROCESS LIBRARY $VOLUME.XPNET.SKELB
SET PROCESS PROGRAM $VOLUME.NV60OBJ.NREFR
SET PROCESS PPD $T1P3
SET PROCESS PRIORITY 145
SET PROCESS CPU 1
SET PROCESS STARTUP AUTOMATIC
SET PROCESS QAT 64
ADD PROCESS P1A^NREFR, UNDER SYSNAME \<sys>, UNDER NODE P1A^NODE
```

```
RESET PROCESS
SET PROCESS BCPU 0
SET PROCESS LIBRARY $VOLUME.XPNET.SKELB
SET PROCESS PROGRAM $VOLUME.NV60OBJ.SMGR
SET PROCESS PPD $T1P4
SET PROCESS PRIORITY 145
SET PROCESS CPU 1
SET PROCESS STARTUP AUTOMATIC
SET PROCESS QAT 64
SET PROCESS SERVICE INVSETL
ADD PROCESS P1A^NSMGR, UNDER SYSNAME \<sys>, UNDER NODE P1A^NODE
```

For more information about adding Processes to the XPNET system, refer to the ***NET24-XPNET Network Control Operations Guide*** or the ***NET24-XPNET Network Control Command Reference Manual***.

File Conversions

BASE24-inventory Voucher Manager Stock Configuration File (NSCF)

The NSCF conversion program was created to convert existing release 6.0 (EMEA Channel Product) NSCF records which have eight occurrences of stock denominations and thresholds to have 20 occurrences of stock denominations and thresholds. The NSCF conversion program must be run to convert the NSCF if an existing customer is running with an NSCF that contains eight occurrences of stock denominations and thresholds.

The NSCF conversion program on the NV60UC04 subvolume can be used to convert AST records from the release 6.0 previous version format to the release 6.0 version 6 format. If modifications to the conversion program are necessary, the program can be modified by editing the CNVNSCFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#cnvnscf.t,name $nscfm,nowait/ cnvnscfm –  
force –list –listreplace
```

Users should update the CNVNSCFR obey file on the NV60UC04 subvolume with the appropriate information. The CNVNSCFR file should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the NV60UC04 subvolume.

Host/Co-Network Impacts

Super Extract Process

Host code changes are required for all BASE24-inventory Voucher Manager users that use OMF Extract to accept and process information for the new NSCF records.

Inventory Refresh Process

The BASE24-inventory Voucher Manager Stock Item File (NSIF) is populated via the BASE24-inventory Voucher Manager Refresh process. The BASE24-inventory Voucher Manager Refresh process takes an input file from the Mobile Operator and partially refreshes the NSIF file with the new stock records. The Host is responsible for providing the refresh input file from the Mobile Operator in the format defined by the BASE24-inventory Voucher Manager Refresh Input Format (NRIF).

Enform Inventory Transaction Log File Extract

The BASE24-inventory Voucher Manager Transaction Log File (NTLF) can be extracted daily via the Enform NTLF query file. This produces a disk file that can be transferred to a PC which the Host may use for settlement purposes.

User/Operational Impacts

Disk Impacts

The new BASE24-inventory Voucher Manager Stock Configuration File (NSCF), BASE24-inventory Voucher Manager Stock Item File (NSIF) and BASE24-inventory Voucher Manager Transaction Log File (NTLF) files impact disk requirements. The total impact varies for each BASE24 user depending on the number of NSCF records and NSIF records defined in the BASE24 system and the number of inventory transactions processed by the BASE24 system.

Inventory Refresh Process

The refresh input file accepted by the BASE24-inventory Voucher Manager Refresh process is an ENSCRIBE TYPE E file with a record size of 80. Operators must be trained in the steps to format the refresh input file from the Mobile Operators into the BASE24-inventory Voucher Manager Refresh Input Format (NRIF) and to perform a refresh of the NSIF file. Information on the performing the refresh function may be found in the ***BASE24-inventory Reference Guide*** publications document.

Housekeeping Functions

Operators must be trained to recognize and respond to the BASE24-inventory Voucher Manager Housekeeping functions of stock monitoring and stock deletion. Reports are produced to assist with the cleanup of the BASE24-inventory Voucher Manager Stock Item File (NSIF). Event (log) messages are produced when stock levels are running low. Information on the housekeeping functions may be found in the ***BASE24-inventory Reference Guide*** publications document.

Enform Extract and Report Generation

Operators must be trained on the steps to execute the Enform NTLF Extract and run the Enform Reports. Refer to the file \$<volume>.NV60RPTS.AAREADME for a description of the steps.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

A: Logical Network Configuration File Changes

Appendix A contains the Logical Network Configuration File (LCONF) changes.

This information includes assigns and params that have been added, deleted or modified with release 6.0 version 6 along with their associated page numbers. Refer to the specified page numbers for explanations of how these assigns and params can or must be set.

Refer to the ***BASE24 Logical Network Configuration File (LCONF) Manual*** for instructions on adding them to the LCONF.

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ASSIGNS

The following assigns were added or modified for release 6.0 version 6. Refer to the specified page numbers for explanations of how these assigns can or must be set.

BASE24-inventory Voucher Manager

INV-EXP-STOCK-CLEANUP-REP-LOC	10-9
INV-GROUP-STOCK-DELETE-REP-LOC.....	10-9
INV-REFR-RPT	10-10
INV-SOLD-STOCK-CLEANUP-REP-LOC	10-10
NRIF	10-10
NSCF.....	10-10
NTLF.....	10-10

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PARAMS

The following params were added or modified for release 6.0 version 6. Refer to the specified page numbers for explanations of how these params can or must be set.

BASE24-inventory Voucher Manager

INV-LN-CUTOVER.....	10-11
INV-LNET-CUTOVER-SERVICE.....	10-11, 10-15
INV-SETL-ATM-CUT-SERVICE	10-11
INV-SETL-LN-CUTOVER-LEAD-TIME.....	10-11
INV-SETL-NTLF-EXT-TO-ALLOCATE	10-11
INV-SETL-NTLF-FILE-RETENTION.....	10-12, 10-13
INV-SETL-POS-CUT-SERVICE.....	10-12
INV-SMGR-STOCK-TYPE	10-12, 10-13

Complex Passwords in Pathway

PASSWORD-COMPLEX	4-15, 4-16
PASSWORD-MAX-CHNG-PER-DAY	4-16
PASSWORD-MAX-STORE	4-17
PASSWORD-NUM-ALPHA	4-17
PASSWORD-NUM-NUMERIC	4-17
PASSWORD-NUM-SPCL	4-17

Cutover Performance Improvements

ATM-AUTH-IDF-DATES-COMPUTE	5-15, 5-17
POS-RTAU-IDF-DATES-COMPUTE.....	6-27, 6-28, 6-29

ISO Host Interface Enhanced Reversal Routing

POS-RVSL-ROUTE	6-50, 6-51
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Mobile Top-Up for POS

POS-SETL-NOTIFYidentifier.....	6-76
POS-SPLIT-TXN-RTG-DEST	6-76

Shared (Wincor) 912 AKDS Signatures

ATM-DH-1000-AKDS-SERIAL-NUM-VRFY	5-39
ATM-DH-AKDS-TSS-TIMER	5-39

Shared (Wincor) NDC+ AKDS Signatures

ATM-DH-AKDS-TSS-TIMER	5-53
ATM-DH-N50-AKDS-SERIAL-NUM-VRFY	5-53

Suppression of POS Issuer Discretionary Data

POS-ISS-AVS-DATA-SUPPRS	6-102
POS-ISS-CVD2-DATA-SUPPRS	6-102
POS-ISS-DISCR-DATA-SUPPRS.....	6-102

XML Telco Interface

SW-ETUO-RETURN-0430	9-20
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B: Host/CO-Network Impacts

Appendix B contains a list of release 6.0 version 6 enhancements that have host/co-network impacts, along with their associated page numbers. Refer to the specified page numbers for more information about these enhancements.

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Enhancements

The following enhancements have host/co-network impacts. Refer to the specified page numbers for more information about these enhancements.

BASE24-inventory Voucher Manager.....	10-17
Complex Passwords in Pathway	4-19
Diebold Bulk Note Acceptance	5-22
Diebold EMV Offline PIN Synchronization.....	5-30
Expand the Number of RTBL & AST Entries.....	6-14
Interac Online Payment	6-47
ISO Host Interface Enhanced Reversal Routing.....	6-51
ISO Host Interface Stored Value Support.....	6-55
Issuer Declined Manually Entered Transactions	6-60
MasterCard Secure Code Support Validation.....	6-65
Mobile Top-Up for POS	6-83
SPDH EMV Transaction Certificates	6-95
Shared (Wincor) 912 AKDS Signatures	5-55
Shared (Wincor) NDC+ AKDS Signatures	5-41
Support For Response Code 85 In ARPC.....	5-62
Suppression of Issuer Discretionary Data.....	6-104

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C: CUSTMACS File Changes

Appendix C contains a list of release 6.0 version 6 enhancements that have CUSTMACS changes, along with their associated page numbers. Refer to the specified page numbers for more information about these enhancements.

BASE24 users should review these changes to determine how the CUSTMACS file should be set for their BASE24 environment before running MAKE to compile their system. More information on all MAKE flags used within BASE24 can be found in the FLGSDOC file located on the BA60MAKE subvolume delivered with BASE24 Release 6.0 Version 6 code.

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Enhancements

The following enhancements have CUSTMACS impacts:

BASE24-inventory Voucher Manager

invtry_genrc_hkp_on	10-8
invtry_genrc_refr_on	10-9
invtry_genrc_setl_on	10-9
invtry_genrc_smgr_on	10-9
invtry_on	10-8
mtu_on	10-9

Boston Communications Group Interface Restructure

base_xml_on	9-12
generic_switch_xml_on	9-12

Mobile Top-Up for POS

mtu_on	6-75
pos_dh_spdh_mtu_on	6-75
pos_tcm_mtu_on	6-75
pos_tcm_on	6-75

Shared (Wincor) 912 AKDS Signatures

atm_dh_c1000_shrd_akd_on	5-38
--------------------------------	------

Shared (Wincor) Diebold 10XX/478X EMV Extension

atm_dh_c1000_win_emv_on	5-46
-------------------------------	------

Shared (Wincor) NDC+ AKDS Signatures

atm_dh_n50_shrd_akd_on	5-52
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Tidel EMV

atm_dh_tidel_emv_on	5-63
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XML Telco Interface

base_xml_on	9-20
generic_switch_xml_on	9-20
xti_on	9-20

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D: Re-Sequenced Files and New Files

Appendix D consists of the following information:

- A table illustrating the new files added to BASE24 for the release 6.0 version 6 enhancements.
- A table illustrating the files resequenced as part of the release 6.0 version 6 enhancements.
- Information regarding the resequencing of the NCR 5xxx Device Handler.

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New Files Added to BASE24 with Version 6

The following table lists the files added by the release 6.0 version 6 enhancements:

Enhancement Name	Subvolume	Filename	Language
Base_axcel_directive changes	BA60SRC	AXCLOBEY	N/A
BASE24 OMF Audit Report	BA60RPT	BARPT03M	N/A
		BARPT03R	N/A
		BARPT03S	COBOL
		BARPT04M	N/A
		BARPT04R	N/A
		BARPT04S	COBOL
		BARPTFM	N/A
		BARPTM	N/A
		BARPTMM	N/A
		OMFTBL	COBOL
BASE24 Pathway Complex Password	BA60DDLD	BADDLDTD	N/A
		SEC08	N/A
		SECG08	N/A
	BA60RPT	BARPT01M	N/A
		BARPT01R	N/A
		BARPT01S	COBOL
		BARPT02M	N/A
		BARPT02R	N/A
		BARPT02S	COBOL
		BARPTFM	N/A
		BARPTM	N/A
		BARPTMM	N/A
		BRCOUTLS	COBOL
	BA60UC09	AAREAD09	N/A
		BAUC09FM	N/A
		BAUC09M	N/A
		BAUC09MM	N/A
		CNVSECM	N/A
		CNVSECR	N/A
		CNVSECS	TAL
BCG Interface Restructure Enhancement	SW60BCGI	BCGIDCD	XML
		BCGITH	TAL
Declining Manually Entered cards for issuer trans	BA60DDLD	CPF08	N/A
	BA60UC09	AAREAD09	N/A
		BAUC09FM	N/A
		BAUC09M	N/A
		BAUC09MM	N/A
		CNVCPFM	N/A
		CNVCPFR	N/A
		CNVCPFS	TAL

Re-Sequenced Files and New Files
New Files Added to BASE24 with Version 6

Enhancement Name	Subvolume	Filename	Language
Expand the Number of RTBL & AST Entries	BA60UC09	AAREAD09	N/A
	BP60DDLD	AST08	N/A
		BPDDLDTD	N/A
	BP60UC04	AAREAD04	N/A
		BPUC04FM	N/A
		BPUC04M	N/A
		BPUC04MM	N/A
		CNVASTM	N/A
		CNVASTR	N/A
		CNVASTS	TAL
		CNVRTBLM	N/A
		CNVRTBLR	N/A
		CNVRTBLS	TAL
	BP60UC09	AAREAD09	N/A
		BPUC09FM	N/A
		BPUC09M	N/A
		BPUC09MM	N/A
		CNVASTM	N/A
		CNVASTR	N/A
		CNVASTS	TAL
		CNVRTBLM	N/A
		CNVRTBLR	N/A
		CNVRTBLS	TAL
Generic XML Switch Enhancement	BA60XML	BAXMLFM	N/A
		BAXMLM	N/A
		BAXMLMM	N/A
		XMLAPPD	TAL
		XMLAPPG	TAL
		XMLAPPM	N/A
		XMLAPPS	TAL
		XMLEMSG	TAL
		XMLEMSM	N/A
		XMLEMSS	TAL
		XMLERRG	TAL
		XMLFILG	TAL
		XMLFILM	N/A
		XMLFILS	TAL
		XMLG	TAL
		XMLM	N/A
		XMLMSGG	TAL
		XMLMSGM	N/A
		XMLMSGS	TAL
		XMLS	TAL
		XMLSQLG	TAL
		XMLSQLM	N/A
		XMLSQLS	TAL
		XMLSTAG	TAL

Re-Sequenced Files and New Files
New Files Added to BASE24 with Version 6

Enhancement Name	Subvolume	Filename	Language
		XMLSTAM	N/A
		XMLSTAS	TAL
		XMLTRCG	TAL
		XMLTRCM	N/A
		XMLTRCS	TAL
		XMLUDTG	TAL
		XMLUDTM	N/A
		XMLUDTS	TAL
		XMLXMLG	TAL
		XMLXMLM	N/A
		XMLXMLS	TAL
	SW60XML	SWCHXM	N/A
		SWXMLFM	N/A
		SWXMLM	N/A
		SWXMLMM	N/A
		SXMLAPPD	N/A
		SXMLAPPG	TAL
		SXMLAPPM	N/A
		SXMLAPPS	TAL
		SXMLDDL	N/A
		SXMLDDL	DDL
		SXMLM	N/A
		SXMLS	TAL
		SXMLSEMG	TAL
		SXMLSEMM	N/A
		SXMLSEMS	TAL
Mobile Top-up Inventory Integration with BASE24	BA60MISC	LCONFV	N/A
	NV60AFT	NVAFTFM	N/A
		NVAFTM	N/A
		NVAFTMM	N/A
		RQNSCFM	N/A
		RQNSCFS	COBOL
		RQNTLFM	N/A
		RQNTLFS	COBOL
		SCRNNSCF	Screen File
		SCRNNTLF	Screen File
		SVNSCFM	N/A
		SVNSCFS	COBOL
		SVNTLFM	N/A
		SVNTLFS	COBOL
	NV60DDL	DDLFNRF	DDL
		DDLFNSCF	DDL
		DDLFSIF	DDL
		DDLFNRLF	DDL
		DDLFNRLF	DDL
		DDLGNSTM	DDL
		NVDDLFM	N/A

Re-Sequenced Files and New Files
New Files Added to BASE24 with Version 6

Enhancement Name	Subvolume	Filename	Language
	NV60EXT	NVDDLMM	N/A
		NVDDLMM	N/A
		NVEXTFM	N/A
		NVEXTM	N/A
		NVEXTMM	N/A
		NVOMFXG	TAL
		NVOMFXM	N/A
		NVOMFXS	TAL
	NV60HKP	HKPDDLMM	N/A
		HKPDDL	DDL
		HKPG	TAL
		HKPM	N/A
		HKPS	TAL
		NVHKPFM	N/A
		NVHKPM	N/A
		NVHKPMM	N/A
	NV60REFR	NREFRG	TAL
		NREFRM	N/A
		NREFRS	TAL
		NVREFRFM	N/A
		NVREFRMM	N/A
	NV60RPTS	AAREADME	DDL
		COUNTSTK	N/A
		ENFNTLF	N/A
		ENFNTLFL	N/A
		ENFSTOCK	N/A
		GONTLF	N/A
	NV60SETL	NSETLG	TAL
		NSETLM	N/A
		NSETLS	TAL
		NVSETLFM	N/A
		NVSETLMM	N/A
	NV60SMGR	NVSMGRFM	N/A
		NVSMGRMM	N/A
		SMGRG	TAL
		SMGRM	N/A
		SMGRS	TAL
	NV60SRC	INVTRYFM	N/A
		INVTRYM	N/A
		INVTRYMM	N/A
		NVCOUTLS	COBOL
		NVSRM	N/A
	NV60UC04	CNVNSCFM	N/A
		CNVNSCFR	N/A
		CNVNSCFS	TAL
		NVUC04FM	N/A
		NVUC04MM	N/A

Re-Sequenced Files and New Files
New Files Added to BASE24 with Version 6

Enhancement Name	Subvolume	Filename	Language
Mobile Top-up POS Acquiring BASE24 Integration	BA60LOGM	PSTCM	N/A
	PS60TCM	PSTCMFM	N/A
		PSTCMMM	N/A
		PTCMD	N/A
		PTCMG	TAL
		PTCMLOGS	TAL
		PTCMM	N/A
		PTCMS	TAL
	PS60TSPD	PSTSPDFM	N/A
		PSTSPDMM	N/A
		SPDHMTUG	TAL
		SPDHMTUM	N/A
		SPDHMTUS	TAL
		TSPDNAMS	TAL
	PS60TTCM	PSTTCMFM	N/A
		PSTTCMMM	N/A
		PTCMMTUG	TAL
		PTCMMTUM	N/A
		PTCMMTUS	TAL
Shared (Wincor) 912 AKDS	AT60AW10	ATAW10FM	N/A
		ATAW10MM	N/A
		C10WAKDG	TAL
		C10WAKDM	N/A
		C10WAKDS	TAL
	AT60DDLD	ATD08	N/A
		ATDDLDTD	N/A
	AT60UC09	AAREAD09	N/A
		ATUC09FM	N/A
		ATUC09M	N/A
		ATUC09MM	N/A
		CNVATDM	N/A
		CNVATDR	N/A
		CNVATDS	TAL
	BA60UC09	AAREAD09	N/A
Shared (Wincor) Diebold 10xx/478x EMV Extension	AT20IBLD	ENTWAPPS	TAL
	AT60IW10	ATIW10FM	N/A
		ATIW10MM	N/A
		C10WEMVG	TAL
		C10WEMVM	N/A
		C10WEMVS	TAL
Shared (Wincor) NDC+ AKDS	AT60AWN5	ATAWN5FM	N/A
		ATAWN5MM	N/A
		N50WAKDG	TAL
		N50WAKDM	N/A
		N50WAKDS	TAL
	AT60DDLD	ATD08	N/A

Re-Sequenced Files and New Files
New Files Added to BASE24 with Version 6

Enhancement Name	Subvolume	Filename	Language
	AT60UC09	AAREAD09	N/A
		ATUC09FM	N/A
		ATUC09M	N/A
		ATUC09MM	N/A
		CNVATDM	N/A
		CNVATDR	N/A
		CNVATDS	TAL
	BA60UC09	AAREAD09	N/A
Test Compatibility of MHI with TSS	PS60DDLD	MHI08	N/A
		PSDDLDTD	N/A
	PS60UC05	TSSMHC FM	N/A
		TSSMHC FR	N/A
Tidel EMV	AT60ITDL	TSSMHC FS	TAL
		ATITDLFM	N/A
		ATITDLMM	N/A
		TIDLEMVG	TAL
		TIDLEMVM	N/A
		TIDLEMVS	TAL
	AT60TIDL	TIDELD	TAL
XML Telco Interface Enhancement (XTI)	SW60XTI	SWXTIM	N/A
		XISODDLM	N/A
		XISODDLS	DDL
		XTIAPPG	TAL
		XTIAPPM	N/A
		XTIAPPS	TAL
		XTICALG	TAL
		XTICALM	N/A
		XTICALS	TAL
		XTICNVG	TAL
		XTICNVM	N/A
		XTICNVS	TAL
		XTIDCD	N/A
		XTIDDLM	N/A
		XTIDDL S	DDL
		XTIFM	N/A
		XTIFMTG	TAL
		XTIFMTM	N/A
		XTIFMTS	TAL
		XTIG	TAL
		XTIM	N/A
		XTIMM	N/A
		XTIRSPS	TAL
		XTIS	TAL
		XTITH	TAL

Files Resequenced with Version 6

The following table lists the files resequenced as part of the release 6.0 version 6 enhancements:

Enhancement Name	Subvolume	Filename	Language
Boston Communications Group (BCGI) Interface	SW60BCGI	BCGIAPPG	TAL
		BCGIAPPM	MAKE
		BCGIAPPS	TAL
		BCGICALG	TAL
		BCGICALM	MAKE
		BCGICALS	TAL
		BCGICNVG	TAL
		BCGICNVM	MAKE
		BCGICNVS	TAL
		BCGIDDLM	MAKE
		BCGIDDLIS	DDL
		BCGIFM	MAKE
		BCGIFMTG	TAL
		BCGIFMTM	MAKE
		BCGIFMTS	TAL
		BCGIG	TAL
		BCGIM	MAKE
		BCGIMM	MAKE
		BCGIS	TAL
		SWBCGIM	MAKE

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Re-sequencing of the N50S File

The following information is in reference to the re-sequencing of the NCR N50S device handler source file. This affects all customers with a license for the NCR 5XXX Device Control (AT-DA031).

ACI re-evaluated BASE24 release 6.0 and determined that it was necessary to re-sequence the N50S file due to change activity for fixes and enhancements.

The AT60N50.N50S file was re-sequenced and made available to customers on October 18, 2005. It was delivered prior to the release of BASE24 release 6.0 version 6 to allow customers the opportunity to add any Customer Specific Modifications (CSMs) into the newly re-sequenced file before a fix needs to be applied to the newly re-sequenced N50S.

Below are detailed instructions for working with the newly re-sequenced source file. Additionally, a special report has been provided that maps the new sequence number to the corresponding former sequence number of each line. This will aid in the insertion of custom code into the newly re-sequenced code.

Working with the Re-sequenced N50S File

Customers who license the NCR 5xxx Device Control fall into three general categories regarding the re-sequencing of the N50S file. Regardless of the method used, any fixes added to the N50S file since the last time fixes have been applied should be reviewed. All dependencies should be checked to verify that other NCR and Non-NCR device handler files are in sync with the N50S file.

Listed below these categories are the general guidelines for use when working with the re-sequenced N50S file:

1. No CSM Code

- Replace the old code with the re-sequenced N50S file.

2. CSM code using the proc replacement methodology

- Create a new N50QS by copying the history section of the old N50QS and the corresponding PROCs from the newly re-sequenced N50S file.
- Choose one of the methods below for moving CSM code into the new N50QS file.

3. Inline CSM code

- Choose one of the methods below for moving CSM code into the new N50S file.

There are three general methods for moving CSM changes into the newly re-sequenced N50S file, with many variations on each method depending upon your environment and personal preferences.

1. Using EDIT or TEDIT - A special mapping file (N50SEQ) has been provided that will aid in the proper placement of CSM code. This file is a copy of the re-sequenced N50S file and will include, on each line of code, both the new sequence number and the corresponding old sequence number for that line of code. This file will not compile and is to be used only as a tool.
 - In the CSM source file, find the sequence number immediately preceding the beginning of each CSM code section
 - Search for this number in the special mapping file to locate the corresponding sequence number in the N50S file
 - Insert your CSM code at the appropriate new sequence number
2. Using CodeWright - During re-sequencing, all redundant blank lines were removed from the source code. The CodeWright difference function does not do a good job of displaying changes side-by-side when multiple blank lines are removed from the source code. Using the CodeWright difference function is probably the least effective method. CodeWright can, however, be used quite successfully by cutting and pasting between screens.
 - Open the CSM code on one screen
 - Open the newly re-sequenced N50S file on a second screen
 - Open the N50SEQ on a third screen to locate corresponding sequence numbers
3. Using BeyondCompare – This is a tool that is very reasonable in cost and makes the migration of CSM code into the newly re-sequenced N50S file very easy. If your code is heavily customized, ACI suggests using the BeyondCompare tool available from Scooter Software (<http://www.scootersoftware.com/>). Scooter offers a 30-day trial version of BeyondCompare that can be used to evaluate the benefits of the tool.

ACI uses BeyondCompare for text file and folder comparisons on a frequent basis. The side-by-side comparison of files that have blank lines removed is superior to that of CodeWright. Using BeyondCompare is by far the quickest method in testing different ways of reinserting custom code into the newly re-sequenced N50S source code.

Following are suggestions for setting up the BeyondCompare environment:

- Strip off the sequence numbers in the CSM'd code
- Open the CSM code in the left window and the re-sequenced code in the right window
- In the menu bar, choose Tools/Pick Rules: select "Everything Else"
- In the menu bar, choose Tools/Edit Current Rules
- In the File Comparison Rules dialog box, choose Importance
- Check the Blank Lines box
- Click the first NEW button
- In the Unimportant Text dialog box, select Category/ Columns. In the Start column box, enter 71 and check the stopping at end of line box
- In the Toolbar, you have options to view all text, just changes, changes with context, and just matching text
- CSM code will appear in RED in the left window. Select lines and move to the re-sequenced code using the toolbar arrow
- Save modified source file

Note: Regardless of the method used, the customer should review any fixes that were added to N50S since that last time they applied fixes to this file. All dependencies should be checked to verify that other NCR and non-NCR device handler files are in sync with the N50S file.

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E: Moved, Renamed, Deleted Files

Appendix E contains a list of files that were Moved, Renamed, or Deleted in BASE24 Release 6.0 Version 6.

No files were moved, renamed or deleted in release 6.0 version 6.

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F: Regional Specific Information

Appendix F contains the information that is required to implement the changes for release 6.0 version 6 that are specific to each region.

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